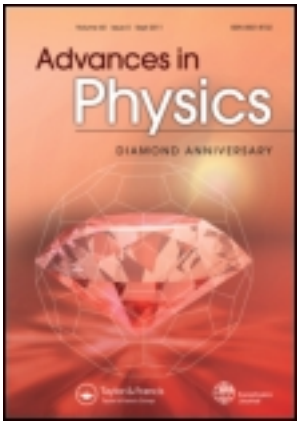
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Supersymmetry and theory of disordered metals

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Supersymmetry and theory of disordered metals

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ABSTRACT

Theoretical studies of the motion of an electron in a disordered system have led to a number of remarkable results especially those concerned with localization and with the importance of dimensionality. The theoretical approaches previously used run into some severe technical difficulties. This article sets out the bases of the recently introduced supersymmetry method which has already, as the article shows, provided unified derivations of the major results in this subject, free of the technical problems. The method reduces the systems it treats to field-theoretical models involving both bosons and fermions.

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§1. Introduction

Nowadays much attention in solid-state physics is given to the study of disordered systems, theoretically and experimentally. The interest in this class of substance is due to some quite new physical properties which they show, different from those of regular crystals. In certain conditions new phases can appear which are not possible in regular crystals. The essential physical differences become apparent in attempting to construct the theory of disordered systems. In the most interesting cases standard methods used in the study of the crystals no longer work. From the mathematical point of view, this is due to the fact that now it is necessary to calculate averaged physical quantities such as the free energy and conductivity, but not the partition function. As a result, the perturbation theory applicable in the limit of the weak disorder was for a long time the only method for the investigation of disordered systems. One-dimensional systems were an exception because one might use the Fokker–Planck equations for them.

In 1975, Edwards and Anderson, when studying a disordered system of spins proposed a new method for the investigation of disordered systems— the so-called replica method. In this method one replaces the disordered system by n systems which are identical to the original. Then, instead of calculating the free energy $F=T\ln z$, one calculates the quantity $F_n=-T[(z^n-1)/n]$. The limit of F_n as $n\to 0$ coincides with the free energy F. It is remarkable that one can average the energy F_n over defects before all other calculations. After the averaging, the partition function of a regular model is obtained and standard methods developed for regular systems can be used. However, the replica method suffers grave shortcomings. Usually the hamiltonians obtained by the replica procedure are very complicated and cannot be solved exactly. Standard approximations which are valid when n is an integer often fail in the limit $n\to 0$. Furthermore the procedure of analytical continuation from integer n to $n=0$ is not simple. Some additional conditions (e.g. at infinity) are necessary for unambiguous continuation. As far as is known to the author, such additional conditions have not yet been discussed. These difficulties often lead to meaningless results when the replica method is applied. This fact was first observed by Sherrington and Kirkpatrick (1975) when they studied spin glasses with long-range interactions. The lack of a justified procedure for analytical continuation $n\to 0$ is the reason why it is not possible to be quite sure of the results when there is no check by an independent method.

In this paper we review the progress in the theory of disordered metals achieved by another method— the supersymmetry method (Efetov 1982 a, b, Efetov and Larkin 1983, Bohr and Efetov 1982). This method is not as general as the replica method and is applicable only to the description of the motion of non-interacting particles in a random potential. It is based on the use of functional integrals over both commuting and anticommuting variables. This makes it possible to represent physical quantities in a form which enables us to average over impurities at the very beginning of calculations (McKane 1980, Parisi and Sourlas 1981). After averaging, the problem is reduced to a field-theory model containing both bosons and fermions and having symmetry (supersymmetry) under the group of transformations mixing bosons and fermions. The models obtained can be studied by the traditional methods applied to regular models. The method based on the fermion–boson representation (called the supersymmetry method) has many advantages though it can be used for a narrower class of problems than the replica method. It is free of the problem of

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analytical continuation. Furthermore the effective hamiltonians contain a number of variables which is not large. They are, therefore, simpler than those in the replica method. Several one-particle problems which can only be solved by the supersymmetry method will be considered here. In addition some results which have been obtained by other methods will be reproduced. It will be seen that, though the supersymmetry method is applicable only for the description of non-interacting particles, it enables us to study a wide class of physical phenomena.

The interest in the study of the behaviour of a particle in a random potential arose mainly after Anderson's original work (Anderson 1958). In this an assertion is made about the localization of states of the particles having energies less than a critical value. In the case of a sufficiently strong disorder the conductivity of a metal is zero. Later it became clear that the localization problem essentially depends on the dimensionality. Mott and Twose (1961) noticed that in a one-dimensional disordered chain the conductivity is zero for an arbitrarily weak disorder. All the electron states in the chain are localized. This result was later obtained rigorously (Berezinsky 1973, Abrikosov and Ryzhkin 1976, Berezinsky and Gor'kov 1979). A stronger assertion about the localization in arbitrarily weakly disordered, sufficiently long, thin wires has been made by Thouless (1977). Abrahams *et al.* (1979) gave arguments in favour of localization for an arbitrarily weak disorder even in two dimensions. The existence of the critical energy separating localized and extended states has been proven in a Cayley tree model (Abou-Chacra *et al.* 1973). One should remark that only the calculations for the chains and those for the Cayley tree model are reliable. In other cases, conclusions about localization were obtained on the basis of qualitative arguments or uncontrolled, though plausible, approximations (Götze 1978, 1979, 1981, Wölfle and Vollhardt 1980 a, b).

Another problem which is closely related to localization is that of energy level statistics for metallic particles with a limited volume. The statistical description was first proposed by Wigner (1951) for the calculation of level correlations in complex nuclei. In such nuclei a large number of particles interact in an unknown way. It is then reasonable to assume that all the interactions have equal probabilities, but a rigorous mathematical definition of the equal probabilities is a rather difficult problem. One of the most convenient ensembles defined by Wigner is the gaussian ensemble. Further development of the Wigner ideas was presented in the works of Porter and Rosenzweig (Porter 1965). A very important step was taken by Dyson (1962) whose statistical hypothesis was that the level distribution in a real system is equivalent to that of the eigenvalues on the unit circumference of the unitary $N \times N$ matrix S, where N is the number of levels in the system. On this basis the level-spacing distribution was calculated (Dyson 1962, Mehta and Dyson 1963).

In metallic particles complex interactions with defects located inside the volume or on the surface are possible even in the absence of interactions between the electrons. In principle, the Dyson arguments can be applied to any complex system and are not restricted to nuclei. Gor'kov and Eliashberg (1965) used them for a calculation of the response to external electromagnetic fields of small metallic particles. In order to calculate this response it is necessary to know the two-level correlation function. Gor'kov and Eliashberg did not calculate this function directly but used that resulting from the Dyson theory. Since the problem of level statistics in a restricted volume is well defined for any given distribution of the defects, it seems to be very interesting to try to derive the two-level correlation function from first principles to check the statistical hypothesis.

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In this review the problems mentioned above are discussed with the help of the supersymmetry method. In § 2, anticommuting classical variables and rules of operations with these quantities are defined. It is shown how one can generalize standard formulae of mathematical analysis and linear algebra to the case when the mathematical quantities concerned contain both commuting and anticommuting variables. In § 3 the model of a disordered metal is formulated. Calculation of kinetic quantities is reduced to the calculation of correlation functions of a generalized non-linear supermatrix σ-model. The effect of magnetic and spin–orbit interactions is considered. In § 4 corrections to the classical diffusion coefficient at high frequencies are calculated. All possible magnetic and spin–orbit interactions are taken into account. In $2+\varepsilon$ dimensions the renormalizability of the σ-model is proved. The Gell–Mann–Low function is calculated in the two-loop approximation.

Level statistics for small metallic particles are studied in § 5. Dyson's statistics are confirmed by direct calculation for the zero-dimensional version of the σ-model. In § 6 we study the conductivity of wires. The one-dimensional σ-model is used. This model is solved by the transfer matrix technique. In order to show how this technique works in models with commuting and anticommuting variables a simpler problem, that of calculating the density of states of a one-dimensional disordered chain, is considered. The same scheme of calculation is used for the wires. It is shown that for any weak disorder all states in the wire are localized and the conductivity is zero. The dielectric permeability is calculated.

In § 7, the results obtained are discussed, unsolved problems which can be studied by the methods developed are formulated, and some experiments are analysed.

§ 2. Mathematical bases of the supersymmetry method

2.1. *What is supermathematics?*

In the supersymmetry method operations with both commuting and anti-commuting variables play the main role. For calculations the mathematical technique which is sometimes called supermathematics is used. In this, the basic concepts of mathematical analysis and algebra are generalized to cover functions depending on these two types of variable.

The use of classical anticommuting variables by Martin (1959) and the introduction of integrals over anticommuting variables by Berezin (1961, 1976) appear to have been the beginning of supermathematics. After these many other mathematical works on this subject have been made. Simultaneously with the purely mathematical activity there appeared applications by Gol'fand and Lichtmann (1972), by Volkov and Akulov (1973) and by Volkov and Soroka (1973) in quantum field theory. After the work by Wess and Zumino (1974), the interest in supermathematics become general. Recently the word 'supersymmetry' has appeared in solid-state physics. Parisi and Sourlas (1979) discovered complex symmetry in a model describing ferromagnets in a random magnetic field. They used the concept of the superspace which includes both conventional and anticommuting co-ordinates. The symmetry obtained by Parisi and Sourlas corresponds to the supersymmetry in quantum field theory.

Later it has been found that field theoretical n-component models with ϕ^4 interactions are equivalent at $n=0$ to those containing both commuting and anticommuting fields (McKane 1980, Parisi and Sourlas 1981). The latter models

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have the symmetry under mixing these two types of the fields without mixing the fields and co-ordinates. This symmetry will also be called supersymmetry though it differs from the supersymmetry under mixing fields and co-ordinates.

There are several reviews in which the properties of mathematical objects containing both commuting and anticommuting variables and operations using them, are described (see, for example, Berezin 1979, Corwin *et al.* 1975, Kostant 1977). In these reviews the general mathematical properties are presented but many of the formulae useful for calculations are written in an implicit form and sometimes are absent altogether. Reading these reviews demands some special mathematical grounding. In this chapter the basic concepts of supermathematics will be presented in a simplified form. It does not demand special knowledge but is necessary for understanding the rest of the review.

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2.2. *Basic definitions*

By definition the classical anticommuting variables χ_i, $i=1,2,\ldots,n$ (the elements of Grassmann algebra) satisfy the following relations (Berezin 1965)

$$\{\chi_i, \chi_j\}=\chi_i\chi_j+\chi_j\chi_i=0 \quad \text{for any } 1\leqslant i,j\leqslant n. \tag{2.1}$$

From the property (2.1) it follows that the square of an arbitrary variable χ_i is zero

$$\chi_i^2=0. \tag{2.2}$$

Owing to eqn. (2.2) any function K of the anticommuting variables is a finite polinomial

$$K(\chi_1,\chi_2,\ldots,\chi_n)=\sum_{\alpha_i=0,1} a(\alpha_1,\alpha_2,\ldots,\alpha_n)\chi_1^{\alpha_1}\chi_2^{\alpha_2}\ldots\chi_n^{\alpha_n}. \tag{2.3}$$

It is very convenient to define for any χ the complex conjugate χ^*. The variables χ_i^* anticommute with each other and with all the variables χ_i. By the complex conjugate of a product of variables $\chi_i,\ldots,\chi_n$ we mean the product of the complex conjugates

$$(\chi_1\chi_2\ldots\chi_n)^*=\chi_1^*\chi_2^*\ldots\chi_n^*. \tag{2.4}$$

Let us define the inverse operation as

$$(\chi_i^*)^*=-\chi_i. \tag{2.5}$$

Equation (2.5) differs in the sign of the right-hand side from the corresponding operation for commuting variables. However, it is reasonable for anticommuting variables; for example, for the quantity $\chi_i^*\chi_i$ we have

$$(\chi_i^*\chi_i)^*=-\chi_i\chi_i^*=\chi_i^*\chi_i. \tag{2.6}$$

Equation (2.6) means that the square of the modulus $\chi^*\chi$ is 'real'.

One can introduce linear transformations in the same way as for conventional numbers.

$$\chi_i=\sum_{k=1}^{n} a_{ik}\rho_k. \tag{2.7}$$

It is not difficult to check that this change of the variables gives:

$$\chi_1\chi_2\ldots\chi_n=\rho_1\rho_2\ldots\rho_n \det a. \tag{2.8}$$

In defining derivatives with respect to anticommuting variables one should distinguish between left and right derivatives. The left (right) derivative of a function Φ with respect to an anticommuting variable χ is defined as the coefficient in the dependence of the increment $\Delta\Phi$ on $\Delta\chi$ standing after (before) $\Delta\chi$. Designating the left derivative by $\vec{\partial}/\partial\chi$ and the right one by $\overleftarrow{\partial}/\partial\chi$ we have by the definition

$$\Delta\Phi = \Delta\chi \frac{\vec{\partial}\Phi}{\partial\chi} = \frac{\overleftarrow{\partial}\Phi}{\partial\chi}\Delta\chi. \tag{2.9}$$

The left derivative of an homogeneous polynomial of nth degree coincides with the right one for odd n and differs in sign from the right derivative for even n.

The anticommutativity of the fields χ_i leads to the anticommutativity of the operations of differentiation with respect to each other, for example

$$+\frac{\partial^2\Phi}{\partial\chi_1\partial\chi_2} = -\frac{\partial^2\Phi}{\partial\chi_2\partial\chi_1}. \tag{2.10}$$

The formulae written above define operations with anticommuting variables and enable us to go to more complex objects containing both commuting and anticommuting variables.

In this section we follow mainly the book of Bogolyubov and Shirkov (1976). The definition of complex conjugation (2.4) and (2.5) appears to have been first suggested by the author (Efetov 1982 a) though it may have been used elsewhere before.

2.3. *Supervectors and supermatrices*

Let us call a sequence of anticommuting χ and of commuting S variables, a supervector Φ

$$\Phi = \begin{pmatrix} \chi \\ S \end{pmatrix}, \quad \chi = \begin{pmatrix} \chi_1 \\ \chi_2 \\ \vdots \\ \chi_n \end{pmatrix}, \quad S = \begin{pmatrix} S_1 \\ S_2 \\ \vdots \\ S_n \end{pmatrix}. \tag{2.11}$$

The transposed supervector Φ^{T} is given by the expression

$$\Phi^{\mathrm{T}} = (\chi^{\mathrm{T}} S^{\mathrm{T}}); \quad \chi^{\mathrm{T}} = (\chi_1, \chi_2, \ldots, \chi_n), \; S^{\mathrm{T}} = (S_1, S_2, \ldots, S_n). \tag{2.12}$$

Addition and multiplication of the supervectors by a number is performed by addition and multiplication of separate components.

The supervector Φ^+ is defined as

$$\Phi^+ = (\Phi^{\mathrm{T}})^*. \tag{2.13}$$

The operation of complex conjugation in eqn. (2.13) is specified by eqns. (2.4) and (2.5). Its action on commuting complex variables is given by the conventional rules. The scalar product of the supervectors is defined by analogy with the scalar product of conventional vectors.

$$\Phi^{i+}\Phi^j = \sum_{\alpha=1}^{n} \chi_\alpha^{i*}\chi_\alpha^j + \sum_{\beta=1}^{m} S_\beta^{i*} S_\beta^j. \tag{2.14}$$

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In eqn. (2.14) the usual matrix product of the supervectors Φ^{i+} (2.12) and Φ^{j} (2.11) is written. The product (2.14) for $i=j$ gives the square of the modulus of a supervector. It is easy to see that this quantity is real

$$(\Phi^{i+}\Phi^{i})^{*}=\Phi^{i+}\Phi^{i}. \tag{2.15}$$

Equation (2.15) defines the length of the supervector. For $i\neq j$, we have

$$(\Phi^{i+}\Phi^{j})^{*}=\Phi^{j+}\Phi^{i}. \tag{2.16}$$

A linear transformation F in the space of supervectors converts a supervector Φ into $\tilde{\Phi}$

$$\tilde{\Phi}=F\Phi. \tag{2.17}$$

The supervector $\tilde{\Phi}$ has the same structure as Φ (n-anticommuting components and m-commuting) only if the matrix F has the following structure.

$$F=\begin{pmatrix} a & \sigma \\ \rho & b \end{pmatrix}. \tag{2.18}$$

In eqn. (2.18) a and b are $n\times n$ and $m\times m$ matrices and consist of commuting variables, σ and ρ are $n\times m$ and $m\times n$ matrices consisting of anticommuting variables. Matrices of the form (2.18) are called the supermatrices. In what follows commuting elements of supermatrices and those of supervectors are designated by Latin letters and anticommuting by Greek ones.

Two different supermatrices multiply according to the conventional rules

$$(FG)_{ik}=\sum_{l=1}^{m+n} F_{il}G_{lk}. \tag{2.19}$$

It follows from eqn. (2.19) that the product of supermatrices is a supermatrix. The supertranspose F^{T} of a supermatrix F is defined with the use of the quadratic form

$$\Phi_1^{\mathrm{T}}F^{\mathrm{T}}\Phi_2=(F\Phi_1)^{\mathrm{T}}\Phi_2. \tag{2.20}$$

Writing out the quadratic forms (2.20) explicitly and using the commutation relations (2.1) one can see that the supermatrix F^{T} is equal to

$$F^{\mathrm{T}}=\begin{pmatrix} a^{\mathrm{T}} & -\rho^{\mathrm{T}} \\ \sigma^{\mathrm{T}} & b^{\mathrm{T}} \end{pmatrix}, \quad (a^{\mathrm{T}})_{\alpha\beta}=a_{\beta\alpha},\ (\sigma^{\mathrm{T}})_{\alpha\beta}=\sigma_{\beta\alpha}, \quad (b^{\mathrm{T}})_{\alpha\beta}=b_{\beta\alpha},\ (\rho^{\mathrm{T}})_{\alpha\beta}=\rho_{\beta\alpha}. \tag{2.21}$$

Equation (2.21) shows that the supertransposition differs from the conventional transposition by the sign in the upper right corner. In the original work of the author (Efetov 1982 a) the notation ST was used for the supertransposition. Below we shall write simply T though this notation is used for the conventional transposition of matrices a, b, σ, ρ. This simplification is reasonable because the conventional transposition will not be used for supermatrices. Furthermore, in conventional matrices all elements are written either as Greek or as Latin letters. In supermatrices commuting variables are designated by Latin letters and anticommuting by Greek ones.

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It follows from the definition (2.20) immediately that

$$(F_1F_2)^{\mathrm{T}}=F_2^{\mathrm{T}}F_1^{\mathrm{T}} \tag{2.22}$$

for any supermatrices F_1 and F_2.

Let us define the hermitian conjugate F^+ in the same way as in the theory of conventional matrices

$$F^+=(F^{\mathrm{T}})^*. \tag{2.23}$$

Note again that the notation + is written instead of $S+$ which was used by Efetov (1982 a). Combining the formulae (2.4) and (2.22) we obtain

$$(F_1F_2)^+=F_2^+F_1^+ \quad \text{and} \quad (F^+)^+=F. \tag{2.24}$$

The latter equality (2.24) shows that the hermitian conjugation is inverse to itself. For the supertransposition operation the analogous assertion does not hold. Supermatrices H satisfying the condition

$$H=H^+ \tag{2.25}$$

will be called hermitian.

By definition, unitary supermatrices u satisfy the following conditions

$$UU^+=U^+U=1. \tag{2.26}$$

It follows from eqns. (2.20), (2.23) and (2.26) that the unitary transformations conserve the length of a supervector

$$(U\Phi)^+U\Phi=\Phi^+U^+U\Phi=\Phi^+\Phi. \tag{2.27}$$

Just as for conventional matrices, supermatrices can be diagonalized. Let us sketch how it can be done for hermitian supermatrices.

We seek eigenvalues λ and eigenvectors Φ of a matrix H

$$H\Phi=\lambda\Phi, \quad \text{where} \quad H=\begin{pmatrix} a & \sigma \\ \sigma^+ & b \end{pmatrix}, \quad \Phi=\begin{pmatrix} \chi \\ S \end{pmatrix}, \quad a=a^+, \quad b=b^+. \tag{2.28}$$

Eliminating the vector S from eqn. (2.28) we obtain

$$(a-\lambda+\sigma(b-\lambda)^{-1}\sigma^+)\chi=0, \tag{2.29}$$

a system of linear equations. The anticommutativity of different χ_i between each other does not play any role. A non-zero solution of the system (2.29) exists provided the following condition is fulfilled

$$\det(a-\lambda+\sigma(b-\lambda)^{-1}\sigma^+)=0. \tag{2.30}$$

The solution of eqn. (2.30) gives the eigenvalues λ_i. Using the properties of the scalar product one can prove that eigenvectors corresponding to different λ_i are orthogonal to each other. If eigenvalues coincide the orthogonalization is performed as usual. All λ_i are real. Therefore any hermitian matrix H can be written as

$$H=U^+\hat{\lambda}U, \tag{2.31}$$

where $\hat{\lambda}$ is a diagonal real matrix, U is the unitary matrix transforming the original basis to the orthonormal basis of the eigenvectors of the matrix H.

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An arbitrary non-hermitian matrix F can be diagonalized using two unitary matrices U and V

$$F = U^{+}\hat{\lambda}V, \quad \hat{\lambda} \text{ is diagonal and real.}$$

Correspondingly the hermitian matrices FF^{+} and $F^{+}F$ have the form

$$FF^{+} = U^{+}\hat{\lambda}^{2}U, \quad F^{+}F = V^{+}\hat{\lambda}^{2}V. \tag{2.32}$$

A very important role in the theory of conventional matrices is played by the concept of the trace of a matrix. Analogously in the theory of supermatrices the very important quantity is the supertrace of a supermatrix. By definition, we have

$$\mathrm{STr}\, F = \mathrm{Tr}\, a - \mathrm{Tr}\, b, \tag{2.33}$$

where F is the supermatrix specified by eqn. (2.18).

The operation STr differs in the sign of the second term in the right-hand side of eqn. (2.33) from that of the conventional trace. Using the anticommutation relations one can prove the following identity

$$\mathrm{STr}\, F_1F_2 = \mathrm{STr}\, F_2F_1 \tag{2.34}$$

for any supermatrices F_1 and F_2.

From eqn. (2.34) the invariance under cyclic permutations follows

$$\mathrm{STr}\,(F_1F_2 \ldots F_n) = \mathrm{STr}\,(F_nF_1 \ldots F_{n-1}). \tag{2.35}$$

Furthermore the evident relation holds

$$\mathrm{STr}\, F = \mathrm{STr}\, F^{\mathrm{T}}.$$

The definition of the supertrace can be found in the review of Berezin (1979). In that review the properties of the superdeterminant of a matrix F were also considered. This superdeterminant is defined as

$$\mathrm{Sdet}\, F = \det(a - \sigma b^{-1}\rho)\det b^{-1}. \tag{2.36}$$

The superdeterminant defined by eqn. (2.36) has the important property of multiplicativity

$$\mathrm{Sdet}\,(F_1F_2) = \mathrm{Sdet}\, F_1\, \mathrm{Sdet}\, F_2. \tag{2.37}$$

The identity (2.37) has been proven by Leites (1975).

From the definitions (2.21) and (2.36), it follows that the superdeterminant does not change under the supertransposition of F

$$\mathrm{Sdet}\, F = \mathrm{Sdet}\, F^{\mathrm{T}}. \tag{2.38}$$

The supertrace and the superdeterminant are related to each other analogously to the conventional trace and determinant

$$\ln \mathrm{Sdet}\, F = \mathrm{STr} \log F. \tag{2.39}$$

To prove eqn. (2.39), it is convenient to use the identity

$$F = \begin{pmatrix} a & \sigma \\ \rho & b \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} 1 & a^{-1}\sigma \\ b^{-1}\rho & 1 \end{pmatrix}. \tag{2.40}$$

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Taking the supertrace of the supermatrix F and using the properties (2.34) and (2.35) we obtain

$$\mathrm{STr}\log F=\mathrm{STr}\log\begin{pmatrix}a & 0\\ 0 & b\end{pmatrix}+\mathrm{STr}\log\left[1+\begin{pmatrix}0 & a^{-1}\sigma\\ b^{-1}\rho & 0\end{pmatrix}\right]$$

$$=\mathrm{Tr}\log a-\mathrm{Tr}\log b+\frac{1}{2}\mathrm{STr}\log\left[1-\begin{pmatrix}a^{-1}\sigma b^{-1}\rho & 0\\ 0 & b^{-1}\rho a^{-1}\sigma\end{pmatrix}\right]. \tag{2.41}$$

Calculating the supertrace of the last logarithm in eqn. (2.41) we have

$$\mathrm{STr}\log F=\mathrm{Tr}\,(\log a-\mathrm{Tr}\log b+\mathrm{Tr}\log(1-a^{-1}\sigma b^{-1}\rho)$$

$$=\mathrm{Tr}(\log a(1-a^{-1}\sigma b^{-1}\rho)+\log b^{-1}). \tag{2.42}$$

Using the definition of the superdeterminant (2.36) and the relation between the trace and the determinant of conventional matrices we reduce eqn. (2.42) to the form (2.39), thus proving it.

The rules of operations with supervectors and supermatrices introduced in this section are convenient because they are similar to those in conventional linear algebra. It simplifies considerably calculations with quantities containing both types of variables.

2.4. *Integrals*

Now let us consider integrals over anticommuting variables. These integrals were first introduced by Berezin (1961). They are defined as follows:

$$\int d\chi_i=\int d\chi_i^*=0,\qquad \int\chi_i d\chi_i=\int\chi_i^* d\chi_i^*=1. \tag{2.43}$$

It is supposed that the differentials $d\chi_i$ satisfy the following anticommutation relations

$$\left.\begin{aligned}\{d\chi_i, d\chi_j\}=\{d\chi_i, d\chi_j^*\}=\{d\chi_i^*, d\chi_j^*\}=0,\\ \{d\chi_i, \chi_j\}=\{d\chi_i^*, \chi_j\}=\{d\chi_i, \chi_j^*\}=\{d\chi_i^*, \chi_j^*\}=0.\end{aligned}\right\} \tag{2.44}$$

By integrals over several variables the repeated integrals are implied. Definitions (2.43) are only compatible with conditions (2.44). The choice of unity in the right-hand side of eqn. (2.43) is arbitrary; one might write any other number.

The accepted definition leads to the following equation for the integral of the function $K(\chi)$ (eqn. (2.3))

$$\int K(\chi)d\chi=a(1,1,\ldots,1), \tag{2.45}$$

where $\chi=(\chi_1,\ldots,\chi_n)$, $d\chi=d\chi_n d\chi_{n-1}\ldots d\chi_1$. Equation (2.45) shows that integration over anticommuting variables is equivalent to differentiation.

For the integrals over anticommuting variables one can introduce formulae for the linear change of variables. Changing the variables from χ to ρ, according to the formula (2.7), and recalling definition (2.43), we come to the following equation

$$d\chi_i=\sum_{k=1}^{n}(a^{-1})_{ik}\,d\rho_k, \tag{2.46}$$

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where the matrix a^{-1} is inverse with respect to a. Correspondingly the transformation of the elementary volume $d\chi$ is given by

$$d\chi = \det a^{-1} d\rho. \tag{2.47}$$

Changing variables according to eqn. (2.7) we can transform integrals as

$$\int K(\chi) d\chi = \det a^{-1} \int K(a\rho) d\rho. \tag{2.48}$$

Analogous expressions can be written for the complex conjugates.

In an integral over anticommuting variables one can shift the variables

$$\int K(\chi + \rho) d\chi = \int K(\chi) d\chi. \tag{2.49}$$

For further calculations the formula for integration of the gaussian exponential is extremely important

$$\int \exp(-\chi^* a \chi) d\chi_1^* d\chi_1 \dots d\chi_n^* d\chi_n = \det a. \tag{2.50}$$

Equation (2.50) can be proved by the change of variables $\rho = a\chi$ which diagonalizes the exponent and by the use of eqn. (2.47). Shifting the variables $\chi' = \chi + a^{-1}\eta$, $\chi^{*\prime} = \chi^* + a^{-1}\eta^*$ one can prove a more general formula

$$\int \exp(-\chi^* a \chi - \chi^* \eta - \eta^* \chi) d\chi^* d\chi \dots d\chi_n^* d\chi_n = (\det a) \exp(\eta^* a^{-1} \eta). \tag{2.50a}$$

The integrals (2.50 and (2.50 *a*) differ from the corresponding integrals over commuting variables by giving $\det a$ instead of $(\det a)^{-1}$. This difference is basic for constructing the supersymmetry method. The formula for averaging with gaussian measure is also important for further calculations

$$\frac{\int \chi_i \chi_k^* \exp(-\chi^* a \chi) d\chi_1^* d\chi_1 \dots d\chi_n^*}{\int \exp(-\chi^* a \chi) d\chi_1^* d\chi_1 \dots d\chi_n^* d\chi_n} = (a^{-1})_{ik}. \tag{2.51}$$

Equation (2.51) follows from eqn. (2.50) after representation of the numerator in the left-hand side of eqn. (2.51) in the form of the derivative of eqn. (2.50) with respect to a_{ki}, and calculation of this derivative. Averaging the product of an arbitrary number of χ_i can be performed according to the Wick theorem. Equation (2.51) coincides with that for integration over conventional numbers.

Now let us consider integrals containing both commuting and anticommuting variables. Let the variables $S_1, S_2, \dots, S_m$ be co-ordinates in a domain B in the space of conventional real numbers and $\chi_1, \chi_2, \dots, \chi_n$ be anticommuting Grassmann variables. For any function $f(\chi, S)$ the following integral is well defined

$$I_{nm} = \int f(\chi, S) d\chi dS, \quad dS = \frac{dS_1 dS_2 \dots dS_m}{\pi^{m/2}}. \tag{2.52}$$

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In the integral (2.52) integration over S is performed over domain B. It is supposed that the integral is convergent when integration over S is performed. For integration over χ such a question does not arise.

It is important to know how to define the integral not only in the case when all S_i are conventional numbers, but also in the case when S_i are general even elements of the Grassmann algebra (e.g. when they are equal to a sum of a conventional number and a product of an even number of the elements χ_i). Let $S_i = S_i(\chi, x)$ be even elements and x_i be co-ordinates in domain B. Then, we suppose that $f=0$ on the boundary of domain B, By definition, the integral I_{nm} is

$$I_{nm}(f) = \int f(\chi, S)\, dS d\chi$$

$$= \int f(\chi, S(\chi, x)) \det\left(\frac{\partial S_i}{\partial x_k}\right) d\chi dx, \quad x \in B. \tag{2.53}$$

Many properties of conventional integrals remain valid for integrals of types (2.52) and (2.53). Provided functions f and g are equal to zero on the boundary of domain B, formulae for partial integration can be written as

$$\int f \frac{\partial g}{\partial S_i} d\chi dS = -\int \frac{\partial f}{\partial S_i} g d\chi dS, \tag{2.54}$$

$$\int f \frac{\vec{\partial}}{\partial \chi_i} g d\chi dS = \int \left(f \frac{\overleftarrow{\partial}}{\partial \chi_i}\right) g d\chi dS. \tag{2.55}$$

Changing variables in a conventional integral leads to the appearance of a Jacobian which is equal to the determinant of the partial derivatives matrix. Analogous formulae exist in supermathematics too. For a function f which is equal to zero on the boundary of B we have

$$\int f(\chi, S) d\chi dS = \int f(\chi(x, \eta), S(x,\eta)) J(\chi, S/\eta, x) d\eta dx. \tag{2.56}$$

The function $J(\chi, S/\eta, x)$ in eqn. (2.56) is the superdeterminant of the partial derivatives matrix.

$$\left.\begin{aligned} J(\chi, S/\eta, x) &= \operatorname{Sdet} R, \\ R &= \begin{pmatrix} p & \alpha \\ \beta & q \end{pmatrix}, \\ p_{ik} &= \frac{\partial S_i}{\partial x_k}, \quad \alpha_{ik} = S_i \frac{\overleftarrow{\partial}}{\partial \eta_k}, \\ \beta_{ik} &= \frac{\partial \chi_i}{\partial x_k}, \quad q_{ik} = \chi_i \frac{\overleftarrow{\partial}}{\partial \eta_k}. \end{aligned}\right\} \tag{2.57}$$

Another equivalent representation for J is

$$J(\chi, S/\eta, x) = (\operatorname{Sdet} \tilde{R})^{-1}, \tag{2.58}$$

where

$$\tilde{R} = \begin{pmatrix} q & \beta \\ \alpha & p \end{pmatrix}.$$

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Often function J is called the Berezinian. The formulae (2.54), (2.55), (2.56) and (2.57) have been obtained in the works by Berezin (1965, 1967) and Pakhomov (1974).

For integration over supervectors an analogue of the expressions (2.50) and (2.51) exists. Using the notations for supervectors and supermatrices introduced in the preceding section and integrating first over the commuting variables and then over the anticommuting ones we obtain

$$\left.\begin{aligned} &\int \exp(-\Phi^+ F\Phi)d\Phi^* d\Phi = \operatorname{Sdet} F \\ &d\Phi^* d\Phi = \frac{1}{\pi^m}\prod_{i=1}^{n} dx_i^* dx_i \prod_{k=1}^{m} dS_k^* dS_k. \end{aligned}\right\} \tag{2.59}$$

Shifting the variables of the integration $\Phi \to \Phi - F^{-1}\Phi_0$, $\Phi^+ \to \Phi^+ - \Phi_0^+ F_1^{-1}$ one can prove the formula for integration of the gaussian exponential with linear terms

$$\int \exp[-\Phi^+ F\Phi - \Phi_0^+\Phi - \Phi^+\Phi_0]d\Phi^* d\Phi = \operatorname{Sdet} F \exp(\Phi_0^+ F^{-1}\Phi_0)$$

and, finally, taking the derivative of eqn. (2.59) with respect to F_{ki} we find the analogue of eqn. (2.51)

$$\frac{\int \Phi_i \Phi_k^+ \exp(-\Phi^+ F\Phi)d\Phi^* d\Phi}{\int \exp(-\Phi^+ F\Phi)d\Phi^* d\Phi} = (F^{-1})_{ik}. \tag{2·60}$$

For averaging a product of a large number of Φ_i one may use Wick's theorem again.

Often it is convenient to change variables in an integral, writing the elementary length $(\Delta l)^2$ in the superspace. Suppose that we calculate the integral I

$$I = \int f(\Phi)d\Phi^* d\Phi \tag{2.61}$$

and that we want to write this integral in terms of the variables $\tilde{\Phi}$ determined by the following relations

$$\Phi = \Phi(\tilde{\Phi}), \quad \Phi^+ = \Phi^+(\tilde{\Phi}). \tag{2.62}$$

Let us write the square of the elementary length $(\Delta l)^2$

$$(\Delta l)^2 = \Delta\Phi^+ \Delta\Phi = \Delta\tilde{\Phi}^+ F \Delta\tilde{\Phi}, \tag{2.63}$$

where

$$F = g^+ g, \quad g_{ik} = \Phi_i \frac{\overleftarrow{\partial}}{\partial\tilde{\Phi}_k}.$$

In eqn. (2.63) the right derivative is written down. This remark is important when the derivative is taken with respect to anticommuting variables. Of course,

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when the derivative is taken with respect to commuting ones, the difference between right and left derivatives does not exist.

On the other hand we know that the change of the variables (2.62) yields the Berezinian J which is equal to

$$J=(\mathrm{Sdet}\,F)^{-1}$$

(c.f. with eqn. (2.58)).

It follows from formulae (2.63) and (2.64) that knowing the square of the elementary length $(\Delta l)^2$ in the superspace written in terms of the increments $\Delta\tilde{\Phi}$ and $\Delta\tilde{\Phi}^+$ in the form (2.63), one can obtain the Berezinian by calculation of the inverse superdeterminant of the metric supertensor F.

In this section, formulae presented in the books of Berezin (1965), Bogolyubov and Shirkov (1976) and in the review of Berezin (1979) were mainly used. Our purpose has been to show that the basic formulae of linear algebra and mathematical analysis can be generalized in supermathematics, practically without change. Calculations can be performed with superquantities using standard rules in the framework of standard methods.

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§ 3. Diffusion modes and the non-linear supermatrix σ-model

3.1. *Choice of the model and averaging over impurities*

The behaviour of a non-interacting electron gas in a random potential will now be studied. This problem is equivalent to that of one-particle motion in a random potential. Nevertheless all formulae will be written in terms of the fermion gas. The behaviour of the system can be described by solutions of the Schrödinger equation

$$H\phi_k=\varepsilon_k\phi_k, \quad H=H_0+H_1, \quad \langle H_1\rangle=0. \tag{3.1}$$

In eqn. (3.1) ε_k and ϕ_k are eigenvalues and eigenfunctions of the hamiltonian H. The term H_0 stands for the kinetic energy $\varepsilon(p)$ of a free electron. In the simplest case this energy is

$$\varepsilon(p)=\frac{p^2}{2m}-\varepsilon_0, \tag{3.2}$$

where m is the electron mass, ε_0 is the Fermi energy.

The term H_1 describes the interaction with the random potential. The angular brackets in eqn. (3.1) stand for the averaging over random realizations of the potential. The potential H_1 can contain the interactions with impurities. The interaction with an external magnetic field can be included in H_0. Effects of magnetic and spin–orbit interactions are considered in § 3.4. In this section only scattering on potential impurities will be taken into account.

It is not possible to solve eqn. (3.1) for an arbitrary potential. However, such a solution is not necessary because, usually, only averaged characteristics are of interest. For calculation of thermodynamic quantities it is enough to know the average density of states given by the following expression

$$\langle\rho(\varepsilon)\rangle=\left\langle\sum_k\phi_k(r)\phi_k^*(r)\delta(\varepsilon-\varepsilon_k)\right\rangle. \tag{3.3}$$

The quantity $\langle\rho(\varepsilon)\rangle$ can be written in terms of the advanced G^{A} and retarded G^{R} Green functions of eqn. (3.1)

$$\langle\rho(\varepsilon)\rangle=\frac{1}{2\pi i}\langle(G^{R}_{\varepsilon}(x,x)-G^{A}_{\varepsilon}(x,x))\rangle=\frac{1}{\pi}\langle\operatorname{Im}G^{R}(x,x)\rangle, \tag{3.4}$$

where

$$G^{R,A}(x,x',\varepsilon)=\sum_{k}\frac{\phi^{*}_{k}(x)\phi_{k}(x')}{\varepsilon-\varepsilon_{k}\pm i\delta}=\sum_{k}\phi^{*}_{k}(x)\phi_{k}(x')G^{R,A}_{k}(\varepsilon). \tag{3.5}$$

In eqns. (3.3)–(3.5), $x=(r,\sigma)$, r is the co-ordinate and σ is the spin variable.

The form (3.4) is convenient for the calculation of the average density of states.

For the study of kinetic characteristics of the system such as the response to an external electromagnetic field it is not sufficient to know the average density of states or Green functions. Kinetic quantities can be calculated provided the averaged retarded density–density correlator is known

$$X(r,t)=\begin{cases} i\langle\langle[\hat{\rho}(r,t),\hat{\rho}(0,0)]\rangle\rangle_{T}, & t>0,\\ 0, & t<0.\end{cases} \tag{3.6}$$

In eqn. (3.6) the symbol $\langle\langle\ldots\rangle\rangle_{T}$ stands for both thermal and impurity averaging. The density operator can be written as usual in the second quantization form

$$\hat{\rho}(r,t)=\sum_{k,k',\sigma}\phi_{k}(x)\phi^{*}_{k'}(x)a^{+}_{k'\sigma}a_{k\sigma}, \tag{3.7}$$

where a^{+} and a are the creation and destruction operators. The standard calculation in eqn. (3.7) yields

$$X(r,t)=\begin{cases} i\left\langle\sum_{k,k'}\exp(i(\varepsilon_{k'}-\varepsilon_{k})t)(n(\varepsilon_{k})-n(\varepsilon_{k'}))\phi_{k}(x)\phi^{*}_{k}(0)\phi_{k'}(0)\phi^{*}_{k'}(x)\right\rangle, & t>0,\\ 0, & t<0.\end{cases} \tag{3.8}$$

where $n(\varepsilon_{k})$ is the Fermi distribution function $n(\varepsilon_{k})=(1+\exp(\varepsilon_{k}/T))^{-1}$. In eqn. (3.8) averaging over the impurities only has to be done. Using definitions (3.5) one can write eqn. (3.8) as

$$X(r,t)=i\int\exp(-i\omega t)(n(\varepsilon)-n(\varepsilon-\omega))K(r,\omega,\varepsilon)\frac{d\omega d\varepsilon}{(2\pi)^{2}}, \tag{3.9}$$

where

$$K(r,\omega,\varepsilon)=\langle G^{A}_{\varepsilon-\omega}(r,0)(G^{R}_{\varepsilon}(0,r)-G^{A}_{\varepsilon}(0,r))\rangle. \tag{3.10}$$

In what follows eqns. (3.9) and (3.10) will be used for calculations of $X(r,t)$. Often it is more convenient to calculate some other correlators. In particular, for calculation of the response to an external electromagnetic field of small metallic particles it is more reasonable to calculate first the energy level spacing distribution $R(\omega)$. In addition to the application to the study of responses this function contains very important information about energy level statistics of the system.

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Let us define the function $R(\omega)$ as

$$R(\omega)=\left\langle \frac{1}{4\omega\nu^2}\sum_{k,m}(n(\varepsilon_k)-n(\varepsilon_m))\delta(\omega-\varepsilon_m+\varepsilon_k)\right\rangle, \tag{3.11}$$

where $\nu=mp_0/2\pi^2$ is the density of levels on the Fermi surface.

The function $R(\omega)$ is equal to the probability of finding two levels at the distance ω.

Equation (3.11) can also be rewritten in terms of the retarded G^{R} and advanced G^{A} Green functions

$$R(\omega)=\frac{1}{8\pi^2\nu^2\omega}\left\langle Re\int_{-\infty}^{\infty}\sum_{m,k}(n(\varepsilon)-n(\varepsilon-\omega))G_m^{\mathrm{A}}(\varepsilon-\omega)(G_k^{\mathrm{A}}(\varepsilon)-G_k^{\mathrm{R}}(\varepsilon))\right\rangle d\varepsilon. \tag{3.12}$$

The impossibility of finding exact solutions of the Schrödinger equation is usually by-passed by expansion in a perturbation series over the random potential (Abrikosov *et al.* 1963) and by further averaging over the impurities. In more complicated problems, where the perturbation theory works badly, the application of more perfect methods is necessary. The representation of the Green functions in the form of functional integrals, which enables us to average over the impurities at the very beginning of calculations, proves to be very convenient. The possibility of such a representation follows from eqns. (2.59) and (2.60) for the integrals over supervectors and from the formula (3.5)

$$G_k^{\mathrm{R,A}}(\varepsilon)=\mp i\int\Phi_{\alpha k}\Phi_{\alpha k}^{+}\exp\left(i\Phi_k^{+}(\pm(\varepsilon-E_k)+i\delta)\Phi_k\right)d\Phi_k^{*}d\Phi_k, \tag{3.13}$$

where

$$\Phi=\begin{pmatrix}\chi\\ S\end{pmatrix}$$

is an arbitrary component of the supervector Φ. In eqn. (3.13), in contrast to eqns. (2.59) and (2.60), χ and S are anticommuting and commuting variables but not the sets of variables. So α can take two values 1 and 2.

The formal absence of a weight denominator in eqn. (3.13) is very important. This absence is due to the fact that now the supermatrix F from eqns. (2.59) and (2.60) is proportional to the unit matrix I. The superdeterminant of a matrix of the form bI, where b is a number, is equal to unity. The possibility of writing the Green functions in the form (3.13) is central in constructing the supersymmetry method. For those for whom the integration over supervectors still remains unfamiliar we explain the absence of the weight in eqn. (3.13) by simpler arguments. Integration of a gaussian exponential over anticommuting variables (2.50) gives, in the right-hand side, $\det a$ instead of $(\det a)^{-1}$. Therefore, when integrating over both types of variable with the same matrix a for both the integrals, these determinants in the denominator cancel each other. If the index α in eqn. (3.13) stands for commuting variables the integral over anticommuting ones plays the role of the weight denominator in the gaussian averaging. If α stands for anticommuting variables the integral over commuting ones plays this role.

Introducing the field variables Φ, Φ^{+} according to the formulae

$$\Phi(x)=\sum_k\Phi_k\phi_k(x),\quad \Phi^{+}(x)=\sum_k\Phi_k^{+}\phi_k^{*}(x) \tag{3.14}$$

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the Green functions $G^{R,A}(x, x', \varepsilon)$ can be rewritten in the form of functional integrals

$$G^{R,A}(x, x', \varepsilon) = \mp \int \Phi_\alpha(x)\Phi_\alpha^+(x') \exp\left(i \int \Phi^+(y)(\pm(\varepsilon - H) + i\delta)\Phi(y)\, dy \right) \mathscr{D}\Phi^* \mathscr{D}\Phi. \quad (3.15)$$

The exponents in eqns. (3.13) and (3.15) are written in a form providing the convergence of the integrals over commuting variables.

Equation (3.13) enables us to average the Green function over the impurities and to calculate the density of states. Such a calculation is performed in § 6.2 for a one-dimensional disordered chain. The study of kinetic quantities is more interesting. For calculation of these quantities one has to average the product of two Green functions (3.10) and (3.13). Calculation of the average of the product $G^R(\varepsilon)G^A(\varepsilon - \omega)$ is the main difficulty. It is this average which will be considered below. Calculation of the average of the product $G^A(\varepsilon)G^A(\varepsilon - \omega)$ in the cases under consideration is not difficult and can be performed in the framework of conventional perturbation theory. In order to write down the average of a product of two Green functions one has to introduce the double set of variables Φ^1 and Φ^2. The appropriate functional integral for the product $G^R(\varepsilon)G^A(\varepsilon - \omega)$ has the form

$$\langle G^A(r, 0, \varepsilon - \omega) G^R(0, r, \varepsilon) \rangle = \left\langle \int \Phi_\alpha^1(r)\Phi_\alpha^{1+}(0)\Phi_\beta^2(0)\Phi_\beta^{2+}(r) \times \exp(-\mathscr{L}) \mathscr{D}\Phi_1^* \mathscr{D}\Phi_1 \mathscr{D}\Phi_2^* \mathscr{D}\Phi_2 \right\rangle, \quad (3.16)$$

$$\mathscr{L} = i \int dy \left[\sum_{m=1}^{2} \Phi^{m+} \left(\left(\varepsilon - \frac{\omega}{2} - H \right)(-1)^{m+1} - \frac{\omega}{2} - i\delta \right) \Phi^m \right]. \quad (3.17)$$

The indices 1 and 2 in eqns. (3.16) and (3.17) relate to the advanced and retarded Green functions, correspondingly, α and β stand for arbitrary components of the supervector Φ. All calculations and formulae can be made more compact if we introduce the eight-component supervectors

$$\psi^m = \begin{pmatrix} \Delta^m \\ v^m \end{pmatrix} \quad \Delta^m = \frac{1}{\sqrt{2}} \begin{pmatrix} \chi^{m*} \\ \chi^m \end{pmatrix}, \quad v^m = \frac{1}{\sqrt{2}} \begin{pmatrix} S^{m*} \\ S^m \end{pmatrix}, \quad m = 1, 2. \quad (3.18)$$

For the supervectors ψ of the form (3.18) one can define, in addition to transposition and hermitian conjugation, the operation of 'charge conjugation'

$$\bar{\psi} = (C\psi)^T, \quad \bar{\psi}^m = (\bar{\Delta}^m, \bar{v}^m). \quad (3.19)$$

In eqn. (3.19) T is the supertransposition, C is the supermatrix of the form

$$C^{mn} = \Lambda^{mn} \begin{pmatrix} c_1 & 0 \\ 0 & c_2 \end{pmatrix} \equiv \Lambda^{mn} C_0, \quad (3.20)$$

where Λ is the diagonal supermatrix with the components $\Lambda_{\alpha\alpha}^{11} = -\Lambda_{\beta\beta}^{22} = 1$, $\alpha, \beta = 1, 2, 3, 4$. The 2×2 matrices c_1 and c_2 have the form

$$c_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3.21)$$

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Here and in what follows the upper indices relate to the advanced and retarded Green functions. The charge-conjugate field $\bar{\psi}$ is related to the hermitian conjugate field ψ^+ as

$$\bar{\psi}=\psi^+\Lambda. \tag{3.22}$$

In terms of the supervectors ψ the lagrangian $\mathscr{L}$ in eqns. (3.16) and (3.17) takes the form

$$\mathscr{L}=i\int\bar{\psi}(r)\left(-\tilde{H}_0+U(r)-\frac{1}{2}(\omega+i\delta)\Lambda\right)\psi(r)\,dr, \tag{3.23}$$

where $U(r)$ is the impurity potential. The operator $\tilde{H}_0$ equals

$$\tilde{H}_0=\varepsilon(-i\nabla_r)-\varepsilon+\frac{\omega}{2}. \tag{3.24}$$

Below, the $\sim$ will be omitted and we shall use for H_0 expression (3.24).

Formulae (3.23) and (3.24) are written for free electrons in the impurity potential in the absence of magnetic and spin–orbit interactions. In this case the spin variable does not play any important role.

It will be supposed for the calculation of averages over impurity positions that the potential $U(r)$ is random and that it is distributed according to the gaussian δ-correlated law

$$\langle U(r)\rangle=0, \qquad \langle U(r)U(r')\rangle=\frac{1}{2\pi\nu\tau}\delta(r-r'), \tag{3.25}$$

where τ is the mean free time.

The distribution (3.25) describes impurities of small sizes with weak interactions. Strictly speaking this distribution corresponds to the limit $V_0\to 0$ as $V_0^2 n$ remains constant, where V_0 is the amplitude of scattering by an isolated impurity, n is the impurity concentration. The value $V_0^2 n$ is inversely proportional to the mean free time τ in eqn. (3.15). In the general case of a non-gaussian distribution the results obtained below remain correct provided an appropriate renormalization of constants is made.

The use of the gaussian distribution enables us to average easily over the impurities in eqn. (3.16) with the lagrangian $\mathscr{L}$ (3.23). This possibility is a consequence of the absence of the weight denominator. After averaging the eqn. (3.16) keeps its form but now the lagrangian $\mathscr{L}$ is written as

$$\mathscr{L}=\int\left[-i\bar{\psi}H_0\psi+\frac{1}{4\pi\nu\tau}(\bar{\psi}\psi)^2-\frac{i(\omega+i\delta)}{2}\bar{\psi}\Lambda\psi\right]dr. \tag{3.26}$$

The lagrangian $\mathscr{L}$ (3.26) is similar to those studied in field theory and belongs to the class of models with the interaction ψ^4. Let us remark that at $\omega=0$ the lagrangian is invariant under rotations of the supervectors in the superspace because it depends on the square of the 'length' $\bar{\psi}\psi$ only. The frequency ω violates this symmetry and plays the role of an 'external field'.

The averaging of the simpler lagrangian (3.15) for the one-particle Green function results in a model with the interaction ψ^4 but with a simpler symmetry.

Models with the interaction ψ^4 can be written for the problem of disordered metals without the use of the functional integrals over commuting and anticommut-

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ing variables provided one applies the replica trick. The equivalence of the models ψ^4 with the bosonic n-component field ψ^4 at $n=0$ and the supervector field was first observed by McKane (1980) and by Parisi and Sourlas (1981). In these works the simplest version of the lagrangians which arises in consideration of average one-particle Green functions of a disordered metal or in polymer problems was discussed.

It is clear that it is not possible to calculate any correlation functions with the lagrangian (3.26) exactly (except for the case of the one-dimensional chain). Further calculations will be performed in the limit of large mean free times τ. It corresponds to the weak interaction in the lagrangian $\mathscr{L}$ (3.26). However, the use of standard perturbation theory is often impossible even in this limit. Before making approximations, necessary for reducing the lagrangian to a simpler form, let us describe the physical problems which will be considered below and the mathematical difficulties related to these problems. The qualitative discussion will help us to understand better the essence of the approximations used in further calculations.

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3.2. *Diffusion modes*

A quantum particle in a random potential can either be localized or go to infinity. According to Anderson (1958) localization is possible for strong disorder only. Later it became clear (Mott and Twose 1961) that in a one-dimensional chain the particle can be localized by arbitrarily weak disorder. These results, obtained by semiqualitative reasoning, did not contradict the formal point of view based on perturbation theory. According to Abrikosov *et al.* (1963) it is enough to sum a certain class of diagrams without intersections to calculate the conductivity. After such a summation the classical formulae for this quantity can be obtained. All diagrams with intersections in two and three dimensions are small provided the mean free path l far exceeds atomic distances. In the case of strong disorder when l is of the order of the atomic distances one has to sum all graphs. In this limit the classical formula for conductivity is no longer correct and one can expect, according to Anderson, localization. In a one-dimensional chain one cannot sum only the diagrams without intersections because the diagrams with intersections have the same order of magnitude even at large l. So, the assertion that electron motion in the one-dimensional chain differs qualitatively from that in two and three dimensions cannot be considered as unclear. The result on localization in a chain has been proved by Berezinsky (1973) by direct summation of all diagrams.

However, it has recently been pointed out that the class of weakly disordered systems in which the electrons are localized is considerably wider. Using scaling arguments, Thouless (1977) predicted that in long thin wires the resistivity increases exponentially with the length of the sample L. The crossover from short wires, with a linear dependence of the resistance on L, to long ones with exponentially large resistivities occurs at some localization length L_c. The resistance of a sample which has the length L_c is approximately equal to $10\,\mathrm{k}\Omega$. In Thouless' reasoning the only quantity determining the behaviour of the system is the total resistivity of the system (or its inverse, g, the conductance). Using this conjecture Abrahams *et al.* (1979) considered the behaviour of a disordered metal in an arbitrary dimension d. They wrote relations for a cube of a size bL in d dimensions by combining b^d cubes of size L into blocks of the size bL. If, according to Thouless' conjecture, the conductance is

the only parameter determining the behaviour of the system then the following equation can be written

$$g(bL)=f(b,g(L)), \tag{3.27}$$

where f is an unknown function. The scaling relation (3.27) can be rewritten in a differential form

$$\frac{d\ln g(L)}{d\ln l}=\beta(g(L)). \tag{3.28}$$

The arguments presented here cannot help to calculate the function β. However, its asymptotes can be determined quite easily. For large g the conventional macroscopic transport theory is applicable. In this range we have

$$g(L)=\sigma L^{d-2}, \tag{3.29}$$

where σ is the conductivity.

It follows from eqn. (3.29) that for large g

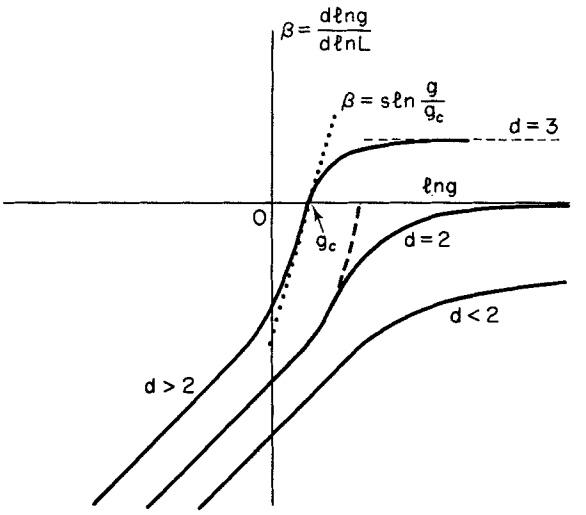
$$\beta(g)=d-2. \tag{3.30}$$

At small g the system is sure to be localized and conductance falls off exponentially

$$g=g_0\exp(-\alpha L) \quad \text{and} \quad \beta(g)=\ln(g/g_0), \tag{3.31}$$

where g_0 is of the order of $e^2/\hbar\sim 0,1(\text{k}\Omega)^{-1}$. Using the asymptotes (3.30) and (3.31) one can plot the function $\beta(g)$ (fig. 1). The basic assumptions used in constructing the function $\beta(g)$ are continuity and monotony. It is seen from the fig. 1 that the behaviour of β strongly depends on the dimensionality of the space. If $d=3$, an

Fig. 1.



Plot of $\beta(g)$ versus $\ln g$ for $d>2$, $d=2$, $d<2$. $g(L)$ is the normalized 'local conductance'. The approximation $\beta=s\ln(g/g_c)$ is shown for $g>2$ (——), this unphysical behaviour necessary for a conductance jump in $d=2$ is shown (– – –).

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unstable fixed point exists where β turns to zero. This point corresponds to the mobility edge. If $d=1$, then $\beta<0$ for any disorder. It corresponds to localization and agrees with the predictions made for chains and wires. Unexpectedly β is everywhere negative, also in two dimensions. This fact signals localization again. Of course, this result applies only if the function β does not have any cusps and is monotonous.

At first glance the conclusion about localization for any arbitrarily weak disorder in two dimensions and in metallic wires contradicts the standard perturbation theory (Abrikosov *et al.* 1963). Summing only the diagrams without intersections one can get, for example, the tensor of the conductivity σ_{ik}

$$\sigma_{ik}=\frac{e^2}{(2\pi)^d}\int\frac{v_iv_k d^dp}{(\varepsilon-p^2/2m+\varepsilon_0+i/2\tau)(\varepsilon-p^2/2m+\varepsilon_0-i/2\tau)}=\frac{e^2}{3}v_0^2\tau\nu\delta_{ik}=\sigma_0\delta_{ik}. \tag{3.32}$$

v_0 is the velocity on the Fermi surface.

The simplest diagram in fig. 2 corresponds to expression (3.32). The product of the average retarded G^R and advanced G^A Green functions is written in the integrand in eqn. (3.32). In calculating these Green functions only the graphs of type fig. 2 (*b*) without intersections were used. An estimation of the contribution of more complex graphs with intersections shows that they can be neglected in two and three dimensions. The contribution of such graphs is small in wires, too, provided the following inequality is fulfilled

$$\tau^{-1}\gg\Delta_t, \tag{3.33}$$

where $\Delta_t=(mS/2\pi)^{-1}$ is the transverse quantization level spacing, S is the cross section. Thus, consideration of a finite number of perturbation theory graphs results, for weak disorder, in the classical formula (3.32) and excludes the possibility of localization.

However, certain sequences of diagrams exist which give divergencies at small frequencies in a low dimension. This fact was first observed by Gor'kov *et al.* (1979) and by Abrahams *et al.* (1980). The simplest sequence of such diagrams is represented in fig. 3. The sum of all these diagrams can be written as

$$\left.\begin{aligned}&-\frac{1}{2\pi\nu\tau}\int\langle G^R(\varepsilon,p)\rangle^2\langle G^A(\varepsilon-\omega,p-k)\rangle^2 v_iv_kK(\omega,k)dp,\\ &K(\omega,k)=\left[1-\frac{1}{2\pi\nu\tau}\langle G^R(\varepsilon,p)\rangle\langle G^A(\varepsilon-\omega,p-k)\rangle\frac{d^dp}{(2\pi)^d}\right]^{-1}.\end{aligned}\right\} \tag{3.34}$$

Calculating the integrals in the limit $\omega, kv_0\ll\varepsilon$, we find for the conductivity

$$\sigma(\omega)=\sigma_0\left(1-\frac{1}{\pi\nu}\int\frac{d^dk}{(2\pi)^d}\frac{1}{-i\omega+\mathscr{D}k^2}\right), \tag{3.35}$$

Fig. 2

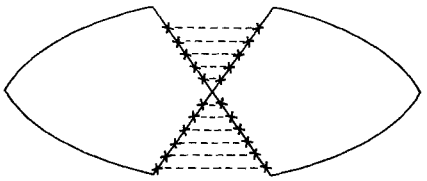


(*a*) (*b*)

Diagrams determining the classical conductivity.

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Fig. 3



The simplest diagram giving the divergent contribution in the conductivity in low dimension as $\omega \to 0$.

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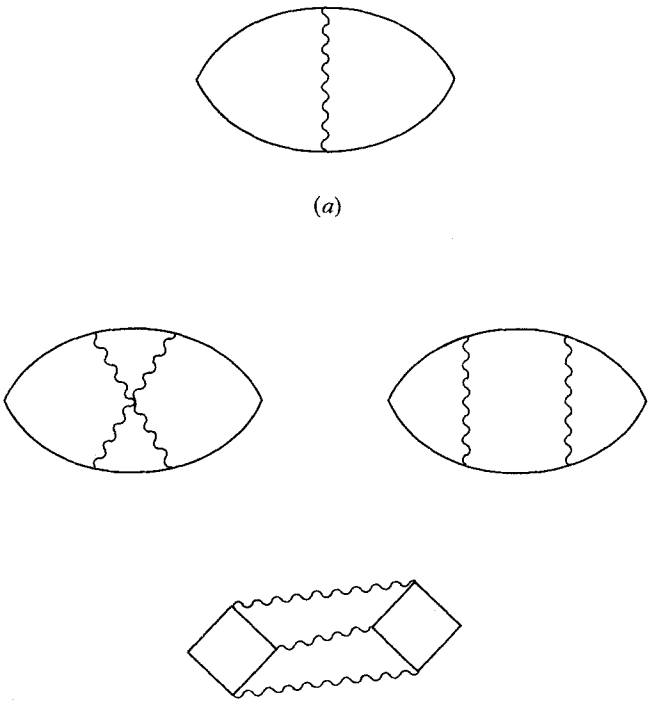
where $\mathscr{D} = v^2\tau/d$ is the diffusion coefficient. Equation (3.35) shows that in one and two dimensions the correction to the conductivity diverges as $\omega \to 0$. In wires one has to integrate over longitudinal momenta and sum over transverse ones. Then, the term corresponding to the zeroth harmonics is divergent as $\omega \to 0$. For this reason the divergencies arise also in films with finite thickness and in samples with limited volume. The latter case corresponds to the zero-dimensional situation. Therefore in these cases formula (3.32) is no longer applicable at small frequencies and the statement about localization in low-dimensional metals does not contradict perturbation theory.

In the model with random hops from one site to another and a large number of orbitals analogous calculations have been performed by Wegner (1979 b) and by Oppermann and Wegner (1979).

The ladder K describing the sequence of the diagrams in fig. 3 can be considered as an effective mode. The spectrum of this mode is of the diffusion form. So we will call these modes *diffusion modes* and will represent them by wavy lines. The sequence is represented by the single diagram in fig. 4 (a). At low frequencies more complicated diagrams containing many diffusion lines are important too. Several diagrams with two or three wavy lines are represented in fig. 4 (b). Of course, all the diagrams discussed can be obtained by expansion in a power series over the interaction of the lagrangian (3.26), using the field-theory formulation of the problem. This expansion would be completely equivalent to the standard perturbation theory (Abrikosov *et al.* 1963). In both these theories a renormalization of dotted lines is absent. In the language of the lagrangian (3.26) it corresponds to different signs of closed fermionic and bosonic loops.

Diagrams containing both electron and diffusion lines have a very important property. In the range of low frequencies $\omega \ll \tau^{-1}$, in integrating the diffusion lines, the momenta $k \sim (\omega/\mathscr{D})^{1/2}$ are essential. The main contribution in integrating the electron lines comes from the domain of moments $||p| - p_0| \sim (\tau v)^{-1}$. This enables us to integrate first the fast electron lines and to reduce the problem to a model of slow-interacting diffusion modes. In principle, this procedure can be done in each graph separately. In the second order in the diffusion lines it has been done by Gor'kov *et al.* (1979) and Hikami (1981). However, the reduction of the problem to interacting diffusion modes using the perturbation theory necessitates consideration of a large number of the graphs. It is a difficult problem. Furthermore, even an exact account of all graphs of the perturbation theory does not cover the region of low frequencies where terms that are non-analytical in the expansion parameter can become important. In particular, the results of § 5 show that the level correlation function for metallic particles cannot be obtained by summing perturbation-theory

Fig. 4



(*a*)

(*b*)

Simplest diagrams with diffusion modes.

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graphs only. So, an attempt to integrate first over the fast-electron variables in the model with the lagrangian (3.26) without the use of an expansion over the interaction, and to reduce the problem to a model with the lagrangian describing the slow-diffusion modes, seems to be reasonable. Such a reduction will be performed in the next section.

Let us emphasize that the diffusion modes arise only when the average of the product of the retarded G^R and advanced G^A Green functions, $\langle G^R(\varepsilon)G^A(\varepsilon\text{-}\omega)\rangle$, is calculated. Calculation of averages such as $\langle G^A(\varepsilon)G^A(\varepsilon-\omega)\rangle$ and $\langle G^R(\varepsilon)G^R(\varepsilon-\omega)\rangle$ in the limit of a weak disorder can be carried out in the framework of standard perturbation theory. The same is valid for calculation of the average one-particle Green functions $\langle G^R\rangle$ and $\langle G^A\rangle$. The direct calculation of first order terms of the perturbation theory shows that the correlator $\langle G^A(\varepsilon,0,r)G^A(\varepsilon-\omega,r,0)\rangle$ falls off rapidly at atomic distances. Therefore, in calculating the slowly decreasing correlator $K(r,\omega,\varepsilon)$ (3.10), this average will not be considered.

3.3. *Reduction of the problem to a non-linear supermatrix σ-model*

In this section an approximation which is rather customary for models with the interaction ψ^4 will be used. In these models the symmetry can be broken in a certain domain of parameters by the existence of non-zero averages $Q\sim\langle\psi\bar{\psi}\rangle$. Usually the

ground state is degenerate. Due to the degeneracy, Goldstone modes appear. The spectrum of the Goldstone modes can be obtained by consideration of slow fluctuations of the collective variable Q. The slowly varying averages and the Goldstone modes exist also in the considered model with the lagrangian (3.26). To get these slowly varying averages let us rewrite the interaction in eqn. (3.26) as

$$\mathscr{L}_{\text{int}}=\frac{1}{4\pi\nu\tau}\sum_{p_1+p_2=p_3+p_4}(\bar{\psi}_{p_1}\psi_{p_2})(\bar{\psi}_{p_3}\psi_{p_4})$$

$$\approx\frac{1}{4\pi\nu\tau}\sum_{p_1,p_2,q<q_0}[(\bar{\psi}_{p_1}\psi_{p_2})(\bar{\psi}_{-p_1+q}\psi_{-p_2-q})+(\bar{\psi}_{p_1}\psi_{p_2})(\bar{\psi}_{-p_2-q}\psi_{-p_1+q})$$

$$+(\bar{\psi}_{p_1}\psi_{-p_1+q})(\bar{\psi}_{p_2}\psi_{-p_2-q})]. \quad (3.36)$$

In the formula (3.36) we have separated the domains with small $q<q_0\ll(\nu\tau)^{-1}$. Domains with large momenta q give a small contribution to the quantities under consideration. Using the charge conjugation operation ((3.19), (3.20)) and the commutation rules one can see that the first term in eqn. (3.36) is equal to the second.

The interaction in (3.36) can be reduced to a quadratic form. This decoupling is performed by gaussian integration. The reduction of the last term in eqn. (3.36) is very simple

$$\exp\left[-\frac{1}{4\pi\nu\tau}\sum_{p_1,p_2,q}(\bar{\psi}_{p_1}\psi_{-p_1+q})(\bar{\psi}_{p_2}\psi_{-p_2-q})\right]$$

$$=\int\exp\left[-\frac{i}{4\tau}\int E(r)\bar{\psi}(r)\psi(r)\,dr-\frac{\pi\nu}{16\tau}\int E^2(r)\,dr\right]\mathscr{D}E$$

$$\times\left(\int\exp\left[-\frac{\pi\nu}{16\tau}\int E^2(r)\,dr\right]\mathscr{D}E\right)^{-1}. \quad (3.37)$$

In eqn. (3.37) we integrate over slowly-varying real functions $E(r)$. Comparing eqns. (3.37) and (3.26) we see that the existence of $E(r)$ results in a non-essential change of energy because $E(r)$ is a slowly-varying real function. Therefore the last term in (3.36) does not lead to any non-trivial phenomena and will not be considered below.

All the other terms in eqn. (3.36) can be decoupled in slowly-varying parts by integration over a matrix only. Taking into account the equality of the first two terms in eqn. (3.36) to each other, and neglecting the third one, the interaction (3.36) can be rewritten as

$$\mathscr{L}_{\text{int}}=\frac{1}{2\pi\nu\tau}\sum_{p_1,p_2,q<q_0}[(\bar{\Delta}_{p_1}\Delta_{p_2})(\bar{\Delta}_{-p_2-q}\Delta_{-p_1+q})+(\bar{v}_{p_1}v_{p_2})(\bar{v}_{p_2-q}v_{-p_1+q})$$

$$+2(\bar{\Delta}_{p_1}\Delta_{p_2})(\bar{v}_{-p_2-q}v_{-p_1+q})]. \quad (3.38)$$

In eqn. (3.38) the fermionic Δ^a and bosonic v^a components of the supervector ψ are written out explicitly. The decoupling of the first two terms in eqn. (3.38) can be carried out by integration over Bose fields. However, for the decoupling of the third term it is necessary to integrate over anticommuting fields. Using the formula for

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integration of the exponential with linear terms over anticommuting variables and that for integration over commuting ones we find

$$\left.\begin{aligned}
&\exp(-\mathscr{L}_{\mathrm{int}})=p_1p_2p_3,\\
&p_1=\int\exp\left(-\frac{1}{2\tau}\sum_{m,n,p,q=1}^{2}\left(\bar{\Delta}^m_p a^{mn}_q\Delta^n_{-p-q}i^{n-m}+\frac{\pi\nu}{4}\operatorname{Tr}a^{mn+}_q a^{mn}_{-q}\right)\right)\mathscr{D}a,\\
&p_2=\int\exp\left(-\frac{1}{2\tau}\sum_{m,n,p,q=1}^{2}\left(\bar{v}^m_p b^{mn}_q v^n_{p-q}i^{n-m+1}+\frac{\pi\nu}{4}\operatorname{Tr}b^{mn+}_q b^{mn}_{-q}\right)\right)\mathscr{D}b,\\
&p_3=\int\exp\left(-\frac{1}{2\tau}\sum_{m,n,p,q=1}^{2}\left(\bar{\Delta}^m_p\sigma^{mn}_q v^n_{-p-q}i^{2n-m-1}+\bar{v}^n_p\sigma^{mn+}_q\Delta^m_{-p-q}i^{m-1}\right.\right.\\
&\qquad\left.\left.+\frac{\pi\nu}{2}(-1)^n\operatorname{Tr}\sigma^{mn+}_q\sigma^{mn}_{-q}\right)\right)\mathscr{D}\sigma.
\end{aligned}\right\}\tag{3.39}$$

In eqns. (3.39) integration is performed over 2×2 matrices a^{mn}, b^{mn} and σ^{mn}. The elements of the matrices a^{mn} and b^{mn} are conventional numbers and those of the matrices σ^{mn} are anticommuting variables. All these matrices satisfy the following conditions

$$\left.\begin{aligned}
a^{mn+}&=a^{nm}=c_1a^{mn\mathrm{T}}c_1^{\mathrm{T}},\\
b^{mn+}&=b^{nm}=c_2b^{mn\mathrm{T}}c_2^{\mathrm{T}},\\
\sigma^{mn+}&=c_2\sigma^{mn\mathrm{T}}c_1^{\mathrm{T}}.
\end{aligned}\right\}\tag{3.40}$$

The matrices a^{mn}, b^{mn} and σ^{mn} satisfying the relations (3.40) can be written out explicitly

$$\left.\begin{aligned}
a^{mn}&=\begin{pmatrix}a_1^{mn}&a_2^{mn}\\-a_2^{mn*}&a_1^{mn*}\end{pmatrix},\qquad b^{mn}=\begin{pmatrix}b_1^{mn}&b_2^{mn}\\b_2^{mn*}&b_1^{mn*}\end{pmatrix},\\
\sigma^{mn}&=\begin{pmatrix}\sigma_1^{mn}&\sigma_2^{mn}\\-\sigma_2^{mn*}&-\sigma_1^{mn*}\end{pmatrix},\qquad a_1^{mn}=a_1^{nm*},\ a_2^{mn}=-a_2^{nm},\\
&\qquad\qquad b_1^{mn}=b_1^{nm*},\ b_2^{mn}=b_2^{nm}.
\end{aligned}\right\}\tag{3.41}$$

The matrices a, b and σ contain 16 independent complex variables, eight variables being bosonic and eight being fermionic. The equality of the numbers of commuting and anticommuting variables enables us to omit the weight denominators because their product is equal to unity. Let us emphasize that all the integrals in eqn. (3.39) are convergent.

The decoupling of the interaction ψ^4 in considering the problem involved by the replica method was first proposed by Wegner (1979) who studied the model with random amplitude of hops from one site to another and a large number of orbitals per site. Wegner's model differs from the model considered here. This difference is not important provided we consider the limit $\varepsilon_0\tau\gg1$ in the model with the random potential and the limit of a large number of the orbitals in the model with random hops. After separation of the slowly-varying parts (3.36), which makes sense if $\varepsilon_0\tau\gg1$, both the models coincide. A careful decoupling within the replica method has been made by Schäfer and Wegner (1980) and Houghton *et al.* (1980) using functional integration over commuting variables, and by Efetov *et al.* (1980) using Green functions written in terms of functional integrals over anticommuting

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variables. The symmetry of matrix a in eqn. (3.41) coincides with the symmetry of the collective variable Q from the work of Efetov *et al.* (1980) and that of matrix b coincides with the symmetry of the corresponding matrix Q from the work of Schäfer and Wegner (1980). Of course, in the replica method one has to deal with $n \times n$ matrices for an arbitrary n. In the supersymmetry method the matrices have finite sizes which are not too large.

Using supermatrices a somewhat cumbersome expression (3.39) can be rewritten in a more compact form. Returning to the supervectors (3.18), we obtain instead of eqn. (3.39)

$$\exp(-L_{\text{int}}) = \int \left[\exp\left(-\frac{1}{2\tau} \int \left(\bar{\psi} Q \psi + \frac{\pi \nu}{4} \operatorname{STr} Q^2 \right) dr \right) \right] \mathscr{D}Q, \tag{3.42}$$

where Q consists of four supermatrices Q^{11}, Q^{12}, Q^{21} and Q^{22}

$$Q = \begin{pmatrix} Q^{11} & Q^{12} \\ Q^{21} & Q^{22} \end{pmatrix}, \tag{3.43}$$

$$\left.\begin{aligned} Q^{11} &= \begin{pmatrix} a^{11} & \sigma^{11} \\ \sigma^{11+} & ib^{11} \end{pmatrix}, \qquad Q^{12} = i \begin{pmatrix} a^{12} & i\sigma^{12} \\ \sigma^{21+} & ib^{12} \end{pmatrix}, \\ Q^{22} &= \begin{pmatrix} a^{22} & i\sigma^{22} \\ i\sigma^{22+} & ib^{22} \end{pmatrix}, \qquad Q^{21} = -i \begin{pmatrix} a^{12+} & \sigma^{21} \\ i\sigma^{12+} & ib^{12+} \end{pmatrix}. \end{aligned}\right\} \tag{3.44}$$

Strictly speaking the form of the matrix Q differs from that of the supermatrices (2.18). However, this matrix can be reduced to the standard supermatrix form by permutations of elements. The same can be said about vector ψ. That is why we call Q and ψ the supermatrix and the supervector. For matrices Q of the form (3.43) the supertrace is written as

$$\operatorname{STr} Q = \operatorname{STr} Q^{11} + \operatorname{STr} Q^{22}. \tag{3.45}$$

Analogously, one can write all the other operation rules for supermatrices of the form (3.43). Of course, the supermatrix Q and the supervector ψ might be written also in the conventional form (2.11), (2.18). However the representations (3.18) and (3.43) are more convenient because they enable one to separate explicitly the blocks (superelements) which relate to the retarded and advanced Green functions. In calculating an average one-particle Green function it would be sufficient to use the superelements Q^{11} or Q^{22} only, which are true supermatrices in the sense of the definition (2.18).

The supermatrix Q in eqn. (3.42) has a certain symmetry. To write down this symmetry let us first define the charge conjugate $\bar{Q}$ of a supermatrix Q as

$$\bar{Q} = C Q^{\mathrm{T}} C^{\mathrm{T}}, \tag{3.46}$$

where C is specified by the formula (3.20) and Q^{T} is

$$Q^{\mathrm{T}} = \begin{pmatrix} Q^{11\mathrm{T}} & Q^{21\mathrm{T}} \\ Q^{12\mathrm{T}} & Q^{22\mathrm{T}} \end{pmatrix}. \tag{3.47}$$

The following equations for arbitrary supermatrices Q_1 and Q_2 and supervectors ψ_1 and ψ_2 can be checked easily

$$\overline{Q_1 Q_2} = \overline{Q_2 Q_1} \quad \text{and} \quad \bar{\psi}_1 Q \psi_2 = \bar{\psi}_2 Q \psi_1. \tag{3.48}$$

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Now one can write down the symmetry of the supermatrix Q mentioned above.

$$Q=\bar{Q}. \tag{3.49}$$

Substituting eqns. (3.42) and (3.36) into (3.26) one can express the lagrangian $\mathscr{L}$ (3.16) in the decoupled form. Using the slowness of variations of the matrix Q it is possible to calculate the integral in formula (3.16) over all the components of the vector ψ. Generalizing equation (2.60) to the case of integration over the supervectors ψ of the form (3.18) and using expressions (3.10), (3.16), (3.26) and (3.42) we obtain

$$K(\omega)=\int g^{12}_{\alpha\beta}(r,r)\,K_{\beta\beta}\,g^{21}_{\beta\alpha}(0,0)\exp\left(-F[Q]\right)\mathscr{D}Q, \tag{3.50}$$

where K is the 4×4 superelement of the form

$$K=\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \tag{3.51}$$

I is the 2×2 unit matrix. The function $g(r,r')$ in eqn. (3.50) is the supermatrix Green function of the following equation

$$\left(-iH_0-\frac{1}{2}i(\omega+i\delta)\Lambda+\frac{Q(r)}{2\tau}\right)g(r,r')=\delta(r-r'), \tag{3.52}$$

where H_0 is determined by eqn. (3.24). The free energy functional $F(Q)$ in eqn. (3.50) is

$$F[Q]=\int dr\left[-\frac{1}{2}\mathrm{STr}\ln\left(-iH_0-\frac{i(\omega+i\sigma)}{2}\Lambda+\frac{Q}{2\tau}\right)+\frac{\pi\nu}{8\tau}\mathrm{STr}\,Q^2\right]. \tag{3.53}$$

The indices α and β in eqn. (3.50) are, as in eqn. (3.16), arbitrary. The repetition of the indices in eqn. (3.50) does not imply a summation. In gaussian averaging in eqn. (3.16) only one type of coupling of the fields Φ has been taken into account. All the other averages arising in decoupling with the use of Wick's theorem fall off rapidly with distance and can therefore be neglected.

Let us mention that the integral over Q with the functional (3.53) in eqn. (3.50) converges as before.

The separation of the averages slowly varying in the space makes sense for large mean free times only if $\tau\varepsilon_0\gg 1$, because only in this case are the omitted terms small. In the limit $\tau\varepsilon_0\gg 1$ in two and three-dimensional samples, and in wires, an essential contribution comes from free energy minima only. Varying the free energy $F(Q)$, putting the first variation to be zero and using the property of the invariance of the supertrace under cyclic permutations (2.35), we find

$$Q=\frac{1}{\pi\nu}\int g_0(p)dp\equiv\frac{1}{\pi\nu}\int\left[i\left(\varepsilon-\frac{\omega+i\delta}{2}+\varepsilon_0-\frac{p^2}{2m}-\frac{1}{2}(\omega+i\delta)\Lambda\right)+\frac{Q}{2\tau}\right]^{-1}dp. \tag{3.54}$$

In obtaining eqn. (3.54) it was assumed that the energy minimum is reached when Q does not depend on co-ordinates. Equation (3.54) can also be obtained using the conventional mean-field approximation. This equation has many solutions. The proper solution can be chosen using the requirement of analycity on ω in the upper half plane. At $\omega\neq 0$ the only solution satisfying this requirement is $Q=\Lambda$. However, this point of the extremum does not belong to the set of the matrices a, b and σ (3.41)

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because the diagonal elements of the matrices ib^{11}, ib^{22} are always imaginary. The analogous discrepancy arises in calculations by the replica method provided functional integrals over commuting variables are used (Schäfer and Wegner 1980). In order to reach the saddle point it is necessary to move the contour of the integration over the matrices b^{aa} into the complex plane. The solution $Q=\Lambda$ chosen by the analycity requirement is the only one which can be reached without intersections of singular points. For $\omega=0$ the solution of eqn. (3.54) is strongly degenerate. The general solution in this case can be written in the form

$$Q=V\Lambda\bar{V}, \tag{3.55}$$

where the supermatrix V satisfies the condition of unitarity.

$$\bar{V}V=1. \tag{3.56}$$

In addition to the condition (3.56) the supermatrix V must obey another constraint to provide equal numbers of independent elements in the matrices Q and V.

The procedure of shifting the contours when integrating over Q can be carried out as follows. Let us move the contours in the integrals over Q^{11} and Q^{22}, keeping Q^{12} and Q^{21} fixed. In fact, one has to take care of the correct shifts in the integrals over b^{11} and b^{22} only. This is due to the fact that matrices a and b are connected with each other by anticommuting variables. Any function of anticommuting variables is a finite polynomial. After the expansion of the exponential (3.50) as a series in the anticommuting variables, the matrices a and b are the only ones remaining in the exponent in the integrand. Then, it is necessary to move the contours in the integrals over the matrices b^{11} and b^{22} only because the matrices a^{11} and a^{22} have the correct form. Shifting in the integrals over b^{11} and b^{22} is carried out by diagonalizing these matrices and by shifting in the integrals over eigenvalues of these matrices. Shifting in the integrals over anticommuting variables can be done with the use of eqn. (2.49). As the result, the general saddle point solution which can be obtained from the solution $Q=\Lambda$ by a continuous transformation is written in the form

$$Q=W+\Lambda(1-W^2)^{1/2}, \quad W=\begin{pmatrix} 0 & Q^{12} \\ Q^{21} & 0 \end{pmatrix}. \tag{3.57}$$

The square root in eqn. (3.57) is defined by Taylor expansion over W^{-2}, Q^{12} and Q^{21} are specified by eqns. (3.44). The supermatrix W has eight independent complex variables.

For the solutions Q of the form (3.57) the following relation holds in addition to the condition (3.49)

$$\bar{Q}=KQ^{+}K; \quad K=\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}, \tag{3.58}$$

where k is the superelement (3.51).

Equation (3.58) can be checked by comparison with the formulae (3.40).

The somewhat strange relation between the hermitian and charge conjugates (3.58) (appearance of the superelement K) is due to the multiplication of the matrices b^{12} and σ^{12} by i in eqns. (3.44). Comparing eqns. (3.58) and (3.55) it is possible to write the additional constraint for the matrix V

$$\bar{V}=KV^{+}K. \tag{3.59}$$

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Thus, in the saddle point the extremal solution (3.55) is degenerate and is related to the transformation group $V(8)/V(4) \times V(4)$. This degeneracy is a consequence of the original degeneracy of the lagrangian (3.26). It is essential that the average product $\langle G^R G^A \rangle$ of the retarded and advanced Green functions be considered. If the products $\langle G^R G^R \rangle$ or $\langle G^A G^A \rangle$ are considered the matrix Λ in eqn. (3.55) must be replaced by the unit matrix. In this case the degeneracy of the extremal solution disappears. The supermatrix Q (3.55) plays the role of an order parameter and contains anticommuting elements. However, these anticommuting elements cannot enter any physical quantities. Calculating the ground-state energy (3.53) with Q, specified by eqn. (3.55), one can see that this energy is equal to zero. Corrections arising from fluctuations near the minimum are expressed in the form of integrals over all the elements and are conventional numbers.

The degeneracy of the ground state at $\omega = 0$ leads to the existence of Goldstone modes. Let us emphasize that these modes arise in calculating $\langle G^R G^A \rangle$ only.

Now we expand the free energy near the minimum. For small deviations ΔQ from the equilibrium value, the free energy is quadratic in these deviations.

$$F[Q] = \frac{1}{16\tau^2} \mathrm{STr} \int \frac{d^d k}{(2\pi)^d} \left\{ \int g_0(p) \Delta Q(k) g_0(p+k) \Delta Q(-k) \frac{d^d p}{(2\pi)^d} + 2\pi\nu\tau \Delta Q(k) \Delta Q(-k) \right\} - \frac{i\omega\pi\nu}{4} \int \mathrm{STr}(\Lambda Q) dr. \quad (3.60)$$

One should distinguish between longitudinal fluctuations of Q changing the eigenvalues of Q and transverse ones when only the matrix V in eqn. (3.55) fluctuates. For the transverse fluctuations the following equation holds

$$Q\Delta Q + \Delta Q Q = 0. \quad (3.61)$$

The longitudinal fluctuations change the free energy considerably. The spectrum of these fluctuations has a gap, so their contribution is small in all the cases considered provided the inequality $\varepsilon_0 \tau \gg 1$ is fulfilled.

Homogeneous transverse fluctuations in the low-frequency limit do not change the free energy at all. The free-energy functional F describing such fluctuations contains only gradients of Q and terms which are linear in the frequencies. Substituting eqn. (3.61) into (3.60) we obtain, for $\omega\tau \ll 1$,

$$F = \frac{\pi\nu}{8} \int \mathrm{STr}\,[\mathscr{D}(\nabla Q)^2 + 2i\omega\Lambda Q] dr, \quad (3.62)$$

where $\mathscr{D} = v_0^2 \tau / d$ is the diffusion coefficient.

The functional (3.62) describes the Goldstone modes. Their existence is a consequence of the spontaneous breaking of the supersymmetry. The system with the free energy (3.62) belongs to the class of non-linear σ-models. The functional (3.62) is similar to that of a ferromagnet. The frequency ω plays the role of an external field. Let us note the formal analogy of eqn. (3.62) with those obtained in the replica trick study (Schäfer and Wegner 1980, Efetov *et al.* 1980). Of course, the structure of the matrices Q is everywhere different.

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Substituting the extremal value of Q (3.55) into eqns. (3.50) and (3.52) and integrating over the momemta we reduce the correlator $K(\omega)$ to the form

$$K(\omega)=(\pi\nu)^2\int Q^{12}_{\alpha\beta}(r)k_{\beta\beta}Q^{21}_{\beta\alpha}(0)\exp(-F[Q])\mathscr{D}Q$$

$$=\frac{(\pi\nu)^2}{64}\int \mathrm{STr}(k(1+\Delta)(1-\tau_3)Q(r)(1-\Lambda)(1-\tau_3)kQ(r'))$$

$$\times\exp(-F[Q])\mathscr{D}Q, \qquad (3.63)$$

where

$$\tau_3=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \qquad (3.63\,a)$$

is the 2×2 matrix in the space of matrices a, b, σ; α and β are arbitrary indices (do not sum over them).

The correlator $K(\omega)$ (3.63) is similar to a two-spin correlation function in a ferromagnet. Using the formalism developed one can rewrite the density of states (3.4) in the form

$$\langle\rho(\varepsilon)\rangle=\frac{\nu}{8}\int \mathrm{STr}\,(k\Lambda Q)\exp(-F[Q])\mathscr{D}Q. \qquad (3.64)$$

Of course, the density of states could be written using the supermatrices Q^{11} and Q^{22} only. Nevertheless eqn. (3.64) is useful because it relates the density of states to the average value of the order parameter Q of the theory involved. In particular, the assertion that the density of states does not have singularities and does not become zero at finite concentrations (Wegner 1981) leads to the conclusion that the average $\langle Q\rangle_F$ with the functional F does not have any singularities and cannot be equal to zero. So, the model under consideration differs from the models of ferromagnets drastically because in the latter the order parameter is zero in the disordered phase.

For calculation of the free diffusion-mode propagator one may consider the first terms of an expansion of Q in eqn. (3.57). Substituting this expansion into eqn. (3.63) we get

$$F_0=\frac{\pi\nu}{8}\int \mathrm{STr}\,[\mathscr{D}(\nabla W)^2-i\omega W^2]dr. \qquad (3.65)$$

From eqns. (3.63) and (3.65) we find

$$K_0(\omega)=\frac{2\pi\nu}{\mathscr{D}k^2-i\omega}. \qquad (3.66)$$

The propagator $K_0(\omega)$ corresponds to the conventional equation for diffusion.

Performing the same expansion for the density of states one can see that the diffusion modes do not contribute to this quantity. One should only substitute the value $Q=\Lambda$ into eqn. (3.64) for calculation of $\langle\rho(\varepsilon)\rangle$.

All the formulae in this chapter have been written for spinless particles. Calculations for real electrons can be carried out in the absence of magnetic and spin–orbit interactions separately for spin 'up' and for spin 'down', adding a factor 2 in eqns. (3.63) and (3.66).

The supermatrix σ-model (3.62) and the density–density correlator (3.63) were first derived by Efetov (1982 a).

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3.4. *Magnetic and spin–orbit interactions*

Only potential impurity scattering has so far been considered. The system has been invariant under the time reversal. Systems of particles with spin have also been invariant under rotations in spin space. The high symmetry of the free energy (3.62) under transformations V (3.55) for $\omega=0$ is related to the invariance of the system under the time reversal and spin rotations.

An external magnetic field and magnetic impurities break the time-reversal symmetry. The spin-rotation symmetry is violated by magnetic impurities and spin–orbit interactions. It is shown below that the magnetic and spin–orbit interactions result in the appearance of external fields of different symmetries in the free energy functional (3.62). These fields break, partially, the invariance (3.55).

The external magnetic field can be taken into account by the replacement $\hat{p}\to\hat{p}-(e/c)\mathbf{A}$ in eqns. (3.2), (3.15) and (3.17). In order to write equations in terms of the supervectors ψ (3.18) one has to use formulae for partial integration

$$\left.\begin{aligned}\int\chi^*\varepsilon\left(-i\nabla-\frac{e}{c}\mathbf{A}\right)\chi\,dr&=\int\left(\varepsilon\left(-i\nabla+\frac{e}{c}\mathbf{A}\right)\chi^*\right)\chi\,dr,\\ \int S^*\varepsilon\left(-i\nabla-\frac{e}{c}\mathbf{A}\right)S\,dr&=\int\left(\varepsilon\left(-i\nabla+\frac{e}{c}\mathbf{A}\right)S^*\right)S\,dr.\end{aligned}\right\}\tag{3.67}$$

As a result eqn. (3.24) for H_0 takes the form

$$H_0=\varepsilon\left(\hat{\mathbf{p}}-\frac{e}{c}\tau_3\mathbf{A}\right)-\varepsilon+\frac{\omega+i\delta}{2},\tag{3.68}$$

where τ_3 is defined by eqn. (3.63 *a*). The appearance of this matrix is due to changing the sign before $\mathbf{A}$ in the partial integration. If the magnetic field is weak its effect can be considered as a small perturbation. To derive the free energy functional we use again eqn. (3.60) with the Green function g determined by eqn. (3.52). Now H_0 in eqn. (3.52) is determined by the formula (3.68). Calculating the integrals in eqn. (3.60) we find

$$F=\frac{\pi\nu}{8}\int\mathrm{STr}\left[\mathscr{D}\left(\nabla Q-\frac{ie}{c}\mathbf{A}[Q,\tau_3]\right)^2+2i\omega\Lambda Q\right]dr.\tag{3.69}$$

The free energy (3.69) contains the commutator $[Q,\tau_3]$ in addition to the gradient term. The matrix τ_3 in eqn. (3.69) breaks the symmetry (3.55) and results in the appearance of a gap in the spectrum of some of the Goldstone modes. Substituting eqns. (3.41) and (3.44) into eqn. (3.69) one can see that only the excitations corresponding to the fields a_1, b_1 and σ_1 remain Goldstone modes. Fluctuations of a_2, b_2 and σ_2 are suppressed by the magnetic field in the long-wavelength limit. Equation (3.69) can be used for both spinless particles and particles with the spin, provided magnetic impurities and spin–orbit interactions are absent. The influence of the external magnetic field on the particle spin is small and is neglected. For calculation of the density–density correlator one can use eqns. (3.63). However, now the indices α and β in the former formula (3.63) must correspond only to the components a_1, b_1 and σ_1 in the matrices (3.41) and (3.44).

For the study of spin interactions it is necessary to use the double set of the components in the supervectors Φ in the equations for the Green functions ((3.13);

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(3.15)). The hamiltonian H in eqn. (3.15) including these interactions is

$$H = H_0 + H_1, \qquad H_1 = U(r) + \mathbf{U}_s(r)\boldsymbol{\sigma} + \sigma[\mathbf{U}_{so} \times \mathbf{p}], \tag{3.70}$$

where H_0 stands for the kinetic energy and the interaction with the external field. The potential $U(r)$ describes as before the interaction with normal impurities, U_s is the interaction with magnetic impurities, and U_{so} stands for the spin–orbit interactions. The components of the vector $\boldsymbol{\sigma}$ in eqn. (3.70) are the Pauli matrices σ_x, σ_y and σ_z. (Note the difference of the notations for the Pauli matrices from those for the elements σ_1 and σ_2 in eqn. (3.41).) It is convenient to define the anticommuting and commuting components of the supervector ψ as

$$\left.\begin{aligned} \Delta^m &= \frac{1}{\sqrt{2}}\begin{pmatrix} \chi^{m*} \\ i\sigma_y\chi^m \end{pmatrix}, & v^m &= \frac{1}{\sqrt{2}}\begin{pmatrix} S^{m*} \\ i\sigma_y S^m \end{pmatrix}, \\ \bar{\psi} &= (C\psi)^{\mathrm{T}}, \qquad c_1 = \begin{pmatrix} 0 & i\sigma_y \\ i\sigma_y & 0 \end{pmatrix}, & c_2 &= \begin{pmatrix} 0 & i\sigma_y \\ i\sigma_y & 0 \end{pmatrix}. \end{aligned}\right\} \tag{3.71}$$

In eqn. (3.71)

$$\chi^m = \begin{pmatrix} \chi_1^m \\ \chi_2^m \end{pmatrix}$$

in accordance with the two possible spin directions. It is supposed that the matrix C is related to the matrices c_1 and c_2 according to eqn. (3.20). All the equations of the preceding sections are unchanged provided the matrix C is specified now by the formulae (3.20) and (3.71). In particular, the symmetry of the 16×16 supermatrix Q is determined by the expressions (3.49) and (3.58) as before.

We suppose, as in the consideration of the effect of the magnetic field, that the interaction with the magnetic and spin–orbit impurities is weak. The effect of these interactions is considered as a small perturbation and can therefore be calculated for each type of interaction independently. Averaging over the random fields of the magnetic impurities U_s gives the additional term in the lagrangian (3.26)

$$\mathscr{L}_s = \frac{1}{12\pi\nu\tau_s}(\bar{\psi}\Sigma\psi)^2, \qquad \Sigma = \boldsymbol{\sigma} \otimes \tau_3, \tag{3.72}$$

where τ_s is the mean time of scattering on magnetic impurities determined as

$$\langle U_s^i(r) U_s^j(r') \rangle = \frac{\delta_{ij}\delta(r-r')}{2\pi\nu\tau_s}.$$

The symbol $\otimes$ stands for the direct product of matrices. Integration over the supervectors ψ gives in the principal approximation the additional term F_s in the free energy (3.62).

$$F_s = -\frac{\pi\nu}{4\tau_s}\int \mathrm{STr}\,[Q, \Sigma]^2 dr, \tag{3.73}$$

where $[\ldots]$ is the commutator. Rewriting the supermatrix Q in terms of the spin quarternions $Q = Q_0 + \mathbf{Q}\boldsymbol{\sigma}$ we reduce eqn. (3.73) to the form

$$F_s = \frac{\pi\nu}{4\tau_s}\int \mathrm{STr}\,(-3[Q_0, \tau_3]^2 + 2(\mathbf{Q}\tau_3)^2 + 2\mathbf{Q}^2) dr. \tag{3.74}$$

Spin blocks of the matrices Q_0 and $\mathbf{Q}$ in eqn. (3.74) are the 2×2 unit matrices.

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One can see from eqns. (3.62) and (3.74) that the excitations corresponding to fluctuations of $\mathbf{Q}$ have a gap. The gap appears also in the excitations corresponding to the fluctuations of a_2, b_2 and σ_2 in Q_0. Therefore the symmetry of the supermatrix Q in the presence of the magnetic field coincides with that in the presence of magnetic impurities. In the long-wavelength limit the excitations with the gap can be neglected. The supermatrix Q describing the remaining Goldstone modes is specified by eqn. (3.41), where one has to use the following expressions for the matrices a, b and σ instead of those in eqns. (3.40)

$$\left.\begin{aligned} a^{mn} &= \begin{pmatrix} a_1^{mn} & 0 \\ 0 & a_1^{mn*} \end{pmatrix}, \qquad b^{mn} = \begin{pmatrix} b_1^{mn} & 0 \\ 0 & b_1^{mn*} \end{pmatrix}, \\ \sigma^{mn} &= \begin{pmatrix} \sigma_1^{mn} & 0 \\ 0 & -\sigma_1^{mn*} \end{pmatrix}, \qquad a_1^{mn} = a_1^{nm*}, b_1^{mn} = b_1^{nm*}. \end{aligned}\right\} \tag{3.75}$$

Equations (3.75) are written for the case of spinless particles in the external magnetic field. In the case of the particles with the spin in the presence of the magnetic impurities one should multiply the elements a_1^{mn}, b_1^{mn} and σ_1^{mn} by the unit matrix, thus obtaining the 16×16 matrix Q. All the other formulae of the preceding section, for example (3.55) and (3.57), remain unchanged.

Of course, one should keep in the supermatrix V in eqn. (3.55) only the part commuting with $\boldsymbol{\Sigma}$.

A weak spin–orbit interaction yields the following term in eqn. (3.26):

$$\mathscr{L}_{so} = \frac{1}{2}\left\langle \left(\int dr \bar{\psi}(r) \boldsymbol{\sigma} [\nabla U_{so} \times \hat{\mathbf{p}}] \psi(r) \right)^2 \right\rangle. \tag{3.76}$$

In contrast to the magnetic case the matrix $\boldsymbol{\sigma}$ instead of $\boldsymbol{\Sigma}$ enters into eqn. (3.76). This is due to additional multiplication by the matrix τ_3 after the partial integration in the transformation from Φ to ψ. Integrating over ψ in the free energy (3.62) we obtain the following term F_{so} describing the spin–orbit interaction

$$F_{so} = -\frac{\pi \nu}{4} \sum_{i=x,y,z} \frac{1}{\tau_{so}^{(i)}} \mathrm{STr}\,[Q, \sigma_i]^2, \tag{3.77}$$

$1/\tau_{so}^{(i)} = 2\pi\nu \langle [\nabla U_{so}(r) \times \mathbf{p}]^{i2} \rangle$ is the inverse mean-free-time of spin–orbit scattering, σ_i in eqn. (3.77) are the Pauli matrices. The angular brackets in eqn. (3.77) stand for both averaging over impurities and averaging over the Fermi surface. Writing the matrix Q as the sum of the singlet Q_0 and triplet Q parts again, one can see that the excitations corresponding to fluctuations of Q have the gap. However, unlike the magnetic case, all fluctuations of Q_0 remain gapless. The symmetry of the singlet part Q_0 of the supermatrix Q is conserved because the spin–orbit interaction does not break the time-reversal symmetry. The matrix elements a, b and σ of the supermatrix Q which describe all the remaining Goldstone modes can be written as

$$\left.\begin{aligned} a^{mn} &= \begin{pmatrix} a_1^{mn} & a_2^{mn} \\ a_2^{mn*} & a_1^{mn*} \end{pmatrix}, \qquad b^{mn} = \begin{pmatrix} b_1^{mn} & -b_2^{mn} \\ b_2^{mn*} & b_1^{mn*} \end{pmatrix}, \\ \sigma^{mn} &= \begin{pmatrix} \sigma_1^{mn} & \sigma_2^{mn} \\ -\sigma_2^{mn} & \sigma_1^{mn} \end{pmatrix}, \quad a_1^{mn} = a_1^{nm*}, \quad a_2^{mn} = a_2^{nm}, b_1^{mn} = b_1^{nm*}, b_2^{mn} = -b_2^{nm}. \end{aligned}\right\} \tag{3.78}$$

In eqns. (3.78) a_i, b_i and σ_i are 2×2 matrices which are proportional to the unit matrix.

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For calculation of the density–density correlator in the presence of magnetic and spin–orbit interactions one can use the formula (3.63) as before, provided the supermatrices $Q(r)$ and $Q(r')$ are substituted by the following expressions

$$Q(y)\to Q\left(1+\frac{\tau}{2}\left[Q,\left(\frac{1}{\tau_s}\Sigma Q\Sigma+\sum_{i=x,y,z}\frac{1}{\tau_{so}^{(i)}}\sigma_i Q\sigma_i\right)\right]\right),\quad y=r,r'. \tag{3.79}$$

The energy F must now contain the terms (3.73) and (3.77). Equation (3.79), as eqns. (3.73) and (3.77), is written in the principal approximation with respect to the magnetic and spin–orbit interactions, $\tau\ll\tau_s$, τ_{so}. The results above show that the magnetic field, the magnetic impurities and the spin–orbit interactions lead to anisotropies with the different symmetries in the free energy and in the density–density correlator. These anisotropies lower the original symmetry. The magnetic impurities have the strongest effect because they break both the time-reversal and spin-rotation invariances. The magnetic field and the spin–orbit interactions have a weaker effect on the symmetry because each breaks only one of these invariances. All the interactions have the same effect on the free energy functional as that obtained in the replica method (Efetov *et al.* 1980).

§4. Interaction of diffusion modes and the renormalization group in $2+\varepsilon$ dimensions

4.1. *Interaction of diffusion modes*

In the preceding chapter the σ-model describing the diffusion modes has been derived. This model is rather complicated. Before studying it for non-trivial cases let us consider the high-frequency limit where the mode interaction is small. In this limit, many results have been obtained by conventional perturbation theory (Oppermann and Wegner 1979, Gor'kov *et al.* 1979, Hikami 1981) and with σ-models obtained by the replica method (Schäfer and Wegner 1980, Efetov *et al.* 1980, Hikami, 1981). The aim of this chapter is to obtain the known results in the limit of small diffusion-mode interaction. This helps demonstrate how the supersymmetry method works. Furthermore, formulae previously derived in different ways will be obtained in a unified manner.

In the gaussian approximation the density–density correlator describes the free diffusion mode (3.66). For calculation of corrections to the correlator which are due to mode interactions one should consider terms of higher order in W. In principle one might expand the supermatrix Q in W using the representation (3.57) and consider the resulting non-linear terms. However, another parametrization of the supermatrix Q exists which is more convenient for the perturbation theory.

$$Q=\Lambda(1+iP)(1-iP)^{-1},\qquad P=\begin{pmatrix}0 & B\\ \bar{B} & 0\end{pmatrix},\qquad \bar{B}=C_0^{\mathrm{T}}B^{\mathrm{T}}C_0, \tag{4.1}$$

where C_0 is specified by eqns. (3.20) and (3.70).

In eqn. (4.1), B is a supermatrix which has the same symmetry as the supermatrix $-iQ^{12}$. In particular, the supermatrix P satisfies the following relation analogous to eqn. (3.58)

$$P=KP^{+}K. \tag{4.2}$$

Of course, eqn. (3.49) follows from eqn. (4.1). In addition the necessary relation $Q^2=1$ is satisfied.

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Substituting eqn. (4.1) into the formulae for the free energy F (3.69), taking into account all the magnetic and spin–orbit interactions ((3.73), (3.77)), one can transform the energy F to the form

$$F=\pi\nu\int \mathrm{STr}\Big[(1+B\bar{B})^{-1}\partial B(1+\bar{B}B)^{-1}\partial\bar{B}+i\omega(1+B\bar{B})^{-1} - \frac{1}{2\tau_s}(1+B\bar{B})^{-1}[B,\Sigma](1+\bar{B}B)^{-1}[\bar{B},\Sigma] - \frac{1}{2}\sum_{i=x,y,z}\frac{1}{\tau_{so}^{(i)}}(1+B\bar{B})^{-1}[B,\sigma_i](1+\bar{B}B)^{-1}[\bar{B},\sigma_i]\Big]dr, \quad (4.3)$$

where $\partial B=\nabla B-(ie/c)\mathbf{A}[B,\tau_3]$.

The density–density correlator ((3.63), (3.79)) can also be rewritten in terms of the supermatrices B. For explicit calculations it is necessary to know the Jacobian (the Berezinian) arising when integrating over the supermatrices B (4.3). The question about the Jacobian is usual in studying σ-models in a certain parametrization. The simplest way to calculate the Jacobian is to write an appropriate quadratic form (cf. with (2.63) and (2.64)). The elementary length in the space of supermatrix elements ΔQ is:

$$\mathrm{STr}\,(\Delta Q)^2=8\,\mathrm{STr}\,((1+B\bar{B})^{-1}\Delta B(1+\bar{B}B)^{-1}\Delta\bar{B}). \quad (4.4)$$

Using the symmetry of the supermatrix B((4.1), (4.2)), one can check that the superdeterminant corresponding to the quadratic form (4.4) is equal to 1. This equality is evident for a simpler quadratic form

$$\mathrm{STr}\,(D\Delta BF\Delta\bar{B}), \quad (4.5)$$

where D and F are diagonal matrices. For calculation of the superdeterminant for this quadratic form it is enough to rewrite this form explicitly using the elements of the supermatrices ΔB and $\Delta\bar{B}$. The commuting elements give the product $\prod_\alpha D_\alpha F_\alpha$ in the numerator of the Berezinian. The anticommuting elements yield the same product in the denominator. It is clear that if D and F are not diagonal but are self-conjugate and, hence, can be diagonalized as, for example, in eqn. (4.4), the result remains unchanged. The equality of the Berezinian to unity makes the parametrization (4.1) very convenient for perturbation theory at high frequencies. For further calculations let us represent the supermatrix B as

$$B=\sum_{a,\alpha}B_{a\alpha}s_\alpha, \quad (4.6)$$

where $s_0=-i$, $s_{1,2,3}=\sigma_{x,y,z}$, $a=\|,\perp$. In eqn. (4.6) the supermatrix $B_\|$ commutes with τ_3, $B_\perp$ anticommutes with τ_3. The condition (4.2) leads to the following relations for the supermatrices $B_{\|\alpha}$, $B_{\perp\alpha}$

$$\tilde{C}^{\mathrm{T}}B_a^{\mathrm{T}}\tilde{C}=kB_a^+, \quad (4.7)$$

where $\tilde{C}=C\sigma_y$. C is specified by the formulae (3.71) as before.

Using the representations (4.6) one can rewrite the quadratic in the part F_2 of the functional F(4.3) as

$$F_2=2\pi\nu\sum_{E,a,\alpha}\mathrm{STr}\,(-i\omega+E_a+X_{a\alpha})B_{a\alpha}^{\mathrm{E}}\overline{B_{a\alpha}^{\mathrm{E}}}. \quad (4.8)$$

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In eqn. (4.8) $B^{E}_{a\alpha}$ are coefficients in the expansion of the functions $B_{a\alpha}(r)$ over the eigenfunctions of the hamiltonian describing a particle in a magnetic field; E_a are eigenenergies

$$E_{\|}=p^2\mathscr{D}, \quad E_{\perp}=\frac{4\mathscr{D}\hbar eH}{c}\left(n+\frac{1}{2}\right)+\mathscr{D}p_H^2, \tag{4.9}$$

where H is the magnetic field and the momentum p_H is parallel to this field.

Calculating the commutators in eqn. (4.3) we obtain for $X_{a\alpha}$:

$$\left.\begin{aligned} X_{\|0}&=0, \qquad X_{\|i}=\frac{4}{\tau_s}+\frac{2}{\tau_{so}}-\frac{2}{\tau_{so}^{(i)}},\\ X_{\perp 0}&=\frac{6}{\tau_s}, \qquad X_{\perp i}=\frac{2}{\tau_s}+\frac{2}{\tau_{so}}-\frac{2}{\tau_{so}^{(i)}}, \end{aligned}\right\} \tag{4.10}$$

where $i=x, y, z$, and $1/\tau_{so}=\sum_{i=x,y,z}1/\tau_{so}^{(i)}$.

The correlation functions in the principal approximation are determined by gaussian integrals. Using the relations (4.7) we have

$$(B_{a\alpha})_{ik}(\bar{B}_{b\beta})_{em}=\frac{1}{4\pi\nu}\cdot\frac{\delta_{ab}\delta_{\alpha\beta}\delta_{im}k_{kl}}{-i\omega+E_a+X_{a\alpha}}. \tag{4.11}$$

The formulae (4.9), (4.10) and (4.11) show in which diffusion modes a gap appears. The gap is absent, as before, in correlations of elements of the supermatrix $B_{\|0}$. In the gaussian approximation the density–density correlator (3.63) is proportional to the correlator $\langle B_{\|0}B_{\|0}\rangle$ which does not have a gap. It is natural that the gap does not appear in the density–density correlator because the break of the time-reversal and spin-rotation symmetries does not influence the particle conservation law which is the reason for the absence of any gap in this correlator.

In the presence of magnetic interactions only excitations corresponding to the supermatrix $B_{\|0}$ remain gapless. It follows from the relation (4.7) that the structure of the supermatrix B coincides with that of the supermatrix Q^{12} ((3.44), (3.75)). If the magnetic interactions are absent but spin–orbit interactions are possible, only excitations corresponding to the supermatrices $B_{\|0}$ and $B_{\perp 0}$ are gapless. In this case the structure is determined by eqns. (3.44) and (3.78).

Now we can calculate the first correction arising from the diffusion-mode interaction. The interaction energy F_{int} in the lowest order in B takes the form

$$F_{\text{int}}=-2\pi\nu\sum_{\substack{a,b,c,d\\ \alpha,\beta,\gamma,\delta}}(e_{\alpha\beta\gamma\delta}+\delta_{\alpha\beta}\delta_{\gamma\delta}+\delta_{\alpha\delta}\delta_{\beta\gamma}-\delta_{\alpha\gamma}\delta_{\beta\delta})$$

$$\times\int \mathrm{STr}\,[\mathscr{D}(\nabla B_{a\alpha}\nabla\bar{B}_{b\beta}B_{c\gamma}\bar{B}_{\alpha\delta}+\nabla\bar{B}_{a\alpha}\nabla B_{b\beta}\bar{B}_{c\gamma}B_{d\delta})-i\omega B_{a\alpha}\bar{B}_{b\beta}B_{c\gamma}\bar{B}_{d\delta}]dr, \tag{4.12}$$

where $e_{\alpha\beta\gamma\delta}$ is the completely antisymmetric tensor ($e_{0123}=1$). In eqn. (4.12) the magnetic field and the impurities are neglected. Considering the same order terms

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one can write the density–density correlator in the form

$$K(\omega)=(\pi\nu)^2\Big\langle \mathrm{STr}\, k(1-\tau_3)(B_{\|0}(r)-\sum_{\substack{a,b,c\\ \alpha,\beta,\gamma}} T_{\alpha\beta\gamma}B_{a\alpha}(r)\bar{B}_{b\beta}(r)B_{c\gamma}(r))$$

$$\times k(1-\tau_3)(\bar{B}_{\|0}(r)-\sum_{\substack{k,l,m\\ \sigma,\rho,\xi}} T^{*}_{\sigma\rho\xi}\bar{B}_{k\sigma}(r')B_{l\rho}(r')\bar{B}_{m\xi}(r'))\Big\rangle, \quad (4.13)$$

where $T_{\alpha\beta\gamma}=\frac{1}{2}\mathrm{Tr}\,(s_\alpha s_\beta^{+} s_\gamma)$.

In eqn. (4.13) the angular brackets stand for averaging with the functional F_2+F_{int}. Expanding over the interaction and calculating the gaussian integrals we find in the main approximation

$$K(\omega,k)=\frac{4\pi\nu}{\mathscr{D}k^2-i\omega}, \quad (4.14)$$

$$\mathscr{D}=\mathscr{D}_0\left(1-\frac{1}{2\pi\nu}\sum_{E\perp,\alpha}\frac{1-2\delta_{\alpha 0}}{E_\perp+X_{\perp\alpha}-i\omega}\right). \quad (4.15)$$

Thus, the diffusion–mode interaction does not change the diffusion form of the density–density correlator but it changes the diffusion coefficient. The residue $4\pi\nu$ in eqn. (4.14) differs from that in eqn. (3.66) by the factor 2 because now particles with spin are considered.

The formula (4.15) contains all the known effects of first order in the diffusion–mode interaction. In the system invariant under time-reversal and spin-rotations the correction to the diffusion coefficient is negative and is proportional to $|\ln\omega\tau|$ in two dimensions (Abrahams *et al.* 1979, Gor'kov *et al.* 1979, Anderson *et al.* 1979). In long thin wires, where the formula (4.15) is also valid, the correction is proportional to $(\omega\tau)^{-1/2}$. Equation (4.15) agrees with the result of the direct summation of diagrams (3.55).

The external magnetic field and the magnetic impurities cut these corrections (Lee 1979, Altshuler *et al.* 1980). In the absence of magnetic interactions, spin–orbit interactions lead to a change in sign of the correction, which becomes positive (Hikami *et al.* 1980), If the spin–orbit interactions are sufficiently weak, the magnetic field increases the conductivity which is connected with the diffusion coefficient by the Einstein relation (negative magnetoresistance) (Altshuler *et al.* 1980). This increase is observed experimentally. Equation (4.15) can also be derived with the σ-model which has been obtained within the replica method (Efetov *et al.* 1980).

Jungling and Oppermann (1979) and Oppermann (1981), considering models with random hopping from site to site and broken time-reversal and spin-rotation invariances, got analogous results.

Study of the small corrections to the diffusion coefficient in low dimensions is applicable for sufficiently high frequencies. For lesser frequencies more complicated methods are necessary. In the next section such an investigation will be carried out by the renormalization group method in $2+\varepsilon$ dimensions.

4.2. *Renormalization group in $2+\varepsilon$ dimensions*

Another example of calculations using the supermatrix σ-model (3.62) is the renormalization group study in $(2+\varepsilon)$ dimensions. Analogous calculations for

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conventional-matrix σ-models, for many types of symmetry, have been made already (Brezin *et al.* 1980). The limit $n \to 0$ for $n \times n$ matrices corresponds to the localization problem (Wegner 1979, Efetov *et al.* 1980). Supermatrix σ-models were first considered by the author (Efetov 1982 a). Following this work, let us prove the renormalizability of the supermatrix model (3.62) for all three types of symmetry of the supermatrix Q determined by eqns. (3.41), (3.57), (3.44), (3.75) and (3.78). The supermatrices Q are 8×8 supermatrices for spinless particles and 16×16 for particles with spin. However, the diffusion of the particles with spin 'up' and spin 'down' can be considered separately provided magnetic and spin–orbit interactions are absent. In this case the 8×8 supermatrix Q and the corresponding σ-model do not differ from those for spinless particles. The particle spin results only in the additional factor 2 in the density–density correlator (3.63). In the presence of magnetic and spin–orbit impurities one should consider the 16×16 supermatrix Q. However, the effective anisotropies corresponding to these impurities partially freeze out excitations. The remaining Goldstone excitations correspond to fluctuations of supermatrices which have spin blocks proportional to the unit 2×2 matrix. All other elements of the spin blocks vanish because the corresponding excitations have a gap. Such supermatrices, of course, are equivalent to 8×8 ones. In this case one can write 8×8 supermatrices in the σ-model (3.62), again. It is necessary only to multiply the free energy F by 2.

Having noted this, we may write the free-energy functional in a unified form

$$F = \frac{1}{t} \int \mathrm{STr}\,[(\nabla Q)^2 + 2i\tilde{\omega}\Lambda Q]\,dr, \tag{4.16}$$

where Q is the 8×8 supermatrix specified by eqns. (3.57), (3.44) or (3.55).

Let the model in which only potential scattering on normal impurities is possible be called Model I. The system with potential scattering in an external magnetic field will be called Model IIa and an addition of magnetic impurities to Model I or Model IIa leads to Model IIb. The system where only potential and spin–orbit scattering are possible will be called Model III. Model I is invariant under time-reversal and spin-rotations. In Model IIa time-reversal invariance is violated but spin-rotation invariance is conserved. In Model IIb both invariances are broken. And, in the Model III, time-reversal symmetry is conserved but spin-rotation invariance is broken. The system with potential and spin–orbit scatterings only, but placed in a magnetic field, relates to the model IIb. In the Model I the matrices a, b and σ are specified by eqns. (3.41), in the Models IIa, IIb, by eqns. (3.75) and in the Model III by eqns. (3.78). The coefficient t in eqn. (4.16) for these models is

$$t = \begin{cases} \dfrac{8}{\pi \nu \mathscr{D}}, & \text{Models I, IIa,} \\ \dfrac{4}{\pi \gamma \mathscr{D}}, & \text{Models IIb, III.} \end{cases} \tag{4.16 a}$$

In all the cases $\tilde{\omega} = \omega/\mathscr{D}$.

If the behaviour of a particle with spin is considered, one should multiply the density–density correlator (3.63) by 2 for Models I and IIa.

The quasi-diagonal structure (3.75) of the supermatrix Q in Models II enables one to reduce the problem to a σ-model with a 4×4 supermatrix.

For calculations in $2+\varepsilon$ dimensions one may use the renormalization group method in the form suggested by Polyakov (1975) for studying a vector σ-model. Let us divide the unitary supermatrix $V(r)$ in eqn. (3.55), into the product of fast $V_0(r)$ and slow $\tilde{V}(r)$ parts

$$V(r)=\tilde{V}(r)V_0(r). \tag{4.17}$$

The matrices $V_0(r)$ and $\tilde{V}(r)$ are supposed to satisfy the conditions (3.56) and (3.59). Substituting eqns. (3.55) and (4.17) into eqn. (4.16) we get

$$F=\frac{1}{t}\int \mathrm{STr}\,[(\nabla Q_0)^2+2[Q_0,\nabla Q_0]\mathbf{\Phi}+[Q_0,\mathbf{\Phi}]^2+2i\tilde{\omega}\bar{\tilde{V}}\Lambda\tilde{V}Q_0]dr,$$

$$Q_0=V_0\Lambda\bar{V}_0,\quad \mathbf{\Phi}=\bar{\tilde{V}}\nabla\tilde{V}=-\bar{\mathbf{\Phi}}. \tag{4.18}$$

Now we integrate in the integral (3.63) over the fast variable $Q_0(r)$ using the free energy functional (4.18). After the integration the energy F in eqn. (4.18) is replaced by the energy F describing the slow fluctuations.

$$\tilde{F}=-\ln\int \exp(-F)\mathscr{D}Q_0. \tag{4.19}$$

For simplification of the calculations it is convenient to choose a certain gauge of the supermatrices $\tilde{V}$ and $\mathbf{\Phi}$. The supermatrix V in eqns. (4.17) and (4.18) is supposed to be close to unity in the vicinity of a co-ordinate r. The slowness of the variations enables us to integrate in different domains independently. In the gauge chosen in the vicinity of the co-ordinate r the superelements Φ^{12} and Φ^{21} are the only ones in the supermatrix Φ which are not equal to zero. These superelements are related to each other by the antiself-conjugacy condition of the supermatrix Φ. The question of separating the fast and the slow variables and that of cutting divergent integrals in an invariant manner is extremely important. It is supposed that momenta of space variations of the supermatrix Q are in the interval $\lambda<k<k_0$. For invariant infrared cutting in the integral over Q_0 one may add in eqn. (4.18) the following term

$$F_{\mathrm{reg}}=-\frac{2}{t}\int \lambda^2\,\mathrm{STr}\,(\Lambda Q_0)dr. \tag{4.20}$$

Cutting integrals divergent at large momenta will be performed by dimensional regularization. According to this approach an analytical continuation is made from $d=2-\delta$ dimensions to $d=2+\varepsilon$ dimensions. For calculation of the integral over Q_0 it is convenient to parametrize the supermatrix Q_0 with the use of eqn. (4.1).

Now, after exclusion of the spin structure, the matrices c_1 and c_2 entering C and C_0 are specified for Models I and II by eqns. (3.20) again. The permutation $c_1\rightleftarrows c_2$ of the matrices c_1 and c_2 in eqn. (3.20) gives the matrices C and C_0 in Model III.

It has been mentioned that integration over the supermatrices B is performed with the unit Berezinian. Using eqn. (4.1), expanding Q_0 over P and retaining, in the

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functional F (4.18), (4.20), the number of terms necessary for obtaining the first two orders, we find

$$\left.\begin{aligned}
&F=F_0+F_0'+F_1+F_2+F_2'+\tilde{F}_0,\\
&F_0=\frac{4}{t}\mathrm{STr}\int[\nabla P)^2+\lambda^2P^2]dr,\\
&F_0'=-\frac{4}{t}\mathrm{STr}\int[2P^2(\nabla P)^2-\lambda^2P^4]dr,\\
&F_1=-\frac{16i}{t}\mathrm{STr}\int P\nabla PP\Phi dr,\\
&F_2=\frac{8}{t}\mathrm{STr}\int[P^2\Phi^2-(P\Phi)^2]dr,\\
&F_2'=-\frac{8}{t}\mathrm{STr}\int\left[P^4\Phi^2+(P^2\Phi)^2-2P\Phi P^3\Phi+\frac{i\omega}{2}\Lambda\tilde{V}\Lambda(P^2-P^4)\tilde{\bar{V}}\right]dr,\\
&\tilde{F}_0=\frac{1}{t}\mathrm{STr}\int[(\nabla\tilde{Q})^2+2i\tilde{\omega}\Lambda\tilde{Q}]dr.
\end{aligned}\right\}\tag{4.21}$$

In eqn. (4.21) the supermatrix $\tilde{Q}$ is $\tilde{Q}=\tilde{V}\Lambda\tilde{\bar{V}}$.

Terms linear in P in eqn. (4.21) are absent because P varies rapidly. It is supposed that $\lambda^2\gg\tilde{\omega}$. Keeping the first order terms in t, and the quadratic terms in Φ, the free energy functional F in eqn. (4.19) can be reduced to the form

$$\tilde{F}=\tilde{F}_0+\langle F_2\rangle_0+\langle F_2'\rangle_0-\frac{1}{2}\langle F_1^2\rangle_0-\langle F_0'F_2\rangle_0.\tag{4.22}$$

In eqn. (4.22) the brackets $\langle\ldots\rangle_0$ stand for averaging with the functional F_0 from eqn. (4.21). The gaussian integrals are calculated according to Wick's theorem. The following formulae which can be checked by direct calculation are useful

$$\langle PMP\rangle_0=\frac{\alpha t}{16(k^2+\lambda^2)}\bar{M}',\quad\text{where}\quad\bar{M}'=C_0^{\mathrm{T}}M^{\mathrm{T}}C_0.\tag{4.23}$$

Equation (4.23) is valid for supermatrices M of the form

$$M=\begin{pmatrix}0 & M^{12}\\ M^{21} & 0\end{pmatrix}.\tag{4.24}$$

For supermatrices N of the form

$$N=\begin{pmatrix}N^{11} & 0\\ 0 & N^{22}\end{pmatrix}$$

the following equation holds

$$\begin{aligned}\langle PNP\rangle_0=\frac{t}{64(k^2+\lambda^2)}&[(1+\alpha^2)(\mathrm{STr}\,N-\Lambda\,\mathrm{STr}\,\Lambda N)\\&+(1-\alpha^2)\tau_3(\mathrm{STr}\,(\tau_3N)-\Lambda\,\mathrm{STr}\,(\Lambda\tau_3N))].\end{aligned}\tag{4.25}$$

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And, at last, for arbitrary supermatrices M_1 and M_2 of the form (4.24) satisfying the conditions $M_1 = M_1'$, $M_2 = M_2'$ the following formula is valid

$$\langle \mathrm{STr}(PM_1)\mathrm{STr}(PM_2)\rangle_0 = \frac{t}{8(k^2+\lambda^2)}\mathrm{STr}(M_1\bar{M}_2'). \tag{4.26}$$

The supermatrices M_i^{12}, M_i^{21}, N^{11} and N^{22} are supposed to have the same structure as the supermatrices B. The coefficient α in eqns. (4.23) and (4.25) depends on the model and is equal to

$$\alpha = \begin{cases} -1, & \text{Model I,} \\ 0, & \text{Model II,} \\ 1, & \text{Model III.} \end{cases} \tag{4.27}$$

Using eqns. (4.23)–(4.27) and calculating the averages in eqn. (4.22) we obtain

$$F = \frac{1}{t}\int\left\{\mathrm{STr}\,(\nabla\tilde{Q})^2\left[1+\frac{\alpha t}{8}\int\frac{d^dk}{(2\pi)^d(k^2+\lambda^2)}\right.\right.$$

$$\left.\left.+\frac{t^2}{64}(1-\alpha^2)\left(\frac{1}{2}-\frac{1}{d}\right)\int\frac{1}{(k_1^2+\lambda^2)(k_2^2+\lambda)}\,\frac{d^dk_1\,d^dk_2}{(2\pi)^{2d}}\right]+2i\tilde{\omega}\,\mathrm{STr}\,\Lambda Q\right\}dr. \tag{4.28}$$

In eqn. (4.28) averaging over angles in the integral arising from $\langle F_1^2\rangle$ has been carried out. All terms which are not logarithmic in two dimensions are neglected.

In this supermatrix σ-model quadratic divergences, usual in conventional σ-models (Brezin *et al.* 1976, Brezin and Zinn-Justin 1976), are absent. Furthermore divergencies arising after expansion of the Jacobian are also absent here. Let us mention an interesting peculiarity of the model under consideration. In this model only the effective temperature is renormalized. The value $\tilde{\omega}/t$ which is proportional to the frequency ω does not change under the renormalization. This property holds in all orders of the renormalization-group calculations and is a consequence of the particle conservation law.

Calculating integrals over momenta in eqn. (4.28) in $2-\delta$ dimensions and making the analytical continuation to $2+\varepsilon$ dimensions one can find the Gell–Mann–Low function in the two-loop approximation

$$\beta(\tilde{t}) \equiv \frac{d\tilde{t}}{d\ln\lambda} = (d-2)\tilde{t} + \alpha\tilde{t}^2 - \frac{1}{2}\tilde{t}^3(1-\alpha^2), \quad \tilde{t} = \frac{t}{4\cdot 2^d\pi(d/2)\Gamma(d/2)}. \tag{4.29}$$

It is seen from eqn. (4.29) that the t^3 term is zero in Models I and III. In strictly two dimensions the solution of eqn. (4.29) can be written as

$$\tilde{t}(\omega) = \frac{\tilde{t}_0}{1+\alpha\tilde{t}_0\ln(1/\omega\tau)}. \tag{4.30}$$

For sufficiently high frequencies the 'temperature' t and hence the diffusion coefficient $\mathscr{D}$ coincide with the bare values.

A decrease of the frequency in Model I results in an increase of t and decrease of $\mathscr{D}$. Unfortunately, it is impossible to reach the low-frequency region because at $t \sim 1$ one must calculate the exact Gell–Mann–Low function. The consideration of terms of first-order in t is no longer valid. The renormalization-group equation for Model I agrees with that obtained by virtue of the scaling hypothesis (Abrahams *et al.* 1979).

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The behaviour $t \to \infty$ as $\omega \to 0$ seems to be quite reasonable. Such behaviour signals electron localization for an arbitrary weak disorder. In Model III the value of t and, hence, the resistivity tends to zero when the frequency ω decreases. In first order of the perturbation theory this effect was first observed by Hikami *et al.* (1980). The appropriate renormalization-group equation was then obtained by the replica method (Efetov *et al.* 1980). Let us emphasize that in this case eqn. (4.30) is applicable everywhere including the region of lowest frequencies. Of course, it is supposed that $t_0 \ll 1$. For $t_0 \gtrsim 1$ the localization behaviour is possible at low frequencies.

In Models II the t^2 term in the Gell–Mann–Low function (4.29) vanishes but the t^3 term is not equal to zero. The sign of this term corresponds to the growth of the resistivity as ω decreases. Apparently, at low frequencies all states are localized. Equation (4.29) for Models I and II was first obtained by Wegner (1979) using the replica method. Wegner used the results obtained by Brezin *et al.* (1980) for matrix σ-models. According to calculations made by Hikami (1981) using replica σ-models the t^4 term in eqn. (4.29) is absent.

In $2+\varepsilon$ dimensions in Models I and II the fixed point t_c, where the Gell–Mann–Low function $\beta(t)$ turns to zero, exists

$$\beta(t_c) = 0. \tag{4.31}$$

In the vicinity of this point the correlation length ξ varies as

$$\xi \sim \xi_0 \left(\frac{t_c - t}{t_c} \right)^{-1/y}, \quad y = -\beta'(t_c). \tag{4.32}$$

Using the formulae (4.32) for the correlation length ξ one can write the equation for the conductivity (Wegner 1976)

$$\sigma = A \frac{e^2}{\hbar} \frac{1}{t_c} \frac{1}{\xi_0^{d-2}} \left(\frac{t_c - t}{t_c} \right)^s, \quad s = \frac{d-2}{y}, \tag{4.33}$$

where A is of the order of unity, the factor $e^2/\hbar\xi_0^{d-2}$ is the Mott conductivity. Equations (4.31)–(4.33) can be used for any dimensionality $d = (2+\varepsilon) > 2$. If ε is small one may use only the first order terms in the Gell–Mann–Low function $\beta(t)$. In this case the fixed points t_c are equal

$$\tilde{t}_c = \begin{cases} d-2 & \text{Model I,} \\ \sqrt{[2(d-2)]} & \text{Model II.} \end{cases} \tag{4.34}$$

It is not difficult to calculate in this approximation the critical indices y and s

$$y = \begin{cases} d-2, \\ 2(d-2), \end{cases} \qquad s = \begin{cases} 1, \\ 1/2. \end{cases} \tag{4.35}$$

In Model III the fixed point is absent provided the disorder is weak and $d-2$ is small. Nevertheless, the mobility edge apparently exists in Model III in three dimensions.

The calculations carried out in this chapter show that using the supermatrix σ-model one can reproduce all the known results in a straightforward and rather simple manner. The difficulties connected with analytical continuations over the number of fields met in the replica method do not exist. Of course, such an

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improvement is not very important for problems which can be treated by perturbation theory or renormalization group calculations, where the replica method is seen always to give correct results. However, in problems that cannot be treated by perturbation theory this advantage is crucial. In the following chapters several problems which cannot be solved by the replica method or other known methods are studied by the supersymmetry method.

§ 5. Energy level statistics in a limited volume

5.1. *Phenomenological theories*

According to the basic principles of quantum mechanics the energy spectrum of a particle in a limited volume is discrete. The positions of the energy levels, and the spacing between them, depend on boundary conditions and interactions in the system. In the simplest cases values for these quantities can be calculated exactly or approximately. However, often the interactions are so complicated that calculations for the levels become impossible. On the other hand the complexity and the variety of the interactions lead to the idea of a statistical description in which information about separate levels is neglected and only averaged quantities are studied. Density of states, level correlations and the like can be so considered. The analogous approach is used in statistical physics where information about separate particles is neglected and only averages over a large number of particles are calculated.

The idea of statistical description for a restricted volume was first proposed by Wigner (1951) for the study of complex nuclei. In such nuclei a large number of particles interacts in an unknown way. It is plausible to assume that all the interactions are possible with an equal probability. According to the Wigner statistical hypothesis all matrix elements H_{mn} of a system having N energy levels are statistically independent and can take any values with equal probabilities. However, this assumption leads to divergent integrals in calculated averages. To make the integrals convergent it is necessary to cut the integrals at large H_{mn} corresponding to strong interactions. The most convenient way to introduce a cutoff into the theory is to use the gaussian ensemble. According to the gaussian statistical hypothesis a physical system having N quantum states contributes with the following statistical weight $\mathscr{D}(H)$

$$\mathscr{D}(H) = A \exp\left[-\sum_{m,n}^{N} \frac{|H_{mn}|^2}{a^2} \right]. \tag{5.1}$$

In eqn. (5.1) a is the cutoff excluding strong interactions.

Of course, now the interactions do not have equal probabilities. The impossibility of defining an ensemble with equal probabilities and the necessity of using a cutoff is the main short-coming of the Wigner theory. Besides, calculations for physical quantities with Wigner's probability distributions are very difficult. A more detailed consideration of these questions is presented by Porter (1965).

The next step in the statistical theory of energy levels was taken by Dyson (1962), who rejected the hamiltonian matrix-element description. In the phenomenological Dyson theory the main object is a unitary $N \times N$ matrix S, where N is the number of the energy levels in the system. This matrix S is related to the hamiltonian of the

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system H in a complicated way. The explicit form of this dependence is not specified. As an example one can write two possible relations

$$S=(1+iH)(1-iH)^{-1}, \qquad S=\exp(iH). \tag{5.2}$$

It is well known that an arbitrary unitary matrix can be diagonalized

$$S=ZS_0Z^{+}, \tag{5.3}$$

where $(S_0)_{ab}=\delta_{ab}\exp(i\theta_a)$, Z is a unitary matrix

$$Z^{+}Z=1. \tag{5.4}$$

According to Dyson's hypothesis the correlation properties of n successive energy levels of the system ($n\ll N$) are statistically equivalent to those of n successive angles θ_a. All the unitary matrices S are assumed to contribute with equal probabilities. Depending on the existence of symmetries in the real system the matrix S has a certain mathematical symmetry.

For systems which are both time-reversal and spin-rotation invariant all wavefunctions can be made real. In this case all the matrix elements of the hamiltonian are real. It is natural to assume that the matrix S for systems having these two types of symmetry is symmetrical. Considering examples (5.2) of relations between matrices S and H one can see that this assumption is reasonable.

Now the matrices in eqn. (5.3) can be made real. Then eqn. (5.4) takes the form

$$ZZ^{\mathrm{T}}=1. \tag{5.5}$$

Real matrices Z obeying the condition (5.5) are called orthogonal. Correspondingly the ensemble of time-reversal and spin-rotation invariant systems is called orthogonal.

If the time-reversal invariance is broken, matrix elements of the hamiltonian H can no longer be made real. Then the unitary matrix S does not have any additional symmetry. The matrix Z is also an arbitrary unitary matrix. The appropriate ensemble is called unitary.

The third possible type of symmetry arises when the system is time-reversal invariant but does not have central symmetry. In this case it is also impossible to make all the matrix elements real. Nevertheless an additional symmetry exists in this case. According to Kramers' theorem all levels in the system remain doubly degenerate. Hence, every eigenvalue of the matrix S must appear twice. Symplectic matrices consisting of real quarternions of the form

$$\begin{pmatrix} a_1 & a_2 \\ -a_2 & a_1 \end{pmatrix} \tag{5.6}$$

have this property. It is clear that the matrix Z is symplectic too. The corresponding ensemble is called symplectic.

A more detailed discussion can be found in the original Dyson work (1962).

Let us discuss the concept of equal probabilities for all matrix elements of the unitary matrix S. One may consider an $N\times N$ complex matrix as a point in the $2N$-dimensional space of real numbers. Then all possible unitary $N\times N$ matrices constitute a hypersurface $\{S\}$ in this space. The equal probability of all matrices S in calculating average quantities implies an equal contribution of each elementary volume on the hypersurfaces $\{S\}$. In other words when calculating averages one

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should integrate over the hypersurface $\{S\}$. In order to perform these calculations one has to choose an appropriate co-ordinate basis. As has been mentioned only correlation functions of eigenvalues of the matrix S are to be studied. Hence, the co-ordinates θ_a and Z (5.3) are convenient. It is not difficult to write the elementary volume in terms of these co-ordinates. First we write as usual the square of the elementary length

$$(dl)^2 = \mathrm{Tr}\,(dSdS^+) = 2\,\mathrm{Tr}\,(\delta Z \delta Z^+ - \delta Z S_0^+ \delta Z^+ S_0) + \sum_a (d\theta_a)^2, \tag{5.7}$$

where $\delta Z = Z^+ \delta Z = -\delta Z^+$. The formula (5.7) can be rewritten in a more convenient form

$$(dl)^2 = \sum_{a,b}^{N} (\delta Z_{ab} \delta Z_{ab}^+) |e^{i\theta_a} - e^{i\theta_b}|^2 + \sum_a (d\theta_a)^2. \tag{5.8}$$

The quadratic form $(dl)^2$ (5.8) is diagonal with respect to the elements δZ_{ab} and $d\theta_a$. It enables us to write down easily the corresponding elementary volume. One should remember that δZ_{ab} are real numbers in the case of the orthogonal ensemble, complex numbers for the unitary ensemble and real quaternions of the form (5.6) for the symplectic one. Therefore the elementary volume is

$$\Delta S = \prod_{a<b} |e^{i\theta_a} - e^{i\theta_b}|^\beta \prod_a d\theta_a \Delta Z. \tag{5.9}$$

$\beta = 1$ for the orthogonal ensemble, $\beta = 2$ for the unitary ensemble and $\beta = 4$ for the symplectic one. The vectors δZ_{ab} constitute the volume ΔZ. In calculating averages of arbitrary functions $f(\theta_1, \theta_2, \ldots, \theta_N)$ one may integrate separately over θ_a and δZ_{ab}. Therefore the volume of the group of the matrices Z does not enter the result of the calculations.

The distribution (5.9) makes it possible to calculate any correlation function of the angles θ_a and hence those of energy levels. The density of states is the simplest correlation function. It follows from the distribution function (5.9) immediately that the density of states is a constant because equation (5.9) contains only differences $\theta_a - \theta_b$.

Two-level correlation functions $R(\omega)$ are more interesting. These functions determine the probability for a level to be at the distance ω from another level. The two level correlation functions have been calculated by Dyson (1962) and by Mehta and Dyson (1963) on the basis of Dyson phenomenology for all three ensembles. The appropriate calculations are not very difficult for the unitary ensemble ($\beta = 2$). However, the calculations for the orthogonal ($\beta = 1$) and symplectic ($\beta = 4$) ensembles required a great deal of mathematical art. The final formulae for the two-level correlation functions can be written as

$$R_{\mathrm{ort}}(x) = 1 - \frac{\sin^2 x}{x^2} - \frac{d}{dx}\left(\frac{\sin x}{x}\right) \int_1^\infty \frac{\sin xt}{t}\, dt, \tag{5.10}$$

$$R_{\mathrm{unit}}(x) = 1 - \frac{\sin^2 x}{x^2}, \tag{5.11}$$

$$R_{\mathrm{sympl}}(x) = 1 - \frac{\sin^2 x}{x^2} + \frac{d}{dx}\left(\frac{\sin x}{x}\right) \int_0^1 \frac{\sin xt}{t}\, dt, \tag{5.12}$$

where $x = \pi\omega/\Delta$, Δ is the mean energy level spacing.

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The phenomenological Dyson and Wigner theories enable us to calculate quantities which can in principle be observed experimentally. At the same time these theories are based on rather strong assumptions which cannot be justified rigorously. In particular, the physical sense of the unitary matrix S is not clear. It is difficult to understand why all unitary matrices must contribute with equal probabilities. The validity of these statistics could be tested only by experiment. Unfortunately the necessary accuracy in experiments on nuclei has not yet been achieved.

Gor'kov and Eliashberg (1965) applied the Dyson theory to systems of a quite different nature. These authors considered small metallic particles. Electron–electron interactions were supposed to be absent and only interactions with impurities were considered. Thus, in their model the interactions with impurities but not electron–electron interactions were random. In order to calculate responses to external electromagnetic field one has to know the level correlation function. Gor'kov and Eliashberg used the expressions given by eqns. (5.10)–(5.12) for this function. They identified the orthogonal ensemble with a system with potential impurity scattering only. The unitary ensemble was used for systems with magnetic impurities or systems placed in a magnetic field. The symplectic ensemble was used for systems with spin–orbit interactions. On the other hand, one can also study statistical properties of energy levels for metallic particles by solving the Schrödinger equation and averaging them over random potential configurations. This approach would enable us to check the basic principles used for the phenomenological description of levels in a restricted volume. The supersymmetry method developed in the preceding sections makes it possible to solve this problem. In what follows (Efetov 1982 b) it will be shown by a direct microscopic calculation that the level correlation functions in metallic particles are correctly described by eqns. (5.10) and (5.11).

5.2. *Zero-dimensional supermatrix σ-model*

Let us consider a metallic particle of a size a. The size a is supposed to be much larger than atomic lengths. The value a can be arbitrarily large provided the zero temperature limit is considered and the lowest excitations are studied. At finite temperatures the size a is supposed to be less than the diffusion length $L_0 \sim (\mathscr{D}\tau_\varepsilon)^{1/2}$, where τ_ε is the non-elastic scattering time, $\mathscr{D}$ is, as before, the diffusion coefficient. Inside the particle there are impurities with random positions. In this model, the two-level correlation function $R(\omega)$ is an interesting quantity characterizing the behaviour of the system. This function has been defined already (3.11). In eqn. (3.11) the sum over all states is written down. It is supposed that states with spin 'up' and spin 'down' are different even in the absence of spin interactions. In real systems the function $R(\omega)$ determines the response to external electromagnetic fields.

We will consider the case

$$\tau^{-1} \gg \Delta. \tag{5.13}$$

The mean level spacing Δ in the absence of spin interactions (Models I and IIa according to the classification in §4.2) is equal to

$$\Delta = (\nu V)^{-1}. \tag{5.14}$$

Equation (5.14) is valid for any shape of the metallic particle except ideally spherical and cylindrical. In the latter cases the spacing between degenerate levels is

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rather large, However, any irregularities, even of an atomic size, will split these degenerate levels and lead to eqn. (5.14). In Models I and IIa the level degeneracy is partially conserved because the electron spin can be arbitrarily directed. Spin interactions also remove this spin degeneracy. As a result, in the models with spin interactions, the mean spacing is one half the previous one

$$\Delta=(2\nu V)^{-1}. \tag{5.15}$$

Inequality (5.13) corresponds to sufficiently strong level mixing. In this limit one may substitute sums of the type $\sum_m(\varepsilon_m-\varepsilon_0+i/\tau)^{-1}$ arising when calculating in eqns. (3.11) and (3.12) by integrals over $\xi=\varepsilon_m-\varepsilon_0$. It enables us to calculate immediately the sum of the product $\langle G^{\mathrm{A}}G^{\mathrm{A}}\rangle$ in eqn. (3.12)

$$\sum_{m,k}\langle G^{\mathrm{A}}_m(\varepsilon+\omega)G^{\mathrm{A}}_k(\varepsilon)\rangle=-(2\pi\nu)^2. \tag{5.16}$$

The simplicity of the calculation of the sum in eqn. (5.16) is due to the absence of the diffusion-mode contribution.

For calculation of the product $\langle G^{\mathrm{R}}G^{\mathrm{A}}\rangle$ one can use the procedure of derivation of the supermatrix σ-model which has been presented in §3. The inequality (5.13) enables us to repeat the derivation of the σ-model without any changes. For the product $\langle G^{\mathrm{A}}G^{\mathrm{R}}\rangle$ in eqn. (3.12) we obtain

$$\left\langle\sum_{m,k}G^{\mathrm{A}}_m(\varepsilon+\omega)G^{\mathrm{R}}_k(\varepsilon)\right\rangle=\frac{(2\pi\nu)^2}{64}\int\mathrm{STr}\,k(1+\Lambda)Q\,\mathrm{STr}\,k(1-\Lambda)Q\exp(-F[Q])\mathscr{D}Q. \tag{5.17}$$

Below, the limiting cases corresponding to Models I, IIa, IIb and III only will be considered. Crossovers from one model to another will not be studied. Therefore the energy $F[Q]$ and the 8×8 supermatrix Q in all the cases are specified by eqns. (4.16), (3.57), (3.44), (3.41), (3.75), (3.78) and (4.16).

The supermatrix σ-model (4.16) can be simplifed considerably provided one considers the lowest excitations. In this limit one may study only those matrices Q which do not vary in the volume.

The energy of non-zero harmonics is large and therefore they can be neglected. In order to find where this approximation is valid we consider the first quantum correction (4.15). In the absence of magnetic and spin–orbit interactions this correction takes the form

$$\frac{1}{2\pi\nu V}\sum_{\mathscr{E}}\frac{1}{\mathscr{E}-i\omega}. \tag{5.18}$$

In eqn. (5.18) the lowest energy $\mathscr{E}_0=0$ corresponds to the zero harmonics. Higher harmonics are of the order of $\mathscr{D}/a^2$. So, in the limit $\omega\ll\mathscr{D}/a^2$ they can be neglected because their contribution is of the order of

$$\frac{a^2}{\nu V\mathscr{D}}\sim\frac{1}{p_0^2al},\quad\text{where } l=v_0\tau.$$

The free energy functional F was derived in the limit $\omega\tau\ll 1$. So the general restriction for the frequency can be written as

$$\omega\ll\min\{\mathscr{D}/a^2,\tau^{-1}\}. \tag{5.19}$$

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The relation between $\mathscr{D}/a^2$ and τ^{-1} can be arbitrary depending on the relation between the mean free path l and the size of the sample a. These restrictions are valid when all sizes of the sample are of the same order. However, in certain cases the non-zero harmonics cannot be neglected even at zero frequency. Let us consider a long metallic cylinder (wire) of length L and cross-section S. The zero-frequency contribution of the non-zero harmonics is of order $L/\nu S\mathscr{D} \sim L/L_c$, where $L_c \sim \nu S\mathscr{D}$.

The non-zero harmonics contribution is small in the limit $L \ll L_c$ only. In the opposite limiting case the approximation which is used in this section is not applicable. In § 6 we will show that L_c is the localization length of the wire. So we can neglect the non-zero harmonics only in the case when electrons can diffuse throughout the sample. For systems where electrons are localized the level theory presented here is not applicable. Equation (4.15) helps us to estimate the parameters of the crossover from Model I to Model II or Model III. In the absence of the magnetic field the contribution of all the corrections in eqns. (4.15) is small provided

$$\Delta \ll \tau_s^{-1}. \tag{5.20}$$

The inequality (5.20) corresponds to Model IIb. Model III is applicable provided the following inequality is fulfilled

$$\tau_s^{-1} \ll \Delta \ll \tau_{so}^{-1}. \tag{5.21}$$

It is clear that Model I is applicable when

$$\tau_s^{-1}, \tau_{so}^{-1} \ll \Delta. \tag{5.22}$$

In weak magnetic fields the energy $E_\perp$ in eqn. (4.15) is of the order of $\mathscr{D}(eH/c)^2a^2$. The characteristic field corresponding to the crossover from Model I to Model IIb is determined as

$$\phi_0 = Ha(\mathscr{D}/\Delta)^{1/2}, \tag{5.23}$$

where $\phi_0 = \pi c/e$ is the flux quantum. It is not difficult to estimate parameters corresponding to all other crossovers. Thus, in many real situations the description in terms of the zero-dimensional σ-model with the three types of symmetry is applicable. It enables us to write the correlation function $R(\omega)$ in the form of a definite integral over the supermatrix

$$\left.\begin{aligned} R(\omega) &= \frac{1}{2} + \frac{1}{128}\,\mathrm{Re}\int \mathrm{STr}(k(1+\Lambda)Q)\,\mathrm{STr}\,(k(1-\Lambda)Q)\exp(-F_0[Q])dQ, \\ F_0 &= +\frac{i\pi}{\Delta}(\omega + i\delta)\,\mathrm{STr}\,(\Lambda Q). \end{aligned}\right\} \tag{5.24}$$

Equation (5.24) is valid for all three models. The mean level-spacing is determined by eqns. (5.14) and (5.15). The symmetry of the supermatrix Q is specified by eqns. (3.57), (3.44), (3.41), (3.75) and (3.78). Thus, the problem of the level-spacing distribution has been reduced to the definite integral over the supermatrix. The integral (5.24) contains many variables and calculation of this integral is not simple. However, variables which make it possible to calculate this integral exist. In the next section it will be shown that the integral (5.24) can be reduced to a definite integral over two (for Models II) or three (for Models I and III) variables.

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5.3. *Reduction to integrals over eigenvalues*

In § 5.1 it has been demonstrated that in certain cases an integral over a matrix can be considerably simplified provided eigenvalues are chosen as the variables of integration. An analogous trick can be used when calculating an integral over the supermatrix Q. The superelement Q^{12} with the symmetry specified by eqns. (3.44), (3.41), (3.75) and (3.78) can be written as

$$Q^{12} = iu \sin\theta \bar{v}, \tag{5.25}$$

where

$$\theta = \begin{pmatrix} \theta_{11} & 0 \\ 0 & \theta_{22} \end{pmatrix}.$$

The elements θ_{11} and θ_{22} are equal to

$$\theta_{11} = \begin{pmatrix} \theta & 0 \\ 0 & \theta \end{pmatrix}, \qquad \theta_{22} = i\begin{pmatrix} \theta_1 & \theta_2 \\ \theta_2 & \theta_1 \end{pmatrix}, \quad 0<\theta<\pi,\ \theta_1>0,\ \theta_2>0, \qquad \text{(Model I)}$$

$$\theta_{11} = \begin{pmatrix} \theta & 0 \\ 0 & \theta \end{pmatrix}, \qquad \theta_{22} = i\begin{pmatrix} \theta_1 & 0 \\ 0 & \theta_1 \end{pmatrix}, \quad 0<\theta<\pi,\ \theta_1>0, \qquad \text{(Model II)}$$

$$\theta_{11} = \begin{pmatrix} \theta_1 & \theta_2 \\ \theta_2 & \theta_1 \end{pmatrix}, \qquad \theta_{22} = i\begin{pmatrix} \theta & 0 \\ 0 & \theta \end{pmatrix}, \quad \theta>0,\ 0<\theta_1<\pi,\ 0<\theta_2<\pi. \qquad \text{(Model III)}$$

The supermatrices u and v in eqn. (5.25) satisfy the following conditions

$$\bar{u}u = 1, \qquad \bar{v}v = 1, \qquad \bar{v} = kv^{+}k, \qquad \bar{u} = u^{+}. \tag{5.26}$$

Let us recall that the charge conjugation in Model II differs from that in Models I and II by the replacement $c_1 \rightleftarrows c_2$ in equation (3.20).

Using eqn. (5.25) the supermatrix Q can be written as

$$Q = U\begin{pmatrix} \cos\hat{\theta} & i\sin\hat{\theta} \\ -i\sin\hat{\theta} & -\cos\hat{\theta} \end{pmatrix}\bar{U}, \qquad U = \begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}. \tag{5.27}$$

One can check that the conditions (5.26) provide the fulfilment of eqns. (3.49) and (3.58). It is very convenient to reduce the integral (5.24) to an integral over the 'eigenvalues' θ and the supermatrices u and v (strictly speaking the variables θ are not the eigenvalues of the supermatrix Q; they look rather like the eigenvalues of the superelement Q^{12}). The use of the variables θ, u, v leads to the appearance of the Berezinian of the transformation. This Berezinian, just as the Jacobian for a conventional matrix, can be written provided the appropriate quadratic form is known (cf (2.63)).

Using eqn. (5.27) we rewrite the quadratic form as

$$\mathrm{STr}\,(dQ)^2 = 2\,\mathrm{STr}\,(d\theta)^2 + 2\,\mathrm{STr}\,(2\delta u \sin\hat{\theta}\,\delta v \sin\hat{\theta} + (\delta u\cos\hat{\theta})^2 + (\delta v\cos\hat{\theta})^2) - 2\mathrm{STr}\,(\delta u)^2 - 2\,\mathrm{STr}\,(\delta v)^2, \tag{5.28}$$

where $\delta u = \bar{u}du = -\delta\bar{u}$, $\delta v = \bar{v}dv = -\delta\bar{v}$. Let us rewrite the expression (5.28) separating in an explicit form the anticommuting Grassmann variables

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$\delta u_{12}, \delta u_{21}, \delta v_{12}, \delta v_{21}$ from the conventional commuting ones $\delta u_{11}, \delta u_{22}, \delta v_{11}, \delta v_{22}$.

$$\begin{aligned}\mathrm{STr}\,(dQ)^2 = 2\,\mathrm{STr}\,(d\theta)^2 + 2\,\mathrm{Tr}\,[&-(\delta u_{11})^2 + (\delta u_{11}\cos\hat{\theta}_{11})^2 - (\delta v_{11})^2\\ &+(\delta v_{11}\cos\hat{\theta}_{11})^2 + 2\delta u_{11}\sin\hat{\theta}_{11}\,\delta v_{11}\sin\hat{\theta}_{11}\\ &+(\delta u_{22})^2 - (\delta u_{22}\cos\theta_{22})^2\\ &+(\delta v_{22})^2 - (\delta v_{22}\cos\hat{\theta}_{22})^2 - 2\delta u_{22}\sin\hat{\theta}_{22}\delta v_{22}\sin\theta_{22}]\\ &+4\,\mathrm{Tr}\,[2\delta u_{12}\sin\hat{\theta}_{22}\,\delta v_{21}\sin\hat{\theta}_{11} - \delta u_{12}\delta u_{21}\\ &+\delta u_{12}\cos\hat{\theta}_{22}\,\delta u_{21}\cos\hat{\theta}_{11} - \delta v_{12}\delta v_{21}\\ &+\delta v_{12}\cos\hat{\theta}_{22}\,\delta v_{21}\cos\hat{\theta}_{11}]. \end{aligned} \tag{5.29}$$

The quadratic form (5·29) consists of two parts. One of them contains only the commuting elements of the matrices δu, δv, $d\hat{\theta}$, the other contains only anticommuting ones. Therefore, the Berezinian is equal to the ratio of the Jacobians corresponding to these two parts. For further calculations it is necessary to write the matrices u and v in an explicit form. Let us represent these matrices as

$$\left.\begin{aligned} &u = u_1 u_2, \quad v = v_1 v_2, \quad \bar{u}_1 u_1 = \bar{u}_2 u_2 = \bar{v}_1 v_1 = \bar{v}_2 v_2 = 1,\\ &u_1 = \begin{pmatrix} 1-2\eta\bar{\eta}+6(\eta\bar{\eta})^2 & 2\eta(1-2\bar{\eta}\eta) \\ -2(1-2\bar{\eta}\eta)\bar{\eta} & 1-2\bar{\eta}\eta+6(\bar{\eta}\eta)^2 \end{pmatrix}, \quad u_2 = \begin{pmatrix} F_1 & 0 \\ 0 & F_2 \end{pmatrix},\\ &v_1 = \begin{pmatrix} 1+2\kappa\bar{\kappa}+6(\kappa\bar{\kappa})^2 & 2i\kappa(1+2\bar{\kappa}\kappa) \\ -2i(1+2\bar{\kappa}\kappa)\bar{\kappa} & 1+2\bar{\kappa}\kappa+6(\bar{\kappa}\kappa)^2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} \Phi_1 & 0 \\ 0 & \Phi_2 \end{pmatrix}. \end{aligned}\right\} \tag{5.30}$$

In eqn. (5.30), the symmetry of the matrices η, κ, F_1, F_2, Φ_1 and Φ_2 depends on the models under consideration. The matrices η for three models are

$$\left.\begin{aligned} &\eta = \begin{pmatrix} \eta_1 & \eta_2 \\ -\eta_2^* & -\eta_1^* \end{pmatrix}, && \text{(Model I)}\\ &\eta = \begin{pmatrix} \eta_1 & 0 \\ 0 & -\eta_1^* \end{pmatrix}, && \text{(Model II)}\\ &\eta = \begin{pmatrix} \eta_1 & \eta_2 \\ -\eta_2^* & \eta_1^* \end{pmatrix}. && \text{(Model III)} \end{aligned}\right\} \tag{5.31}$$

The matrices $\bar{\eta}$ coincide with the hermitian conjugates η^+. (One should remember that $(\eta_i^*)^* = -\eta_i$.) The matrices κ have the same structure as the matrices η.

Due to the diagonal form of the matrices η and κ in Model II, only terms which are linear and quadratic with respect to η and κ are not zero in the matrices u_1 and v_1 (5.30). Of course, in all cases, $\bar{u}_1 u_1 = \bar{v}_1 v_1 = 1$. The representation of the matrices u_1 and v_1 in the form (5.30) is not occasional. The coefficients in (5.30) provide the fulfilment of the equations

$$\int \prod_{i,k} (\delta v_{12})_{ik} = \int \prod_{i,k} (\delta u_{12})_{ik} = 0.$$

If these equations the products of all independent elements of the matrices δu and δv are written down. The fact that these integrals turn to zero allows us to consider the

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elements $(\delta v_{12})_{ik}$ and $(\delta u_{12})_{ik}$ as Grassmann differentials. The matrices u_2 and v_2 can be parametrized as

$$\left.\begin{aligned}
&F_1=(1-iM)(1+iM)^{-1},\\
&F_2=\exp(i\phi\tau_3),\quad \Phi_1=1,\quad \Phi_2=\exp(i\chi\tau_3), && \text{(Model I)}\\
&F_1=\exp(i\phi\tau_3), F_2=\exp(i\chi\tau_3),\quad \Phi_1=\Phi_2=1, && \text{(Model II)}\\
&F_1=\exp(i\phi\tau_3),\\
&F_2=(1-iM)(1+iM)^{-1},\quad \Phi_1=\exp(i\chi\tau_3),\quad \Phi_2=1 && \text{(Model III)}
\end{aligned}\right\} \quad (5.32)$$

In the formulae (5.32) the matrices τ_3 and M are equal to

$$\tau_3=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix},\quad M=\begin{pmatrix}m & m_1^*\\ m_1 & -m\end{pmatrix}, \quad (5.33)$$

m, ϕ and χ are real numbers.

Substituting eqns. (5.30), (5.31) and (5.32) into eqn. (5.29) one can rewrite the quadratic form through the independent variables η, κ, M, ϕ and χ. The anticommuting variables η and κ are separated from the commuting ones. Therefore, the elementary volume $[dQ]$ can be written in the form of the product

$$[dQ]=J_1J_2dR_1dR_1dR_2d\theta,$$

where J_1dR_1 is the elementary volume in the space of anticommuting variables, J_2dR_2 is the elementary volume in the space of anticommuting ones. Writing Jacobians according to the quadratic forms for the commuting and anticommuting elements we find

$$\left.\begin{aligned}
&J_1^{(\mathrm{I})}=\frac{1}{2^{20}}(\cosh(\theta_1+\theta_2)-\cos\theta)^{-2}(\cosh(\theta_1-\theta_2)-\cos\theta)^{-2},\\
&dR_1^{(\mathrm{I})}=d\eta_1d\eta_1^*d\eta_2d\eta_2^*d\kappa_1d\kappa_1^*d\kappa_2d\kappa_2^*,\\
&J_2^{(\mathrm{I})}=\frac{2^{12}}{\pi^4}\frac{|\sin\theta|^3|\sinh\theta_1||\sinh\theta_2|}{(1+m^2+|m_1|^2)^3},\\
&dR_2^{(\mathrm{I})}=dmdm_1dm_1^*d\phi d\chi,\quad d\theta^{(\mathrm{I})}=d\theta d\theta_1d\theta_2,\\
&J_1^{(\mathrm{II})}=\frac{1}{2^8}(\cosh\theta_1-\cos\theta)^{-2},\qquad dR_1^{(\mathrm{II})}=d\eta_1d\eta_1^*d\kappa_1d\kappa_1^*,\\
&J_2^{(\mathrm{II})}=\frac{2}{\pi^2}|\sin\theta||\sinh\theta_1|,\qquad dR_2^{(\mathrm{II})}=d\phi d\chi,\quad d\theta^{(\mathrm{II})}=d\theta d\theta_1,\\
&J_1^{(\mathrm{III})}=\frac{1}{2^{20}}(\cosh\theta-\cos(\theta_1+\theta_2))^{-2}(\cosh\theta-\cos(\theta_1-\theta_2))^{-2},\\
&dR_1^{(\mathrm{III})}=d\eta_1d\eta_1^*d\eta_2d\eta_2^*d\kappa_1d\kappa_1^*d\kappa_2d\kappa_2^*,\\
&J_2^{(\mathrm{III})}=\frac{2^{12}}{\pi^4}\frac{|\sinh\theta|^3|\sin\theta_1||\sin\theta_2|}{(1+m^2+|m_1|^2)^3},\\
&dR_2^{(\mathrm{III})}=dmdm_1dm_1^*d\phi d\chi,\quad d\theta^{(\mathrm{III})}=d\theta d\theta_1d\theta_2.
\end{aligned}\right\} \quad (5.34)$$

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Substituting eqns. (5.27 and (5.30)–(5.34) and integrating over u and v we reduce eqns. (5.24) for the correlation functions to the form

$$\left.\begin{aligned} R^{(I)}(x) &= 1+\mathrm{Re}\int_{-1}^{1}\int_{1}^{\infty}\int_{1}^{\infty}\frac{(\lambda_1\lambda_2-\lambda)^2(1-\lambda^2)\exp\left[i(x-i\delta)(\lambda-\lambda_1\lambda_2)\right]d\lambda d\lambda_1 d\lambda_2}{(\lambda_1^2+\lambda_2^2+\lambda^2-2\lambda\lambda_1\lambda_2-1)^2}, \\ R^{(II)}(x) &= 1+\frac{1}{2}\mathrm{Re}\int_{-1}^{1}\int_{1}^{\infty}\exp\left[i(x-i\delta)(\lambda-\lambda_1)\right]d\lambda d\lambda_1, \\ R^{(III)}(x) &= 1+\mathrm{Re}\int_{1}^{\infty}\int_{-1}^{1}\int_{0}^{1}\frac{(\lambda-\lambda_1\lambda_2)^2(\lambda^2-1)\exp\left[i(x-i\delta)(\lambda_1\lambda_2-\lambda)\right]d\lambda d\lambda_1 d\lambda_2}{(\lambda^2+\lambda_1^2+\lambda_2^2-2\lambda_1\lambda_2\lambda-1)^2}, \end{aligned}\right\} \quad (5.35)$$

where $x=\pi\omega/\Delta$.

When deriving eqn. (5.35) the following changes of variables were used.

$$\left.\begin{aligned} &\lambda=\cos\theta, \quad \lambda_1=\cosh\theta_1, \quad \lambda_2=\cosh\theta_2, && \text{(Model I)} \\ &\lambda=\cos\theta, \quad \lambda_1=\cosh\theta_1, && \text{(Model II)} \\ &\lambda=\cosh\theta, \quad \lambda_1=\cos\theta_1, \quad \lambda_2=\cos\theta_2, && \text{(Model III)} \end{aligned}\right\} \quad (5.35\,a)$$

In order to take the limit $x\to 0$ in eqn. (5.35) one must first calculate the integral for non-zero x and only after that take the limit. Taking the limit $x\to 0$ directly in the integrand leads to divergent integrals.

The study carried out in this section shows that the integration over the group of the supermatrix Q can be reduced to integrals over the eigenvalues. Such a reduction is possible because the energy F_0 (5.24) depends only on these eigenvalues.

5.4. *The two-level correlation function and response to electromagnetic fields*

Integrals (5.35) completely solve the problem of calculation of the two-level correlation function (3.11) in all three cases. It turns out that the integrals (5.35) can be calculated analytically in all three models. This assertion is evident for Model II but is not trivial for cases I and III. We show how the integral for the Model I can be calculated. (The integral for Model III is calculated analogously)

$$I(x)=\int_{-1}^{1}\int_{1}^{\infty}\int_{1}^{\infty}\frac{(\lambda_1\lambda_2-\lambda)^2(1-\lambda^2)\exp\left[i(x-i\delta)(\lambda-\lambda_1\lambda_2)\right]}{(\lambda_1^2+\lambda_2^2+\lambda^2-2\lambda\lambda_1\lambda_2-1)^2}d\lambda d\lambda_1 d\lambda_2. \quad (5.36)$$

Making the Fourier transformation

$$\tilde{I}(t)=\int_{-\infty}^{\infty}I(x)\exp(ixt)dx$$

we obtain

$$\tilde{I}(t)=2\pi\int_{-1}^{1}\int_{1}^{\infty}\int_{1}^{\infty}\frac{(\lambda_1\lambda_2-\lambda)^2(1-\lambda^2)\delta(\lambda-\lambda_1\lambda_2+t)}{(\lambda_1^2+\lambda_2^2+\lambda^2-2\lambda\lambda_1\lambda_2-1)^2}d\lambda d\lambda_1 d\lambda_2. \quad (5.37)$$

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Using the variables $u=\lambda_1\lambda_2$, $z=\lambda_1^2$ we reduce the integral $\tilde{I}(t)$ (5.37) to the form

$$\tilde{I}(t)=\pi t^2\int_1^\infty\int_1^{u^2}\frac{(1-(u-t)^2)\theta(u-t+1)\theta(1-u+t)z\,dz\,du}{(z^2+z(t^2-u^2-1)+u^2)^2}, \tag{5.38}$$

where

$$\theta(\tau)=\begin{cases}1, & \tau>0,\\ 0, & \tau<0.\end{cases}$$

The integrand in eqn. (5.38) is a rational function of z. It enables us to integrate over z. We have as the result

$$\left.\begin{aligned}
&\tilde{I}(t)=\tilde{I}_1(t)+\tilde{I}_2(t),\\
&\tilde{I}_1(t)=2\pi\int_1^\infty\frac{\theta(u-t+1)\theta(1-u+t)(u^2-1)}{(u+t+1)(u+t-1)}\,du,\\
&\tilde{I}_2(t)=2\pi t^2\int_1^\infty\frac{\theta(u-t+1)\theta(1-u+t)(t^2-u^2-1)}{(1+u-t)^{1/2}(1+u+t)^{3/2}(1-u+t)^{1/2}(u+t-1)^{3/2}}\\
&\quad\times\left(\tan^{-1}\frac{t^2-u^2-1}{(4u^2-(t^2-u^2-1)^2)^{1/2}}-\tan^{-1}\frac{u^2+t^2-1}{(4u^2-(t^2-u^2-1)^2)12}\right)du.
\end{aligned}\right\} \tag{5.39}$$

Further calculation for the integral $\tilde{I}_1(t)$ is rather simple. Integration for $\tilde{I}_2(t)$, in spite of the complicated integrand, can also be performed analytically. Let us use the identity

$$\frac{d}{du}\left[\frac{(1+u-t)^{1/2}(1-u+t)^{1/2}}{(1+u+t)^{1/2}(t+u-1)^{1/2}}\right]=\frac{2t(t^2-u^2-1)}{(1+u-t)^{1/2}(1+u+t)^{3/2}(1-u+t)^{1/2}(u+t-1)^{3/2}}.$$

This identity makes it possible to integrate by parts in the integral $\tilde{I}_2(t)$ (5.39). Performing this procedure we find

$$\tilde{I}_2(t)=\pi t\int_1^\infty\frac{\theta(u-t+1)\theta(1-u+t)(3u^2-t^2+1)}{(1+u+t)(u+t-1)u}\,du. \tag{5.40}$$

Integrating the rational functions in the integrals $\tilde{I}_1(t)$ (5.39) and $\tilde{I}_2(t)$ (5.40) we obtain

$$\tilde{I}(t)=t\begin{cases}\dfrac{4}{t}-\ln\dfrac{t+1}{t-1}, & t>2,\\ 2-\ln(t+1), & 0<t<2,\\ 0, & t<0.\end{cases} \tag{5.41}$$

Finally, performing the inverse Fourier-transformation, we have

$$R^{(\mathrm{I})}(x)=R_{\mathrm{ort}}(x), \tag{5.42}$$

where $R_{\mathrm{ort}}(x)$ is the correlation function for the Dyson orthogonal ensemble (5.10). Integrating in formulae for $R^{(\mathrm{II})}$ and $R^{(\mathrm{III})}$ we find

$$R^{(\mathrm{II})}(x)=R_{\mathrm{unit}}(x), \tag{5.43}$$

$$R^{(\mathrm{III})}(x)=R_{\mathrm{sympl}}(x). \tag{5.44}$$

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Thus, the two-level correlation functions for Models I, II and III exactly coincide with those for the Dyson orthogonal, unitary and symplectic ensembles. This coincidence is seen in the final result only. Nothing common arises in intermediate steps of the calculations. In the Dyson theory the formulae (5.10)–(5.12) are obtained after calculation of complicated determinants. In the theory developed above these formulae are obtained as a result of calculation of the integrals (5.35). Nevertheless, this coincidence cannot be considered as accidental. Unfortunately a confirmation of the Dyson hypotheses at earlier steps of the calculation is absent as yet.

In the limits $x \to 0$ and $x \to \infty$ the functions $R(x)$ (5.10)–(5.12) have quite different asymptotes.

$$\left.\begin{aligned} R_{\text{ort}}(x) &\approx \frac{\pi}{6}x, \quad x \to 0; \quad 1 - \frac{1}{x^2} + \frac{1}{x^4}(1+\cos^2 x), \quad x \to \infty, \\ R_{\text{unit}}(x) &\approx \frac{x^2}{3}, \quad x \to 0; \quad 1 - \frac{\sin^2 x}{x^2}, \quad x \to \infty, \\ R_{\text{sympl}}(x) &\approx \frac{x^4}{135}, \quad x \to 0; \quad 1 + \frac{\pi}{2}\frac{\cos x}{x}, \quad x \to \infty. \end{aligned}\right\} \tag{5.45}$$

Distant correlations are strongest in the symplectic ensemble and weakest in the orthogonal one. Equations (5.45) demonstrate the existence of level repulsion ($R(x) \to 0$ as $x \to 0$). This effect is again strongest in the symplectic ensemble.

The study carried out above shows that a class of systems exists which can be described by the Dyson statistical theory. Nevertheless, the Dyson statistics are not universal. It is known that in a one-dimensional chain of a length $L > l$, where l is the mean free path, the level spacing distribution is not described by Dyson's formulae (5.10) –(5.12) (Molchanov, 1979, Malkin 1981, Gor'kov *et al.* 1982). In particular, level repulsion is absent. Apparently, the Dyson statistics are also not applicable for strongly disordered metals $\tau\varepsilon_0 \gtrsim 1$. In all these cases the theory developed above is not valid. It seems to be reasonable to assume that the Dyson statistics will be valid when a particle can diffuse throughout the sample. In strongly disordered metals and one-dimensional chains the particle is localized and does not feel boundaries. Therefore at a sufficiently large distance from one localization centre to another, a level with practically the same energy can be found. This leads to the disappearance of level repulsion. The disorder above was supposed to be due to impurities inside the sample. It would appear that the results remain valid for pure samples with a random shape of surface.

Though the two-level correlation function is rather complicated, the density of states in the model involved is equal to ν, because the diffusion modes do not contribute to this quantity. Let us emphasize that eqns. (5.10)–(5.12) cannot be obtained by the summation of perturbation-theory graphs. These graphs yield a series in x^{-1}, where $x = \pi\omega/\Delta$. Oscillating terms of the type $\cos 2x/x^{2n}$ arising in eqns. (5.10)–(5.12) cannot be obtained within perturbation theory because they are non-analytical in x^{-1}. An application of the σ-models obtained by the replica method (Schäfer and Wegner 1980, Efetov *et al.* 1980) to the study of the problem also leads to insuperable mathematical difficulties because calculation of the encountered integral over the $n \times n$ matrix Q obeying the constraint $Q^2 = 1$ is very difficult. So, the supersymmetry method is now the only one which makes it possible to solve the energy-level statistics problem.

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As has been mentioned above, the two-level correlation function determines such quantities as the response to an electromagnetic field, magnetic susceptibility, etc. Using Dyson's results for the orthogonal, unitary and symplectic ensembles, Gor'kov and Eliashberg (1965) calculated these quantities. These authors considered a metallic granule of size a in a weak electric field $eEa \ll \Delta$. For their calculations they used the Kubo formula

$$\langle d_\alpha \rangle = i \int \langle\langle [d_\alpha(t), d_\beta(t')] \rangle\rangle E_\beta(t') dt', \tag{5.46}$$

where d_α is the α-component of the dipole momentum, $[\ldots,\ldots]$ is the commutator. The angular brackets $\langle\langle \ldots \rangle\rangle$ stand for both thermodynamic and impurity averaging. Calculating the thermodynamic average we find

$$d_{\alpha\omega} = \frac{e^2 E_{\beta\omega}}{2} \sum_{k,l} \left\langle \frac{\{\tanh[(\varepsilon_k - \varepsilon_0)2T] - \tanh[(\varepsilon_l - \varepsilon_0)/2T]\}}{\varepsilon_k - \varepsilon_l + \omega + i\delta} r^\alpha_{kl} r^\beta_{lk} \right\rangle, \tag{5.47}$$

where $d_{\alpha\omega}$ is the Fourier-component of the dipole momentum.

In eqn. (5.47) r^α_{kl} is the matrix element of the co-ordinate over the states k and l. The summation is performed over all the states of the system. The angular brackets in eqn. (5.47) stand for averaging over the impurities only. The formula (5.47) enables us to obtain the electric susceptibility

$$\chi_\omega = \chi_0 - \frac{e^2\omega}{6V} \sum_{k,l}^{\chi\omega} \left\langle \frac{\tanh[(\varepsilon_k - \varepsilon_0)/2T] - \tanh[(\varepsilon_l - \varepsilon_0)/2T]}{\varepsilon_k - \varepsilon_l} \frac{r_{kl} r_{lk}}{(\varepsilon_k - \varepsilon_l)^2 - (\omega + i\delta)^2} \right\rangle, \tag{5.48}$$

where χ_0 is the static susceptibility which is equal to

$$\chi_0 = \frac{e^2}{6V} \sum_{k,l} \left\langle \frac{r_{kl} r_{lk} (\tanh[(\varepsilon_k - \varepsilon_0)/2T] - \tanh[(\varepsilon_l - \varepsilon_0)/2T])}{\varepsilon_k - \varepsilon_l} \right\rangle = \frac{e^2 a^2 m p_0}{5\pi^2}.$$

V is the volume of the granule.

In the limit $\omega \ll v/a$ the averaging of the matrix elements $r_{kl} r_{lk}$ can be performed independently of every other. Recalling the definition of the correlation function $R(\varepsilon)$ (3.11), and calculating the average of the product of the matrix elements $r_{kl} r_{lk}$ in the same way as Gor'kov and Eliashberg, we find

$$\chi_\omega = \chi_0 - \frac{1}{3} e^2 \beta \nu A, \quad A = \frac{\omega^2}{\Delta} \int_0^\infty \frac{R(\varepsilon)\, d\varepsilon}{\varepsilon^2 - (\omega + i\delta)^2}. \tag{5.49}$$

The coefficient β (5.49) depends on the relation between the mean free path and the size of the sample a. In limiting cases this coefficient is

$$\beta = \begin{cases} \dfrac{139}{200} \dfrac{1}{2\pi\nu} \dfrac{1}{v}, & l \gg a, \\ \dfrac{12}{25} \dfrac{1}{2\pi\nu} \dfrac{a}{\mathscr{D}}, & l \ll a. \end{cases}$$

Equation (5.49) describes the relation between the susceptibility in the complex domain of frequencies of the metallic particle and the two-level correlation function.

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Calculation of the integral (5.49) can be done easily. Separating the real part A, from the imaginary, we find (Gor'kov and Eliashberg 1965, Devaty and Sievers 1980):

for the orthogonal ensemble:

$$A_1^{\text{orth}} = 1 - \frac{\sin x}{2x} + xCi(x)\frac{d}{dx}\left(\frac{\sin x}{x}\right),$$

$$A_2^{\text{orth}} = x - \frac{\sin^2 x}{x} + x\left(Si(x) - \frac{\pi}{2}\right)\frac{d}{dx}\left(\frac{\sin x}{x}\right);$$

for the unitary ensemble:

$$A_1^{\text{unit}} = 1 - \frac{\sin 2x}{2x}, \quad A_2^{\text{unit}} = x - \frac{\sin^2 x}{x};$$

for the symplectic ensemble:

$$A_1^{\text{sympl}} = 1 - \frac{\sin^2 x}{2x} + xSi(x)\frac{d}{dx}\left(\frac{\cos x}{x}\right),$$

$$A_2^{\text{sympl}} = x - \frac{\sin^2 x}{x} + xSi(x)\frac{d}{dx}\left(\frac{\sin x}{x}\right);$$

$$Si(x) = \int_0^x \frac{\sin t}{t}\,dt, \quad Ci(x) = -\int_x^\infty \frac{\cos t\,dt}{t}. \tag{5.50}$$

The variable x is defined by eqns. (5.35). Both the real and imaginary parts are oscillating functions of the frequency. The amplitude of the oscillations decreases in a power law as the frequency increases.

The polarizability α of a metallic particle is connected with the susceptibility by a simple equation (Rice *et al.* 1973, Perenboom *et al.* 1981)

$$\alpha = \frac{V\chi}{1 + L\chi}, \tag{5.51}$$

where L is the depolarization factor.

Equations (5.50) and (5.51) determine the real and imaginary parts of the dielectric permeability of a granular material provided all the granules have equal volumes. Thus, the problem of kinetics of small metallic particles has been completely solved in this chapter.

§6. Localization in wires

6.1. *One-dimensional σ-model and wires*

Recently Thouless (1977) predicted electron localization in long metallic wires. An analogous assertion has been made by Abrahams *et al.* (1979). The arguments involved were based on the scaling hypothesis. It was not important whether a chain or a thick wire was considered. At the same time, from the formal point of view, these models are quite different. For the one-dimensional chain very-well-developed methods exist (Mott and Twose 1961, Berezinsky 1973, Abrikosov and Ryzhkin

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1976, Berezinsky and Gor'kov 1979) which allow one to calculate the frequency-dependence of the conductivity.

On the basis of the Berezinsky method, Weller *et al.* (1982) considered N coupled chains and showed that at low frequencies the equations describing the problem can be reduced to those of Berezinsky. Therefore the conductivity turns to zero as $\omega \to 0$. This model was also considered by Dorokhov (1982) who calculated the dependence of the localization length on the number of chains. The N-conducting-channels model was considered by Anderson *et al.* (1980) using the Landauer method. These authors also made the assertion about localization. In principle, the models used in these works are valid for the description of wires. However, the number of the chains or channels cannot be large (or alternatively, the mean free path must be large), otherwise one cannot begin with the description in terms of separate chains. It corresponds to sufficiently thin or sufficiently pure wires.

In the following the opposite limit of a sufficiently thick wire is considered. The following condition is supposed to be fulfilled:

$$\tau \ll Sm, \tag{6.1}$$

where S is the cross-section and m is the electron mass.

The inequality (6.1) corresponds to the case when the inverse scattering time far exceeds the transverse-quantization level spacing. In this limit classical diffusion exists at small distances. The system is three dimensional almost in all respects. Only the geometry of the sample is one dimensional. It is supposed as before that the mean free path is much larger than atomic lengths. The inequality (6.1) allows us to derive the supermatrix σ-model ((3.62), (4.16)). An analogous condition was necessary for the derivation of the σ-model for a restricted volume. Provided the frequency ω is low enough, $\omega \ll \min\{\mathscr{D}/S, \tau^{-1}\}$, all non-zero transverse harmonics can be neglected. At high frequencies corrections to the diffusion become three dimensional and their contribution is small. Therefore, the one-dimensional σ-model is adequate and is considered in this section.

For studying one-dimensional models the well-developed transfer-matrix method exists (Feynmann 1948, Vaks *et al.* 1966) which enables one to reduce the calculation of a functional integral to the solution of a Schrödinger equation. The supermatrix σ-model ((3.62), (4.16)) is rather complicated because it contains a large number of both commuting and anticommuting variables. Before deriving the Schrödinger equation for the σ-model it is useful to consider a simple problem of calculating the average Green function and the density of states for a one-dimensional disordered chain. This problem can be studied with the use of integrals over supervectors. Such a consideration will clarify the principal calculation scheme for the one-dimensional σ-model.

6.2. *Derivation of the Green function for a disordered chain by integrating over commuting and anticommuting variables*

In this section the averaged Green function and the density of states for a disordered chain are calculated (Bohr and Efetov 1982). The exact solution of this model has long been known (Frish and Lloid 1960, Halperin 1965, Thouless 1975); the treatment in this section is presented because it throws some light on methodology.

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The one-particle Green function has been written down in the form of an integral over supervectors (3.15). Averaging over the random gaussian potential (3.24) we obtain

$$\langle G_\varepsilon(x_1, x_2)\rangle = i\int \mathscr{D}\Phi^*\mathscr{D}\Phi\Phi_\alpha(x_1)\Phi^*_\alpha(x_2)\exp\left[-\int_0^L \mathscr{L}dx\right], \tag{6.2}$$

where

$$\left.\begin{aligned} \mathscr{L} &= a\Phi^+(\varepsilon - H_0 - i\delta)\Phi + \frac{\gamma}{2}(\Phi^+\Phi)^2, \\ H_0 &= -\frac{1}{2m}\frac{d^2}{dx^2}, \quad a = i, \quad \gamma = \frac{1}{2\pi\nu\tau}. \end{aligned}\right\} \tag{6.3}$$

L is the length of the sample; α in eqn. (6.2) can take values 1 and 2.

Further evaluation of the functional integral (6.2) can be made by the transfer-matrix technique. The inclusion of anticommuting variables demands no essential changes. Following the method of Feynmann (1948) we obtain for $x_1 < x_2$

$$\langle G_\varepsilon(x_1, x_2)\rangle = i\int \Gamma(0, x_1; R_0, R_1)\chi_1\Gamma(x_1, x_2; R_1, R_2) \times \chi_2^*\Gamma(x_2, L; R_2, R_0)dR_0dR_1dR_2, \tag{6.4}$$

where we have defined

$$R \equiv \{\chi, \chi^*, S, S^*\} \quad \text{and} \quad dR = \frac{1}{\pi}d\chi^*d\chi dS^*dS.$$

The kernel Γ appearing in eqn. (6.4) satisfies the Schrödinger equation

$$\left(-\frac{\partial}{\partial x} + \mathscr{H}(R)\right)\Gamma(x, x'; R, R') = \delta(x - x')\delta(R - R'), \tag{6.5}$$

where

$$\mathscr{H} = -\frac{2m}{a}\left(\frac{\partial^2}{\partial S^*\partial S} + \frac{\partial^2}{\partial\chi^*\partial\chi}\right) + a\varepsilon(|S|^2 + |\chi|^2) + \frac{\gamma}{2}(|S|^2 + |\chi|^2)^2, \quad |\chi|^2 \equiv \chi^*\chi.$$

The δ-function in eqn. (6.5) is of the form

$$\delta(R - R') = -\delta(S_1 - S_1')\delta(S_2 - S_2')|\chi - \chi'|^2, \tag{6.6}$$

where $S_1 = \mathrm{Re}\, S$ and $S_2 = \mathrm{Im}\, S$.

Equation (6.6) appears after taking the limit $\alpha \to 0$ of the expression

$$\frac{1}{\pi}\exp\left\{-\frac{|S - S'|^2}{\alpha} - \frac{|\chi - \chi'|^2}{\alpha}\right\}.$$

The δ-function defined by eqn. (6.6) has the properties of normal δ-functions. For example

$$\int f(R')\delta(R - R')dR' = f(R)$$

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for any function f. The kernel Γ satisfying eqn. (6.5) can be expanded over orthonormal eigenfunctions Ψ_n of the Schrödinger equation

$$\mathscr{H}\Psi_n = E_n\Psi_n \tag{6.7}$$

as

$$\Gamma(x, x'; R, R') = \theta(x'-x)\sum_n \Psi_n(R)\Psi_n^*(R')\exp(-E_n(x'-x)). \tag{6.8}$$

Since Ψ can contain terms linear in χ or χ^* the order of Ψ and Ψ^* is important and for evaluation of Ψ^* eqn. (2.5) must be kept in mind. By normalization we mean

$$\int |\Psi_n|^2 dR = \int \Psi_n^*\Psi_n\, dR = 1. \tag{6.9}$$

The expansion (6.8) is valid for hermitian $\mathscr{H}$ corresponding to a being real. We shall assume that a is real and positive for the present and only in the end make the analytical continuation $a \to i$. A similar procedure was used by Thouless (1975).

When eqn. (6.8) is inserted into eqn. (6.4) we obtain

$$\langle G_\varepsilon(x'-x)\rangle = i\sum_{m,n} |J_{mn}|^2 \exp[LE_n - (x'-x)(E_m - E_n)], \tag{6.10}$$

where $J_{mn} = \int \Psi_n^*(R)\chi\Psi_m(R)\, dR$. For $x = x'$ we obtain the density of states as the imaginary part of

$$G(0) = i\sum_n \int |\Psi_n|^2|\chi|^2 dR \exp(-LE_n) \quad \text{as } a \to i. \tag{6.11}$$

Any function of Grassmann variables is a polynomial of finite degree with respect to the latter. Hence Ψ_n can be expanded as

$$\Psi_n(S, S^*, \chi, \chi^*) = \Psi_n^{(0)}(S, S^*) + \Psi_n^{(+)}(S, S^*)\chi + \Psi_n^{(-)}(S, S^*)\chi^* + \Psi_n^{(2)}(S, S^*)\chi^*\chi. \tag{6.12}$$

Inserting eqn. (6.12) into eqn. (6.7) yields the equations

$$\left.\begin{aligned}
\left(-\frac{2m}{a}\frac{\partial^2}{\partial S^*\partial S} + V(|S|^2)\right)\Psi_n^{(0)} + \frac{2m}{a}\Psi_n^{(2)} &= E_n\Psi_n^{(0)},\\
\left(-\frac{2m}{a}\frac{\partial^2}{\partial S^*\partial S} + V(|S|^2)\right)\Psi_n^{(2)} + \frac{dV}{d|S|^2}\Psi_n^{(0)} &= E_n\Psi_n^{(2)},\\
\left(-\frac{2m}{a}\frac{\partial^2}{\partial S^*\partial S} + V(|S|^2)\right)\Psi_n^{(\pm)} &= E_n\Psi_n^{(\pm)},
\end{aligned}\right\} \tag{6.13}$$

where $V(|S|^2) = a\varepsilon|S|^2 + \gamma/2|S|^4$. In eqns. (6.13) the last equation containing only $\Psi^{(\pm)}$ decouples from the other two and coincides with the Schrödinger equation for a two-dimensional anharmonic oscillator. We now introduce polar variables r and ϕ through

$$S = r\exp(i\phi) \quad \text{and} \quad \frac{\partial^2}{\partial S^*\partial S} = \frac{1}{4}\left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial\phi^2}\right) \tag{6.14}$$

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and separate r and ϕ by the usual statement

$$\psi_{n,l}=\exp(il\phi)u_{n,l}. \tag{6.15}$$

The solution corresponding to the ground state can be found by a symmetry argument. Supposing that the ground state wavefunction depends only on the 'length' in the superspace

$$\Psi_0=\Psi_0((|S|^2+|\chi|^2)^{1/2}) \tag{6.16}$$

and comparing (6.16) with (6.12) we obtain

$$u_0^{(2)}=\frac{1}{2r}\frac{du_0^{(0)}}{dr}. \tag{6.17}$$

Substitution of eqn. (6.17) into eqn. (6.13) enables us to reduce the first equation of (6.13) to the form

$$-\frac{m}{2a}\left(\frac{d^2}{dr^2}-\frac{1}{r}\frac{d}{dr}\right)u_0^{(0)}+V(r)u_0^{(0)}=Eu_0^{(0)}. \tag{6.18}$$

The second equation of (6.13) now coincides with the derivative of the first one. Equation (6.18) has the lowest eigenvalue $E_0=0$. The eigenfunction $u_0^{(0)}(r)$ of the ground state can be expressed in terms of the Airy function as

$$u_0^{(0)}(r)=\mathrm{Ai}\left(\frac{1}{2}r^2\left(\frac{a}{m}\right)^{1/3}(2\gamma)^{1/2}+2a\varepsilon\left(\frac{a}{m}\right)^{1/3}(2\gamma)^{-2/3}\right). \tag{6.19}$$

For $L\to\infty$ only the term with $E_n=0$ contributes in (6.10) and so the Green function becomes

$$\langle G_\varepsilon(x-x')\rangle=i\sum_m |J_{0m}|^2\exp(-|x'-x|E_m). \tag{6.20}$$

Using eqns. (6.12), (6.14) and (6.17) we can rewrite the matrix element J_{0m} as

$$J_{0m}=\frac{1}{N}\int_0^r ru_0^{(0)}(r)\,\phi_m(r)dr, \tag{6.21}$$

where

$$N^2=\int|\Psi_0|^2dr\int|\Psi_m^+|^2dr=\frac{1}{2}(u_0^{(0)}(0))^2\int|\phi_m|^2rdr.$$

Here $u_0^{(0)}$ is given by eqn. (6.19) and $\Psi_m=\phi_m\chi^*$ where ϕ_m is the mth eigenfunction of the two-dimensional anharmonic oscillator with $l=0$ and energy E_m.

In eqn. (6.21) the following property of the normalization integral for the function Ψ_0 was used

$$\int\Psi_0^2\,dr=\int\left(u_0(r)+\frac{du_0}{dr}\frac{|\chi|^2}{2r}\right)^2dr=u_0^2(0). \tag{6.22}$$

Equations (6.20) and (6.21) completely solve the problem under consideration. The average Green function (6.20) is expressed through the eigenfunctions and eigenvalues of the two-dimensional anharmonic oscillator. At large distances only the ground state contributes.

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For evaluation of the density of states one may retain in eqn. (6.11), as previously, only one term corresponding to the ground state

$$G(0)=i\int|\Psi_0|^2|\chi|^2dr=-2i\frac{\int_0^\infty r(u_0^{(0)}(r))^2dr}{(u_0^{(0)}(0))^2}. \tag{6.23}$$

Integrating in (6.23) by parts and using eqn. (6.18) we reduce eqn. (6.23) to the form

$$G(0)=\frac{i\gamma m}{2}\frac{d^2}{d\varepsilon^2}\ln\mathrm{Ai}\left(\exp\left(\frac{2i\pi}{3}\right)\varepsilon\left(\frac{2}{m\gamma^2}\right)^{1/2}\right). \tag{6.24}$$

The imaginary part of eqn. (6.24) gives the density of states.

For arbitrary $|x-x'|$ the function $\langle G_\varepsilon(x-x')\rangle$ can be written in another form. Performing the Fourier-transformation in eqn. (6.4) we rewrite this equation in the form

$$\langle G_\varepsilon(k)\rangle=i\sum_n\int\Psi_0(R)\chi(P_k(R)+P_{-k}(R))dr\exp(-E_nL), \tag{6.25}$$

where

$$P_k(R)=\int\Gamma(R,R')\chi'^*dR'.$$

Ψ_0 is determined by eqns. (6.15), (6.16) and (6.19). The function $P_k(R)$ satisfies the equation

$$(ik+\mathscr{H}(R))P_k(R)=\chi^*\Psi_n^*(R), \tag{6.26}$$

where $\mathscr{H}$ is determined by eqn. (6.5). We seek the solution of eqn. (6.26) in the form

$$P_k=\chi^*f_k. \tag{6.27}$$

Inserting eqn. (6.27) into eqns. (6.25) and (6.26) we obtain, as $L\to\infty$,

$$G_\varepsilon(k)=2i\frac{\int_0^\infty u_0^{(0)}(r)(f_k(r)+f_{-k}(r))rdr}{(u_0^{(0)}(0))^2}. \tag{6.28}$$

The function $u_0^{(0)}(r)$ in eqn. (6.28) is given by eqn. (6.19). The function $f_k(r)$ satisfies the inhomogeneous equation

$$ikf_k-\frac{1}{2a}\frac{1}{r}\frac{d}{dr}\left(r\frac{df_k}{dr}\right)+V(r)f_k=u_0^{(0)}(r). \tag{6.29}$$

Of course, the representations (6.20), (6.21) and (6.28), (6.29) are equivalent. Using this simple example it has been shown that the transfer-matrix technique worked as well in the case of the inclusion of anticommuting variables. The results (6.20), (6.21), (6.24), (6.28) and (6.29) are known very well (Frish and Lloid 1960, Halperin 1965, Thouless 1975). The scheme of the study of this simple problem enables us to solve a more complicated problem of localization in wires.

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6.3. *Reduction of the one-dimensional σ-model to partial differential equations*

Now let us return to the problem of electron behaviour in wires. In §6.1 it has been shown that this system could be described by the one-dimensional supermatrix σ-model. We suppose, as in the preceding section, that the parameters of the problem correspond to Models I, IIa, IIb and III. The density–density correlator K (3.63) can be rewritten in the form valid for all these models

$$K(x_1, x_2) = -\frac{2\pi^2 \nu \tilde{\nu}}{S} \int Q_{13}^{12}(x_1) Q_{31}^{21}(x_2) \exp(-F[Q]) \mathscr{D}Q, \tag{6.30}$$

$$F[Q] = \frac{\pi \tilde{\nu}}{8} \mathrm{STr} \int [\mathscr{D}(\nabla Q)^2 + 2i\omega \Lambda Q] dx, \tag{6.31}$$

where

$$\tilde{\nu} = \begin{cases} \nu S, & \text{Models I and IIa,} \\ 2\nu S, & \text{Models IIb and III} \end{cases}$$

and S is the cross-section.

The symmetry of the supermatrix Q is specified, as before, by eqns. (3.67), (3.44), (3.41), (3.75) and (3.78). In the first equation of (3.63) the indices α and β of the superblocks Q^{12} and Q^{21} were supposed to be arbitrary. In eqn. (6.30) these indices are fixed and the certain values 1 and 3 are taken. The one dimensionality of the model allows us to find the exact solution (Efetov and Larkin 1983). As in deriving the average Green function in the preceding section we reduce the functional integral (6.30) to a definite integral over the supermatrices Q

$$K(x_1, x_2) = -2\pi^2 \nu \tilde{\nu} \int \Gamma(0, x_1; Q_0, Q_1)(Q_1)_{13}^{12} \Gamma(x_1, x_2; Q_1, Q_2) \times (Q_2)_{31}^{21} \Gamma(x_2, L; Q_2, Q_0)\, dQ_0 dQ_1 dQ_2. \tag{6.32}$$

The kernel Γ in eqn. (6.32) satisfies the Schrödinger equation (cf with (6.5))

$$\left(-\frac{\partial}{\partial x} + \mathscr{H}(Q)\right) \Gamma(x, x'; Q, Q') = \delta(x - x') \delta(Q - Q'). \tag{6.33}$$

The δ-function $\delta(Q-Q')$ appears after taking the limit $\alpha \to 0$ of the following expression

$$\exp\left(-\frac{\mathrm{STr}(Q-Q')^2}{\alpha}\right).$$

The hamiltonian $\mathscr{H}$ can be written in a differential form just as in the preceding section. However, such a representation would require a certain parametrization of the matrix Q. Resulting formulae are cumbersome and an appropriate treatment in difficult. To simplify calculations we shall retain for a while the integral form corresponding to the Schrödinger equation. In this representation the hamiltonian is determined by the equation

$$\left(\mathscr{H} - \frac{i\pi\tilde{\nu}}{4}(\omega + i\delta)\,\mathrm{STr}\,\Lambda Q\right)\Phi(Q) = \lim_{\alpha \to 0} \frac{1}{\alpha} \int \exp\left[-\frac{\pi \tilde{\nu}}{8\alpha} \mathrm{STr}\, \mathscr{D}(Q-Q')^2\right](\Phi(Q') - \Phi(Q)) dQ, \tag{6.34}$$

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$\Phi(Q)$ is an arbitrary matrix. Evaluating the integral in the limit $\alpha \to 0$ in the right-hand side of eqn. (6.34) would give the differential form of the hamiltonian.

Now let us expand the kernel Γ over the orthonormal eigenfunctions Ψ_n of the hamiltonian $\mathscr{H}$

$$\Gamma(x, x'; Q, Q') = \theta(x'-x) \sum_n \Psi_n(Q) \Psi_n^*(Q') \exp[-E_n(x'-x)]. \tag{6.35}$$

The normalization condition is again

$$\int \Psi_n^*(Q) \Psi_n(Q) dQ = 1. \tag{6.36}$$

The order for the functions $\Psi_n(Q)$ and $\Psi_n^*(Q)$ in eqns (6.35) and (6.36) is important, as before.

Inserting eqns. (6.35) and (6.36) into eqn. (6.32) and taking the Fourier-transform with respect to the co-ordinate we obtain

$$K(\omega, k) = -2\pi^2 \nu \tilde{\nu} \sum_n \int \Psi_n(Q) Q_{13}^{12} (P_{k31}^{(n)}(Q) + P_{-k31}^{(n)}(Q)) \exp(-E_n L) dQ, \tag{6.37}$$

where L is the length of the sample.

The function $P_k^{(n)}(Q)$ is related to Γ as

$$P_k^{(n)}(Q) = \int \Gamma(x_1 - x_2; Q, Q') Q'^{21} \Psi_n(Q') \exp[(-ik + E_n)(x_1 - x_2)] dQ' d(x_1 - x_2). \tag{6.38}$$

Multiplying eqn. (6.33) by $Q'^{21} \Psi_n(Q')$ and integrating both sides of this equation over Q' we find after taking the Fourier-transform

$$(ik - E_n + \mathscr{H}) P_k^{(n)}(Q) = Q^{21} \Psi_n(Q). \tag{6.39}$$

As we have mentioned, $\Psi_n(Q)$ and E_n are eigenfunctions and eigenvalues of the Schrödinger equation

$$\mathscr{H} \Psi_n = E_n \Psi_n. \tag{6.40}$$

For further calculations it is convenient to use, as in the previous section, the polar co-ordinates (5.25). Values of Q' close to those of Q are only essential in the integral (6.34). In this limit the supertrace of $(Q-Q')^2$ in the exponent can be written as

$$\mathrm{STr}\,(Q-Q')^2 = \Delta F_0 + \Delta F_1,$$

where

$$\left.\begin{aligned} &\Delta F_0 = 2\,\mathrm{STr}\,(\Delta\theta)^2, \\ &\Delta F_1 = 2\,\mathrm{STr}\,[(\Delta u \cos\tilde{\theta})^2 + (\Delta v \cos\tilde{\theta})^2 + 2\Delta u \sin\tilde{\theta} \Delta v \sin\tilde{\theta} - (\Delta u)^2 - (\Delta v)^2], \\ &\Delta\tilde{\theta} = \theta' - \theta, \quad \Delta u = -\Delta\bar{u} = -\ln(\bar{u}u'), \Delta v = -\Delta\bar{v} = -\ln(\bar{v}v'), \quad \tilde{\theta} = \frac{\theta + \theta'}{2}. \end{aligned}\right\} \tag{6.41}$$

The quadratic form (6.41) is written with the precision necessary for obtaining second derivatives in the Schrödinger equation. For infinitesimal increments eqns. (6.41) coincide with eqn. (5.28).

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The absence of weight denominators is due to the use of integrals over both commuting and anticommuting variables. In the limit $L\to\infty$ only the ground state in the sum over n in eqn. (6.32) contributes. The right-hand side of eqn. (6.32) is finite and does not turn into zero if only the ground state energy E_0 is zero. As in the preceding section, we assume that the wavefunction of the ground state Ψ_0 has a high symmetry under transformations in the space of the supermatrices Q and depends only on the 'radial' variables θ. For such functions the normalization conditions take the form (cf with (6.22)).

$$\int \Psi^2(\hat{\theta})dQ=\Psi^2(0)=1. \tag{6.42}$$

Equation (6.42) can be proved in the following way. Let us consider the integral

$$I(\beta)=\int f(\hat{\theta})\exp\left[\beta\,\mathrm{STr}\,(\Lambda Q)\right]dQ, \tag{6.43}$$

where $f(\hat{\theta})\to f(0)\neq 0$ as $\theta\to 0$. It seems to be convenient to calculate the integral (6.43) using polar co-ordinates (5.25). However, the arising Berezinian (5.34) is singular when $\hat{\theta}\to 0$. So, the integral over $\hat{\theta}$ becomes divergent. On the other hand the integrand does not depend on supermatrices u and v from eqn. (5.27) thus providing the vanishing of the contribution when integrating over anticommuting variables. Therefore we have an uncertainty of the type $0\cdot\infty$. However, integration over $\hat{\theta}$ in the integral $\partial I(\beta)/\partial\beta$ does not lead to divergencies. Further integration over the anticommuting variables gives zero,

$$\frac{\partial I(\beta)}{\partial\beta}=0. \tag{6.44}$$

Therefore we get

$$I(\beta)=C, \tag{6.45}$$

where C is independent of β. To calculate C we may consider the limit $\beta\to\infty$. In this limit $f(\hat{\theta})$ in eqn. (6.43) can be replaced by $f(0)$. Calculating the integral by the perturbation methods of §4, which are valid in the limit $\beta\to\infty$, we come to the conclusion that

$$\int f(\hat{\theta})\exp\left[\beta\,\mathrm{STr}\,(\Lambda Q)\right]dQ=f(0) \tag{6.46}$$

for an arbitrary β and f decreasing at infinity. From the identity (6.46) eqn. (6.42) can be easily obtained.

Now assuming that the ground state eigenfunction Ψ_0 depends only on $\hat{\theta}$ we can derive the explicit Schrödinger equation for it. So we put $\Phi(Q)=\Psi_0(\hat{\theta})$ in eqn. (6.34) and integrate over u and v. For the explicit evaluation the parametrization (5.30) can be used. In the limit $\alpha\to 0$ all the arising integrals in (6.34) are gaussian with respect to Δu and Δv. Calculating these integrals we reduce the right-hand side of eqn. (6.34) to the form

$$\lim_{\alpha\to 0}\frac{1}{\alpha}\left(\frac{\tilde{\nu}\mathscr{D}}{2\alpha}\right)^q\int\exp\left[-\frac{\pi\tilde{\nu}\mathscr{D}}{4}\,\mathrm{STr}\,(\Delta\hat{\theta})^2\right]J(\hat{\theta}')J^{-1}(\tilde{\hat{\theta}})(\Psi_0(\hat{\theta}')-\Psi_0(\hat{\theta}))d\hat{\theta}', \tag{6.47}$$

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where

$$q=\begin{cases}\dfrac{3}{2}, & \text{Models I and III,}\\[2ex] 1, & \text{Models IIa and IIb.}\end{cases}$$

The Berezinian $J(\theta')$ in eqn. (6.47) arises after changing the variables (5.25), $J^{-1}(\tilde{\theta})$ is the result of integration of the exponentials (6.34) and (6.41).

Expanding $J(\theta')$, and $J^{-1}(\tilde{\theta})$ and $\Psi_0(\theta')$ over $\theta'-\theta$ and calculating gaussian integrals we find in the limit $\alpha\to 0$

$$\mathscr{H}_0\Psi_0=0, \tag{6.48}$$

$$\mathscr{H}_0=-\frac{1}{2\pi\tilde{\nu}\mathscr{D}}\left[\frac{1}{J_\lambda}\frac{\partial}{\partial\lambda}\left(J_\lambda|1-\lambda^2|\frac{\partial}{\partial\lambda}\right)+\sum_m\frac{1}{J_\lambda}\frac{\partial}{\partial\lambda_m}\left(J_\lambda|1-\lambda_m^2|\frac{\partial}{\partial\lambda_m}\right)\right]-i(\omega+i\delta)\pi\tilde{\nu}|\prod_m\lambda_m-\lambda|. \tag{6.49}$$

The hamiltonian (6.49) is written down through variables λ, λ_1 and λ_2 (5.35 *a*). The index m in eqn. (6.49) can take the values 1 and 2 in Models I and III and is equal to 1 in Model II. The Berezinian J_λ corresponds to the use of the variables λ, λ_1 and λ_2 and is equal to

$$J_\lambda=\frac{|1-\lambda^2|}{(\lambda_1^2+\lambda_2^2+\lambda^2-2\lambda\lambda_1\lambda_2-1)^2}, \qquad \text{(Models I and III)} \tag{6.50}$$

$$J_\lambda=\frac{1}{(\lambda_1-\lambda)^2}. \qquad \text{(Model II)} \tag{6.51}$$

The Berezinian (6.50) can be obtained from eqn. (5.34) by integrating over $\Delta\phi$, $\Delta\chi$, Δm, Δm_1 and Δm_1^* and using the variables Δu and Δv instead of the variables $\Delta\eta$ and $\Delta\kappa$.

As we have mentioned, only the ground state contributes in the sum (6.37). So, in order to calculate the density–density correlator $K(\omega, k)$ we have to solve eqn. (6.39) with the ground state wavefunction $\Psi_0(\tilde{\theta})$ in the right-hand side and the zero ground state energy $E_0=0$. The matrix $P_k^{(0)}$ satisfying this equation can be sought in the form

$$P_k=-iuR_k\bar{v}2\pi\tilde{\nu}\mathscr{D}, \quad R_k=\begin{pmatrix}R_{11k} & 0\\ 0 & R_{22k}\end{pmatrix}, \tag{6.52}$$

$$R_{11k}=\begin{pmatrix}(1-\lambda^2)^{1/2}f_k & 0\\ 0 & (1-\lambda^2)^{1/2}f_k\end{pmatrix}, \qquad R_{22k}=i\begin{pmatrix}(\lambda_1^2-1)^{1/2}\lambda_2 f_{1k} & (\lambda_2^2-1)^{1/2}\lambda_1 f_{2k}\\ (\lambda_2^2-1^{1/2}\lambda_1 f_{2k} & (\lambda_1^2-1)^{1/2}\lambda_2 f_{1k}\end{pmatrix},$$

(Model I)

$$R_{11k}=\begin{pmatrix}(1-\lambda^2)^{1/2}f_k & 0\\ 0 & (1-\lambda^2)^{1/2}f_k\end{pmatrix}, \qquad R_{22k}=i\begin{pmatrix}(\lambda_1^2-1)^{1/2}f_{1k} & 0\\ 0 & (\lambda_1^2-1)^{1/2}f_{1k}\end{pmatrix},$$

(Model II) (6.53)

$$R_{11k}=\begin{pmatrix}(1-\lambda_1^2)^{1/2}\lambda_2 f_{1k} & (1-\lambda_2^2)^{1/2}\lambda_1 f_{2k}\\ (1-\lambda_2^2)^{1/2}\lambda_1 f_{2k} & (1-\lambda_1^2\lambda_2 f_{1k}\end{pmatrix}, \quad R_{22k}=i\begin{pmatrix}(\lambda^2-1)^{1/2}f_k & 0\\ 0 & (\lambda^2-1)^{1/2}f_k\end{pmatrix}.$$

(Model III)

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When deriving $\mathscr{H}_0$ (6.48) we used the independence of the ground state wavefunction Ψ_0 of the matrices u and v. The form (6.52) for the function P_k containing only the first powers of u and v allows the derivation of the appropriate hamiltonian for the function P_k depending on $\hat{\theta}$ only. We substitute $\Phi(Q)=P_k$ into eqn. (6.34) and integrate first over Δu and Δv. Using the expansion, following from eqns. (6.41)

$$u'=u\left(1-\Delta u+\frac{(\Delta u)^2}{2}\right), \quad v'=v\left(1-\Delta v+\frac{(\Delta v)^2}{2}\right) \tag{6.54}$$

we reduce the right-hand side of eqn. (6.34) to the form

$$v\left(\mathscr{H}_0 R_k-\frac{1}{2}\langle(\Delta v)^2\rangle_1 R_k-\frac{1}{2}R_k\langle(\Delta u)^2\rangle_1+\langle\delta v R_k\delta u\rangle_1\right)\bar{u}. \tag{6.55}$$

The symbol $\langle\ldots\rangle_1$ stands for the following averaging

$$\langle\ldots\rangle_1=\frac{\int(\ldots)\exp(-F_1)d(\Delta u)d(\Delta v)}{\int\exp(-F_1)d(\Delta u)d(\Delta v)}.$$

The matrices u and v in eqn. (6.55) decouple from the variable $\hat{\theta}$. Therefore, calculating the averages in eqn. (6.55) and substituting this equation into eqn. (6.34) yields differential equations containing only the variables θ, θ_1 and θ_2. This procedure is similar to that used in quantum mechanics problems with problems with a central field. In such problems angular variables decouple but give rise to additional terms in radial parts. In Model II, there are only two independent variables, λ and λ_1, and two unknown functions, f_k and f_{1k}. Performing the appropriate calculations we find

$$\left.\begin{aligned}2\pi\tilde{\nu}\mathscr{D}(ik+\mathscr{H}_0)f_k+2\lambda\frac{\partial f_k}{\partial\lambda}+2(\lambda_1^2-1)J_\lambda(f_k-f_{1k})=\Psi_0,\\ 2\pi\tilde{\nu}\mathscr{D}(ik+\mathscr{H}_0)f_{1k}-2\lambda_1\frac{\partial f_{ik}}{\partial\lambda_1}+2(1-\lambda^2)J_\lambda(f_{1k}-f_k)=\Psi_0.\end{aligned}\right\} \tag{6.56}$$

The corresponding equations for Models I and III are more complicated. After somewhat cumbersome calculations we get

$$\left.\begin{aligned}&2\pi\tilde{\nu}\mathscr{D}(ik+\mathscr{H}_0)f_k+2(-1)^s\lambda\frac{\partial f_k}{\partial\lambda}+4(\lambda_2|\lambda_1^2-1|\beta(\lambda_1,\lambda,\lambda_2)(f_k-f_{1k})\\ &\qquad+\lambda_1|\lambda_2^2-1|\beta(\lambda,\lambda_2,\lambda_1)(f_k-f_{2k}))=\Psi_0,\\ &2\pi\tilde{\nu}\mathscr{D}(ik+\mathscr{H}_0)f_{ak}-2(-1)^s\left(\lambda_a\frac{\partial f_{ak}}{\partial\lambda_a}+\frac{|\lambda_b^2-1|}{\lambda b}\frac{\partial f_{ak}}{\partial\lambda_b}\right)\\ &+4\left(\frac{\lambda_a|\lambda_b^2-1|}{\lambda_b}\beta(\lambda_a,\lambda_b,\lambda)(f_{bk}-f_{ak})+\frac{|1-\lambda^2|}{\lambda_b}\beta(\lambda_a,\lambda,\lambda_b)(f_{ak}-f_k)\right)=\Psi_0.\end{aligned}\right\} \tag{6.57}$$

In eqns. (6.57) $s=0$ for Model I and $s=1$ for Model III. The indices a and b can take the values 1 and 2, $a\neq b$. The function of three variables $\beta(x,y,z)$ is equal to

$$\beta(x,y,z)=\frac{-2xy+z-z^3+zx^2+zy^2}{(x^2+y^2+z^2-2xyz-1)^2}. \tag{6.58}$$

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Inserting eqn. (6.52) into eqn. (6.37) and integrating over u and v we reduce the density–density correlation K to the form

$$K(\omega,k)=\tilde{K}(\omega,k)+\tilde{K}(\omega_1-k), \tag{6.59}$$

$$\tilde{K}=4\pi^3\nu\tilde{\nu}^2\mathscr{D}\int_1^\infty\int_{-1}^1\Psi_0 J_\lambda((\lambda_1^2-1)f_{1k}+(1-\lambda^2)f_k)d\lambda_1 d\lambda, \quad \text{(Model II)} \tag{6.60}$$

$$\tilde{K}=4\pi^3\nu\tilde{\nu}^2\int\int\int\Psi_0 J_\lambda(|\lambda_1^2-1|\lambda_2^2 f_{1k}+|\lambda_2^2-1|\lambda_1^2 f_{2k}+|1-\lambda^2|f_k)d\lambda_1 d\lambda_2 d\lambda,$$

(Models I and III) (6.61)

In eqn. (5.61) integration is performed over the domain $1<\lambda_1<\infty$, $1<\lambda_2<\infty$ and $-1<\lambda<1$ for Model I and $-1<\lambda_1<1$, $0<\lambda_2<1$ and $1<\lambda<\infty$ for Model III. Equations (6.42), (6.48), (6.49), (6.56), (6.57), (6.59), (6.60) and (6.61) completely solve the problem of quantum particle kinetics in a long metallic wire for all types of symmetry. Unfortunately eqns. (6.48), (6.49), (6.56) and (6.57) are very complicated and can be solved analytically only in the limiting cases of high, $\omega\ll 1/\tilde{\nu}^2\mathscr{D}$, and low, $\omega\ll 1/\tilde{\nu}^2\mathscr{D}$, frequencies. In the intermediate region results can be only numerically obtained. In the next section the limiting cases of high and low frequencies will be considered.

6.4. *Density–density correlator and dielectric permeability*

Let us first consider the limit of high frequencies, $\omega\gg(\tilde{\nu}^2\mathscr{D})^{-1}$. Of course, for calculations in this domain the use of the transfer-matrix method is not necessary. The perturbation theory developed in the §4 is much more convenient and is applicable in any dimension. However, the calculation for high frequencies gives the possibility of checking the equations obtained and shows how the transfer-matrix method works. At high frequencies the essential values λ, λ_1 and λ_2 are close to unity in all three models. Therefore, it is convenient to use the following variables

$$\left.\begin{aligned}
&\lambda_1=1+x_1, \quad \lambda_2=1+x_2, \quad \lambda=1-x, && \text{(Model I)}\\
&\lambda_1=1+x_1, \quad \lambda=1-x, && \text{(Model II)}\\
&\lambda_1=1-x_1, \quad \lambda_2=1-x_2, \quad \lambda=1+x, && \text{(Model III)}
\end{aligned}\right\} \tag{6.62}$$

When integrating over x_1, x_2 and x in eqns. (6.60) and (6.61), the main contribution comes from $x_1\sim x_2\sim x\sim(\mathscr{D}\tilde{\nu}^2\omega)^{-1/2}\ll 1$. In the principal approximation one may integrate in eqns. (6.60) and (6.61) over all positive x_1, x_2 and x. Expanding in eqns. (6.49), (6.50) and (6.57) over small x_1, x_2 and x and then solving these equations in this limit we find

$$\left.\begin{aligned}
&\Psi_0=\exp[-(x_1+x)\gamma], \quad f_k\approx f_{1k}=\frac{1}{2\pi\tilde{\nu}\mathscr{D}}\frac{1}{ik+(-i\omega/\mathscr{D})^{1/2}}\exp(-(x_1+x)\gamma)\\
&\qquad\qquad\qquad

we find in the principal approximation for all the models

$$K(\omega, k)=\frac{4\pi\nu}{-i\omega+\mathscr{D}k^2}. \tag{6.64}$$

Equation (6.64) describes the classical diffusion and coincides with eqn. (3.66). (Recall that the density–density correlator K (6.64) is calculated for particles with spin but the correlator (3.66) was calculated for spinless particles. It results in the extra factor 2 in eqn. (6.64).)

Now let us consider a more interesting case, $\omega \ll (\tilde{\nu}^2\mathscr{D})^{-1}$. In this limit the main contribution comes from the domain $\lambda_1 \gg 1$, $\lambda_2 \gg 1$, and $\lambda \sim 1$ in Model I, $\lambda_1 \gg 1$ and $\lambda \sim 1$ in Model II and $\lambda_1, \sim 1$, $\lambda_2 \sim 1$ and $\lambda \gg 1$ in Model III. In these domains the variables of eqns (6.49), (6.56) and (6.57) and integrals (6.60) and (6.61) can also be considerably simplified. In formulae (6.49), (6.56) and (6.60) one may keep only the function f_{1k} and neglect the function f_k for Model II. In formulae (6.49), (6.57) and (6.61) one may retain only the functions f_{1k} and f_{2k}, for Model I and f_k for Model III. In the principal approximation we seek solutions of the equations in the form

$$\left.\begin{aligned} &\Psi_0=\Psi_0(\lambda_1\lambda_2), \quad f_{1k}=f_{2k}=\Phi_k(\lambda_1\lambda_2), && \text{(Model I)}\\ &\Psi_0=\Psi_0(\lambda_1), \quad f_{1k}=\Phi_k(\lambda_1), && \text{(Model II)}\\ &\Psi_0=\Psi_0(\lambda), \quad f_k=\Phi_k(\lambda). && \text{(Model III)} \end{aligned}\right\} \tag{6.65}$$

Changing the variables according to the formulae

$$\left.\begin{aligned} &z=-i\omega\pi^2\tilde{\nu}^2\mathscr{D}\lambda_1\lambda_2, && \text{(Model I)}\\ &z=-i\omega 2\pi^2\tilde{\nu}^2\mathscr{D}\lambda_1, && \text{(Model II)}\\ &z=-i\omega 2\pi^2\tilde{\nu}^2\mathscr{D}\lambda && \text{(Model III)} \end{aligned}\right\} \tag{6.66}$$

and calculating integrals over the remaining variables we obtain instead of (6.49), (6.56), (6.57), (6.59), (6.60) and (6.61) the following equations

$$\left.\begin{aligned} &-z\frac{d^2\Psi_0}{dz^2}+\Psi_0=0,\\ &-\frac{d}{dz}\left(z^2\frac{d\Phi_k}{dz}\right)+ikL_c\Phi_k+z\Phi_k=\Psi_0(z),\\ &K(\omega,k)=\frac{4\pi\nu A(k)}{-i\omega}, \quad A(k)=\int_0^\infty(\Phi_k(z)+\Phi_{-k}(z))\Psi_0(z)dz. \end{aligned}\right\} \tag{6.67}$$

In eqns. (6.67) the length L_c is equal to

$$L_c=\begin{cases}\pi\tilde{\nu}\mathscr{D}, & \text{Model I},\\ 2\pi\tilde{\nu}\mathscr{D}, & \text{Models II and III}.\end{cases} \tag{6.68}$$

Recalling eqns. (6.31) for $\tilde{\nu}$ we can rewrite L_c in an explicit form valid for all the models

$$L_c=\begin{cases}\pi\nu S\mathscr{D}, & \text{Model I},\\ 2\pi\nu S\mathscr{D}, & \text{Model IIa},\\ 4\pi\nu S\mathscr{D}, & \text{Models IIb and III}.\end{cases} \tag{6.69}$$

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Thus, in all the considered models the density–density correlator is determined by the same equations. Only the characteristic length L_c is different. The solutions $\Psi_0(z)$ and $\Phi_k(z)$ have to decrease at infinity. Furthermore, the function $\Psi_0(z)$ must satisfy the boundary condition

$$\Psi_0(0)=1. \tag{6.70}$$

The condition (6.70) is a consequence of the fact that at $\omega=0$ eqn. (6.49) has the exact solution $\Psi(\lambda_1,\lambda_2,\lambda)=1$obeying the normalization condition (6.42). Then the limit $\omega\to 0$ in terms of the variables z (6.66) corresponds to the condition (6.70). Furthermore the, integral for $A(k)$ in eqn. (6.67) must be convergent provided the main contribution comes from the domain $z\sim 1$. Therefore, the function Φ_k must be integrable. Other possible functions Φ_k, for which the domain $\lambda,\lambda_1,\lambda_2\sim 1$ is essential in the integrals (6.60) and (6.61) give a contribution of a higher order in ω.

Equations (6.67) with the boundary conditions pointed out above exactly coincide with the low-frequency limit of the Berezinsky equations written for the problem of particle kinetics in a one-dimensional disordered chain (Berezinsky 1973) provided the length L_c in eqns. (6.67) is substituted by the mean free path l. Solving eqns. (6.67) we find

$$\left.\begin{aligned}
&\Psi_0(z)=2\sqrt{z}\,K_1(2\sqrt{z}),\\
&\Phi_k(z)=\frac{2}{\sqrt{z}}\Bigg[I_{\sqrt{(1+4i\kappa)}}(2\sqrt{z})\int_{2\sqrt{z}}^{\infty}\xi K_{\sqrt{(1+4i\kappa)}}(\xi)K_1(\xi)\,d\xi\\
&\qquad +K_{\sqrt{(1+4i\kappa)}}(2\sqrt{z})\int_0^{2\sqrt{z}}\xi I_{\sqrt{(1+4i\kappa)}}(\xi)K_1(\xi)\,d\xi\Bigg].\\
&\kappa=kL_c.
\end{aligned}\right\} \tag{6.71}$$

K_α and I_α are the Bessel functions. The functions Ψ_0 and Φ_k (6.71) decrease at infinity. The function Ψ_0 satisfies the condition (6.70), the function Φ_k is integrable at small z.

Substituting eqns. (6.71) into the integral (6.67) and carrying out the same calculations as those in the work of Gogolin *et al.* (1975) we obtain for the density–density correlator at large distances, $x\gg L_c$, and times, $t\to\infty$,

$$K(\varepsilon,t\to\infty,x)\equiv p_\infty(x)=\frac{1}{4\sqrt{\pi L_c}}\left(\frac{\pi^2}{8}\right)^2\left(\frac{4L_c}{|x|}\right)^{3/2}\exp\left(-\frac{|x|}{4L_c}\right). \tag{6.72}$$

Equation (6.72) shows that the correlator p_∞ decreases exponentially with the localization length L_c specified by formulae (6.69). This fact proves the localization of all states in the system.

In the range of small momenta $k\ll L_c^{-1}$ the function $A(k)$ takes the form

$$A(k)=1-4\zeta(3)k^2L_c^2, \tag{6.73}$$

where $\zeta(x)$ is the Riemann ζ-function. The function $A(k)$ determines at small momenta the dielectric permeability ε

$$\varepsilon=-4\pi e^2\nu\left.\frac{d^2A(k)}{dk^2}\right|_{k=0}=32\zeta(3)e^2\nu L_c^2. \tag{6.74}$$

The formulae (6.72)–(6.74) have the same form as those in the model of a one-dimensional disordered chain. However, now the localization length considerably exceeds the mean free path l, $L_c\sim l(p_0^2s)$.

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In the limit of high frequencies, $\omega \gg \mathscr{D}/L_c^2$, where the classic diffusion formula (6.64) is applicable, the conductivity σ, according to the Einstein relation, is equal to

$$\sigma = 2\nu e^2 \mathscr{D}. \tag{6.75}$$

Decreasing the frequency ω leads to a smooth crossover from eqn. (6.75) to eqn. (6.74).

The dependence on the frequency is different for different models. In particular, this dependence is not monotonous in Model III, where the first correction to the conductivity at high frequencies is positive (4.15).

The dependence of the localization length L_c on magnetic and spin–orbit interactions (6.69) results, according to eqn. (6.74), in an appropriate dependence of the dielectric permeability. One can estimate the order of the magnitude of interactions corresponding to the crossover between Models I, IIa, IIb and III by evaluating, as in the § 5, the first correction to the diffusion coefficient. The external magnetic field leads to the crossover from Model I, which is invariant under the time reversal and spin rotations. to Model IIa. In the latter model the dielectric permeability is four times as large as the one in Model I. This crossover corresponds to fields H of the order of $H \sim \phi_0(\sqrt{S}L_c)^{-1}$, where $\phi_0 = \pi\hbar c/e$ is the flux quantum. If a sufficiently high concentration of magnetic or spin–orbit impurities is added (Models IIb and III)

$$\tau_s^{-1},\ \tau_{so}^{-1} \gg \mathscr{D}/L_c^2, \tag{6.76}$$

the dielectric permeability according to formulae (6.69) and (6.74) is 16 times as large as the one in a wire without any magnetic or spin–orbit impurities.

§ 7. Discussion

The supersymmetry method developed in the preceding sections seems to be a powerful tool for studying disordered metals. This method made it possible to reduce the problems with disorder to a regular model which can be investigated by well-developed methods of solid-state physics and field theory. In §§ 5 and 6 the problems of level statistics in a limited volume and of localization in long wires have been solved for the first time. The former problem is interesting not only for the theory of disordered metals. The study of stochastic processes in quantum and classical systems is receiving much attention (see, for example Zaslavsky 1979), but the Dyson statistical hypothesis has not yet been confirmed. Therefore, the problem of level statistics in small metallic particles solved above is a serious confirmation of the applicability of Dyson statistics for many quantum systems. An assertion has been made above that the Dyson statistics are not applicable for systems with localization. Nevertheless, it is possible to assume that low-frequency limiting behaviour as ω, ω^2 and ω^4 for the orthogonal, unitary and symplectic ensembles is correct for such systems also, provided the frequency is small enough, i.e. $\omega \ll \Delta \exp(-L/L_c)$, where Δ is the mean level spacing and L is the size of the system. In this limit an electron can travel throughout the sample. This universality agrees with the universality of the type ω^h, where h is the Kolmogorov entropy (see, for example, Zaslavsky 1979). However, the entropy h has not been calculated so far for non-trivial cases. The results obtained in § 5 enable one to calculate this entropy and determine possible universality classes.

The localization in wires is also interesting because it occurs even for large mean free paths in a system which is almost in all respects three dimensional. Only the

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geometry of the sample is one dimensional. Due to this property wires differ from chains. However, as shown in § 6, the low-frequency behaviour is the same for both wires and chains, only the characteristic localization lengths are different. Magnetic and spin–orbit interactions influence the localization length, conserving the form of the correlation functions. Of course, the behaviour of electrons at higher frequencies is different in different models.

It should be noted that the derived σ-model is obtained in the limit of weak disorder. So, the σ-model is formally applicable only to the study of low dimensional systems: zero dimensional, one dimensional and two dimensional, where interesting effects exist for a weak disorder. In conventional three-dimensional systems the diffusion modes give, in the limit $\tau\varepsilon_0 \gg 1$, only small corrections to the classical diffusion. The interesting phenomenon of the Anderson transition can be studied in the framework of the continuum σ-model in $2+\varepsilon$ dimensions for small ε only.

However, the method developed can be applied after small changes to real granular materials. Such substances consist of metallic particles covered by a layer of an oxide (see, for example, Abeles *et al.* 1975, Perenboom *et al.* 1981). There is a probability of tunnelling from one granule to another. Substances with different tunnelling probabilities between granules and different granule sizes are now available experimentally. The case of one granule has already been considered in § 5. Tunnelling from one granule to another leads to coupling between different granules. The free energy of this system can be written as (cf (3.62) and (5.24))

$$F = \mathrm{STr} \sum_{i,j} \int \left[-J_{ij} Q_i Q_j + \frac{i\pi\omega}{4\Delta} \Lambda Q_i \delta_{ij} \right]. \tag{7.1}$$

In eqn. (7.1) the coupling constant J_{ij} is proportional to $|T_{ij}|^2/\Delta^2$, where T_{ij} is the amplitude of tunnelling from a granule i to granule j and Δ is the mean spacing in an isolated granule. The model described by eqn. (7.1) is the σ-model on a lattice. One can expect that this model undergoes the Anderson transition from the metal to the dielectric at $J = J_c \sim 1$. The transition in this system is more convenient for studying because it is the 'pure' Anderson transition. Fluctuations of the density of states in the model (7.1) are absent, the density of states being independent of J. The model (7.1) and corresponding granular materials warrant detailed investigations.

Two-dimensional models with potential scattering and magnetic interactions (Models I, IIa and IIb according to the classification of § 4) remain unsolved. The renormalization-group equations lead to logarithmic poles and are not applicable when the 'effective temperature' becomes of the order of unity. Apparently, the two-dimensional localization problem is complicated because even a simpler two-dimensional vector σ-model has not been solved. Only Model III with spin–orbit interaction, where the renormalization-group equations lead the 'zero-charge' situation, is an exception. In this case the renormalization group equations solve the problem completely.

In this review only systems of non-interacting particles were considered. In real metals the Coulomb interaction and interactions with phonons are very important. Phonons destroy the quantum interference and diminish corrections to the classical diffusion. The phonon contribution can be neglected at low temperatures but the Coulomb interaction is always important (Altshuler and Aronov 1979, Altshuler *et al.* 1980). Attempts to write down renormalization-group equations and to study the behaviour near the mobility edge has already been made (McMillan 1981, Lee 1982)

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but involve qualitative discussion only. In two dimensions the effect of the Coulomb interaction in the model with broken time-reversal symmetry was considered by the replica method by Finkelstein (1982). In that work the renormalization-group equations were derived and were shown to be dependent on the interaction. Unfortunately calculations with the interaction are very complicated. The problem of localization of interacting particles is still far from being solved. However, one can hope that the theory of non-interacting particles describes correctly characteristic properties of disordered metals. The theory of interacting particles is a matter for the future.

Consideration of all work on disordered metals was not the aim of this review. Many theoretical and experimental studies not directly related to the principal subject of the review are not mentioned. There are many experimental studies of the corrections to the classical diffusion. According to the formula (4.15) the correction to the diffusion coefficient becomes large, in low dimensions, at low frequencies. Phonons cut this correction. The value $-i\omega$ has to be replaced by $-i\omega+\tau_\varepsilon^{-1}$, where τ_ε^{-1} is the inverse time of scattering on phonons. This value is proportional to T^p, where T is temperature and the exponent p depends on details of phonon dynamics. Experimentally one can study the dependence of the conductivity on temperature and magnetic field. In two dimensions the first correction to the conductivity is proportional to the logarithm of the temperature. The logarithmic increase of the resistivity with decreasing temperature was first observed experimentally in metallic films by Dolan and Osheroff (1979). A similar change of the resistivity was observed in inversion layer systems (Kawaguchi *et al.* 1978, Bishop *et al.* 1980, Uren 1980). A very interesting effect has been predicted by Altshuler *et al.* (1981). According to their calculations the resistivity of a metallic cylinder oscillates with the changing magnetic field inside this cylinder. This prediction is also based on consideration of the first correction in eqn. (4.15) for the sample geometry involved. This effect is analogous to that of Little and Parks (1962) in superconductors but is possible in dirty normal metals. The oscillating dependence of the conductivity on the magnetic field was soon observed experimentally (Sharvin and Sharvin 1981).

In long wires the resistivity must increase with decreasing temperature too. However, now it is a power-low dependence. Such behaviour has been observed many times (Giordano *et al.* 1979, Giordano 1979, 1981, Chaudhari and Habermeier 1980, White *et al.* 1982) but unfortunately, all such measurements were carried out at temperatures between 1 and 10 K, for which the corrections to the classical diffusion coefficient are small. It would be very interesting to carry out measurements at low temperatures when the wire should become a dielectric. According to formulae (6.56) and (6.61) the dielectric permeability ε is related to the diffusion coefficient $\mathscr{D}$. This coefficient can be measured at high temperatures or frequencies when electrons can diffuse freely in the sample. An observation of the dependence of the dielectric permeability on the external magnetic field would also be interesting. It is easier to carry out such observations for thin strongly disordered wires because in this case the corrections to the classical diffusion (4.15), with the replacement $\omega+i\tau_\varepsilon^{-1}$, become large not only for very low temperatures. The dielectric state in dirty films can be obtained only with greater difficulty because the localization length is exponentially large. In this case exponentially low temperatures are necessary.

The possibility of an experimental observation of level correlations has been discussed (Devaty and Sievers 1980, Perenboom *et al.* 1981). The mean level spacing Δ in real granular materials corresponds to the far infrared range of

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frequencies. Therefore, attempts to observe the oscillations in the absorption in this region, which would correspond to the level spacing, have been made (Granqvist *et al.* 1976, Granqvist 1978). However, the oscillations have not been observed. This fact can be explained easily. In real granular materials the granules have different sizes and, hence, different level spacing. This results in smearing the levels in the absorption spectrum. Estimates by Devaty and Sievers (1980) show that even for very small deviations of sizes from the average all the oscillations disappear. The experimental methods of preparing granular metals do not yet give uniform sizes with sufficient precision and study of the absorption of a separate granule is also impossible now.

To summarize, it has been shown that the supersymmetry method enables one to study problems which are interesting both from the theoretical and the experimental points of view. I hope this method will be used to solve many other problems in the theory of disordered metals.

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